

UNIVERSITÀ DEGLI STUDI DI MESSINA

DIPARTIMENTO DI SCIENZE MATEMATICHE E INFORMATICHE,
SCIENZE FISICHE E SCIENZE DELLA TERRA
Bachelor's Degree in Computer Science L31
Curriculum: Data Analysis

Tiered Edge AI Architecture for Cooperative UAV-Based Industrial Fire Detection

Supervisor:
Prof. Giovanni Merlino

Candidate:
Mohammadamin Jahanimajd

Academic Year
2025/2026

Abstract

Resource-constrained unmanned aerial vehicles cannot assume cloud connectivity or identical computing platforms when monitoring visually complex industrial environments. This thesis develops a tiered edge architecture in which a Sentinel performs initial hazard detection, temporal filtering and target designation, while a Scout performs secondary visual hazard verification. The study combines source-aware dataset curation, a frozen six-model comparison, conversion-fidelity checks, device-specific integration, ROS 2/DDS messaging, stationary two-node trials and development simulation. In the implemented physical path, camera capture, inference, inter-device messaging and final decision handling remained on the two edge nodes; the laboratory laptop was an operator and evidence interface rather than an inference server.

The model comparison did not identify a universal winner. YOLOv8n-416 led validation AP50–95 and F1, whereas EfficientDet led the one-time test comparison; all candidates lost F1 from validation to test, and their grouped-bootstrap intervals overlapped. The historical deployed pair executed on both edge platforms. In retained camera-free benchmarks, the final 640-pixel Sentinel TensorRT pipeline averaged 58.151 ms and the Scout TFLite pipeline averaged 999.431 ms. Bounded stationary trials produced correlated Scout confirmations and rejections, with mean alert-to-final-decision time of approximately 1.44–1.46 s in accepted positive conditions. Two 300-event communication campaigns returned all application-level replies, with median round-trip times near 5–6 ms. A stationary complete-stack run exceeded 30 minutes on both devices without process death, OOM, thermal shutdown or logger loss. Its corrected 36-event probe returned every correlated verdict, but its 2.618 s p95 alert-to-verdict time failed the registered 2 s target and Sentinel swap increased by 0.922 MiB. A separate 1,000-event ROS 2 campaign under live inference load delivered every correlated reply with 7.984 ms p95 round-trip

time. Five-repetition interface, stale-frame, unavailable-process and late/duplicate-verdict faults reached their specified fail-closed outcomes. A hash-linked Gazebo run completed one confirmed-hazard mission from patrol to Scout verification, alarm publication and landing using distinct airframe proxies.

The evidence supports role division and bounded 30-minute concurrent operation across the tested heterogeneous nodes under stationary laboratory conditions. It does not establish improved accuracy over a common baseline or compliance with every registered runtime criterion. Temporal-filter effects, formal mission reliability, external transfer, geographic localization and physical two-vehicle flight remain unestablished. The final physical evidence concerns visible flame appearance in RGB imagery; physical smoke and industrial-site operation were not evaluated. The thesis therefore contributes a traceable prototype and a prioritized validation path, not a claim of deployment readiness.

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Acronyms

AI	Artificial intelligence
AP	Average precision
DDS	Data Distribution Service
FPS	Frames per second
ONNX	Open Neural Network Exchange
RGB	Red, green and blue
ROS	Robot Operating System
RTT	Round-trip time
UAV	Unmanned aerial vehicle

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CHAPTER 1

Introduction

1.1 Motivation and application context

Industrial facilities contain spatially distributed assets, restricted areas, and conditions in which a potential fire must be assessed quickly. Aerial robots can extend the field of view of a monitoring system and can approach areas that are inconvenient or unsafe for a person to inspect. Their use does not remove the sensing problem, however. An RGB camera records visual appearance rather than temperature or combustion products, and an airborne computer must process that stream within strict limits on memory, computation, power, heat dissipation, payload and wireless connectivity.

Sending every camera frame to a remote server would make hazard detection dependent on link capacity and remote availability. Edge inference places the detector beside the camera and permits a compact event record to be transmitted instead. Published work on edge-oriented UAV fire detection consequently assesses predictive quality together with runtime and energy or resource cost [1]. Research on collaborative UAV edge intelligence similarly treats computation, storage and link availability as joint design constraints rather than assuming an unlimited central processor [2]. These studies motivate local and cooperative processing, but their numerical results cannot be transferred to the hardware, data or two-stage decision process evaluated in this thesis.

This work studies a heterogeneous two-vehicle system. The *Sentinel* performs initial visual hazard detection on an NVIDIA Jetson Nano. The *Scout* performs secondary visual hazard verification on a ModalAI VOXL 2. The design aims to reserve the Scout

for candidates that persist at the Sentinel and to preserve a traceable relationship among the initial candidate, the verification request, the returned verdict and the final decision. The title's phrase "industrial fire detection" identifies the target application domain; it does not imply validation at an operating industrial facility. The final physical target is visible flame, while broader industrial sensing and physical smoke verification remain outside the evaluated scope.

1.2 Problem statement

A single detector on a constrained UAV faces competing objectives. Lower model capacity or resolution may reduce runtime cost but can remove visual detail. A permissive threshold may retain weak hazards but can also pass fire-like backgrounds. Requiring persistence across frames can suppress transients, but it can delay or discard short events. A second verifier can reject some candidates, but it introduces another model, camera, computer, communication path and timeout condition. The architectural benefit is therefore an experimental question rather than an automatic consequence of adding a second vehicle.

The problem addressed here is to determine how far a resource-aware division of responsibilities can support visual fire detection and verification on two heterogeneous edge platforms. The study must compare detector choices, measure the selected runtimes on the devices that execute them, examine sustained operation with concurrent services, and determine what the temporal and tiered verification stages do to event retention and decision time. It must also test the two-vehicle mission logic under defined simulation conditions and examine whether predictive performance transfers beyond the development data.

This formulation deliberately separates an engineering prototype from an operational safety system. RGB predictions do not provide physical fire ground truth. Secondary verification by another RGB model is an independent visual assessment, not proof of combustion. The laboratory system is stationary, and the simulation uses modeled vehicles and scenario coordinates. The thesis therefore evaluates a bounded research prototype and records the evidence required before broader deployment claims could be considered.

1.3 Terminology and system interpretation

Three terms in the title require precise interpretation.

Tiered. Detection and verification responsibilities are assigned to distinct, ordered stages across the two edge platforms. The Sentinel detects a candidate and decides whether to request verification; the Scout returns an independent verdict; the Sentinel records one correlated terminal decision. The term describes

role specialization and decision flow; it does not imply a multi-level command organization or feature-level neural-network fusion.

Cooperative UAV pair. The prototype contains two cooperating aerial platforms that exchange typed messages over a shared network. This is the smallest multi-vehicle configuration needed to study role specialization. It does not support conclusions about scaling to a larger fleet or swarm.

Industrial. Industrial describes the intended monitoring context and its resource and safety constraints. The retained evidence comes from offline datasets, exact-device measurements, stationary laboratory trials and simulation. No industrial-site deployment was conducted.

Throughout the thesis, *hazard detection* means the Sentinel's initial visual identification and *hazard verification* means the Scout's secondary assessment. The *temporal filter* is the consecutive-frame rule applied before an alert is published. The title's *tiered edge AI architecture* denotes this division of responsibilities between the two edge platforms and the ordered detection–verification decision flow. "Fire" in the final live evaluation refers to a visible flame. Historical fire/smoke artifacts are reported only as development history or offline evidence; physical smoke verification is outside the validated live scope.

1.4 Research questions and objectives

The investigation is organized around five research questions.

RQ1. What predictive-performance and resource trade-offs do the candidate detectors exhibit, and which are supported for each edge platform?

RQ2. Can the selected configurations operate sustainably with the required concurrent services?

RQ3. How do temporal filtering and Scout verification affect event retention, false alerts and decision latency?

RQ4. Does the implemented two-vehicle mission behave correctly and handle faults in the tested simulation?

RQ5. How does predictive performance transfer to eligible external UAV and industrial-like datasets?

Table 1.1 converts these questions into objectives and identifies where each is answered. An unanswered question remains an evidence gap; it is not assigned a favourable result by inference.

Table 1.1: Research-question objectives and thesis traceability.

Question	Operational objective	Principal chapters
RQ1	Compare predictive metrics, artifact fidelity and exact-device resource behaviour without mixing laptop and edge measurements.	Chapters 4, 6–8
RQ2	Evaluate the selected configurations with the camera, communication and logging services required by the workflow over a defined duration.	Chapters 5–8
RQ3	Compare single-frame, temporally filtered and verified decisions on admissible labelled units, including the latency introduced by each stage.	Chapters 5–8
RQ4	Exercise nominal mission branches and preregistered faults in the simulator after its readiness gates are satisfied.	Chapters 5–8
RQ5	Evaluate frozen models on external data admitted through provenance, overlap and annotation checks.	Chapters 4 and 6–8

1.5 Contributions

The thesis makes four evidence-bounded contributions. First, it specifies a two-node tiered edge architecture in which a recall-oriented Sentinel stage and a precision-oriented Scout stage exchange compact, correlated decision records. Second, it develops an auditable protocol with explicit experiment, trial and event identities; monotonic timing; model hashes; timeouts; terminal decisions; and append-only logging. Third, it establishes a comparison framework that connects dataset-controlled detector screening to runtime feasibility on the intended devices while preserving the distinction between historical and final deployment artifacts. Fourth, it integrates the decision process with a safety-constrained two-vehicle simulation architecture and defines gates that prevent development demonstrations from being reported as formal reliability evidence.

These contributions concern architecture, implementation and evaluation method. Claims about effectiveness are made only where retained measurements support them. In particular, the study does not claim universal reliability, zero false positives, autonomous industrial flight, geographic hazard localization or physical smoke verification.

1.6 Scope and evidence boundaries

The main evidence categories are kept separate. Offline predictive evaluation tests frozen image splits. Exact-device experiments measure particular model artifacts and runtime configurations. Stationary laboratory trials demonstrate the integrated physical detection–verification path under recorded conditions. Simulation evaluates

mission software under modeled dynamics and scenario inputs. External-data evaluation, when its admission conditions are met, concerns transfer to those datasets only. A result in one category is not presented as evidence in another.

The physical evaluation uses stationary vehicles with their propellers removed. The final live hazard is a handheld visible flame. Image-space box coordinates describe the Sentinel camera observation; they are neither geographic coordinates nor a position in the Scout camera frame. Integrated AI-controlled flight, common-world localization, physical-smoke trials, industrial-site trials and systems containing more than two vehicles are outside the validated scope.

The deployed Sentinel ROS workspace contains a 640-pixel TensorRT engine. An earlier benchmark used a different 416-pixel engine. Their input and output shapes and artifact identities differ, so measurements from the earlier engine are not treated as performance measurements of the final deployment. The same identity rule applies to every model and runtime comparison in later chapters.

1.7 Thesis organization

Chapter 2 reviews visual hazard detection, lightweight detectors, edge optimization, cooperative perception, temporal processing and ROS 2/DDS. Chapter 3 derives the requirements and presents the tiered architecture. Chapter 4 documents dataset preparation, detector development and deployment artifacts. Chapter 5 describes the software, device and communication implementation. Chapter 6 defines the experimental and statistical methods, and Chapter 7 reports accepted results. Chapter 8 answers the research questions and examines limitations and threats to validity. Chapter 9 states the evidence-supported conclusions and the work required for broader validation.

CHAPTER 2

Background and Related Work

This chapter establishes the concepts needed to assess the proposed system. The literature is compared by sensing task, computational setting and validation conditions. Results obtained on different datasets or devices are not treated as a common numerical benchmark for this thesis.

2.1 Visual fire and smoke detection

RGB fire detection is a visual recognition task. A camera provides colour, texture, boundary and motion cues, but it does not directly measure heat, combustion gases or chemical products. Bright reflections, lamps, sunsets and displayed fire imagery can resemble flame, whereas a distant flame may occupy too few pixels to preserve useful structure. Smoke creates a different appearance problem because it can be diffuse, semi-transparent and strongly affected by background and illumination. A visual prediction must therefore be described as evidence of a fire-like appearance rather than physical ground truth.

UAV-oriented fire and smoke research has identified early small targets and transparent smoke as difficult cases and has used multi-scale processing and augmentation to address them [3]. That work supports the need to examine target scale and environmental conditions. Its data, model and test setting differ from those used here, so its reported accuracy is not a baseline for the present system.

Domain shift is central to this application. Training images may contain large, well-exposed outdoor fires, while a deployment camera observes small flames, glare, clouds, reflective metal or indoor scenes. Closely related frames or derived copies split

between training and evaluation can also inflate a performance estimate. Evidence from another imaging domain demonstrates how subject-level leakage can produce optimistic deep-learning evaluation [4]. The specific leakage mechanism differs here, but the principle motivates source-family grouping, duplicate checks and frozen splits in Chapter 4.

The final physical scope of this thesis is visible flame. Historical datasets and models include smoke, but no controlled physical-smoke campaign was conducted. Smoke-video playback can expose the detector to displayed imagery; it cannot establish performance on airborne smoke. This distinction is needed whenever fire and smoke are discussed together in the literature.

2.2 Lightweight object detectors

Object detection assigns class scores and spatial boxes to candidate objects. The original YOLO formulation cast this as one end-to-end regression from an image to boxes and class probabilities [5]. The YOLOv8 software used in this project is a later engineering family. Ultralytics documents its training, validation, inference and export interfaces, but no formal YOLOv8 architecture paper exists [6]. The original YOLO paper is therefore used for historical single-stage context and the official software record identifies the implementation family; neither source is used as performance evidence for the trained thesis checkpoints.

SSD predicts class scores and box offsets over default boxes at multiple feature-map scales [7]. MobileNetV2 supplies a mobile-oriented backbone based on inverted residual blocks and linear bottlenecks, and its evaluation explicitly relates predictive quality to operations, parameter count and latency [8]. EfficientDet combines a weighted bidirectional feature pyramid with compound scaling of resolution, depth and width to address multiple resource budgets [9]. These design goals justify comparing SSD MobileNet V2 FPNLite and EfficientDet-Lite0 with YOLOv8n and YOLOv8s. They do not predict which trained implementation will perform best on the curated fire data or on either edge device.

Input resolution and model capacity create coupled trade-offs. A larger input can retain more pixels for a small object but increases the tensor and the work performed by many layers. A larger model can represent more complex patterns but increases memory, computation and often latency. Runtime libraries, preprocessing, output decoding and accelerator support can reverse an ordering suggested by parameter count alone. The candidate comparison in this thesis therefore treats model family, size, resolution, exported graph and execution backend as explicit configuration variables.

2.3 Edge inference and model optimization

Edge deployment separates at least three artifacts: a trained checkpoint, an interchange or framework representation, and a runtime-specific optimized artifact. Export success does not prove numerical fidelity, and a valid file does not prove that the intended accelerator executes every operation. The result must be checked against a frozen preprocessing and decoding contract and then measured on the target device.

TensorRT constructs an optimized engine for a specified network, hardware and precision configuration. NVIDIA documents reduced-precision execution and the target-specific nature of serialized engines [10]. In this project, TensorRT FP16 is relevant to the Sentinel, while TensorFlow Lite is used for the accepted Scout runtime. TensorFlow distinguishes float16 weight conversion from full integer quantization; actual execution still depends on the available delegates and kernels [11]. An artifact labelled FP16 or quantized is therefore not, by itself, evidence of accelerator use or end-to-end speed.

Latency also requires a declared boundary. *Inference-call latency* measures a model invocation after its input is available. *Backend pipeline latency* may add preprocessing and decoding. *Camera-to-decision latency* additionally includes capture, queues, temporal filtering, communication, Scout processing and finalization. Frames per second computed as the reciprocal of mean inference duration describe potential invocation rate; they do not automatically equal camera or decision throughput. Chapters 6 and 7 preserve these distinctions.

The Jetson Nano and VOXL 2 also differ in architecture and software support. NVIDIA specifies the Jetson Nano module with a quad-core Arm Cortex-A57 CPU, a 128-core Maxwell GPU and 4 GB LPDDR4 memory [12]. ModalAI specifies the VOXL 2 around the QRB5165 with eight CPU cores, 8 GB LPDDR5, an Adreno 650 GPU and a neural-processing accelerator [13]. Manufacturer specifications describe available resources; they do not predict application latency, sustained temperature or power. Those quantities require measurements of the exact artifact and concurrent services.

2.4 Cooperative UAV perception and tiered verification

Cooperative perception can exchange raw observations, intermediate features or final decisions. These choices retain different amounts of information and impose different bandwidth, synchronization and fusion requirements. Work on adaptive feature fusion illustrates this early/intermediate/late distinction in connected-vehicle LiDAR perception [14]. Its spatial scene-fusion results do not transfer to this camera-based two-drone prototype, but the representation distinction is useful.

Collaborative UAV edge-intelligence proposals allocate computation and storage across a swarm while considering node and link failures [2]. Hierarchical aerial-computing research has also examined cooperative resource allocation and task offloading [15]. The present architecture addresses a narrower problem. It does not partition a neural network or offload part of an inference graph. Each vehicle executes a complete detector and exchanges a compact decision-level message.

Decision-level exchange reduces communication volume and decouples the two model runtimes, but it discards feature information that could support joint fusion. It also requires the two cameras to observe a relevant scene at the right time. In the evaluated stationary workflow, a Sentinel alert triggers a full-frame Scout inference. The Sentinel box is retained as source-image metadata; it is not projected into the Scout image or a world map. The Scout therefore supplies a second visual decision but not geometric localization.

The tiered workflow assigns different intended priorities to its stages. The Sentinel is intended to retain possible hazards for review, while the Scout is intended to reject unsupported candidates before a final confirmation. Whether this reduces false alerts while retaining true events depends on the data, thresholds and error dependence between the two RGB models. The extra verification and network path also adds latency and new failure modes. RQ3 tests this trade-off rather than assuming a benefit.

2.5 Temporal filtering

A frame-wise detector can respond to a transient fire-like pattern that does not persist. Video fire detection has therefore combined frame-level evidence over time. Kong *et al.* used temporal smoothing with a frame-level fire classifier and reported a trade-off between sensitivity, delay and false alarms [16]. Their classifier, datasets and smoothing method differ from this project, so their timing parameters and numerical results are not transferred.

The mechanism studied here is a three-consecutive-frame rule. A qualifying detection increments the current evidence count; loss of the qualifying event resets it. An alert becomes eligible when the count reaches three. If the per-frame events were independent and the probability of a false candidate in one frame were p, an idealized three-frame probability would be p^3. Real video frames are temporally correlated, however, so this expression is not an estimator for the deployed system. Retained sequences and event-level labels are required to measure suppression, retention and delay.

2.6 ROS 2, DDS and correlated decisions

ROS 2 uses a middleware abstraction over DDS/RTPS implementations. It exposes quality-of-service policies including history, depth, reliability, durability, deadline and liveliness, and publisher/subscriber policies must be compatible for communication [17]. ROS 2 can use different middleware vendors, so implementation and compatibility are deployment details rather than interchangeable labels [18].

A reliable DDS setting does not prove that the application produced one valid answer within a deadline. The application must identify an event, reject unrelated or late replies, define timeout behaviour and record a terminal outcome. The protocol in this thesis consequently carries experiment, trial and event identifiers and requires one accepted final decision for each published alert.

Wireless ROS 2 studies with large payloads have identified fragmentation, retransmission timing and buffer bursts as possible failure mechanisms [19]. Those results concern a different payload and network setting. They motivate explicit message size, load and runtime reporting but do not establish the behaviour of the compact SwarmNet event messages used here. Likewise, MAVLink is designed for constrained robotic links [20], but protocol design does not validate the project's untested physical mission interface.

2.7 Comparative synthesis and research gap

Table 2.1 summarizes how the principal sources inform this thesis and why their validation settings cannot replace project-specific experiments.

Table 2.1: Comparison of related work and its role in this thesis. Reported findings remain conditional on each source's data, hardware and setting.

Source	Main topic	Validation setting	Use and limitation here
Fan *et al.* [3]	UAV fire/smoke detection	UAV remote-sensing imagery	Motivates small-target and smoke challenges; different data and model.
Vazquez *et al.* [1]	Lightweight wildfire detection	Transfer-learned models with edge/resource evaluation	Motivates joint predictive/resource analysis; no numerical transfer.
Liu *et al.* [7], Sandler *et al.* [8]	Single-stage and mobile detection	Generic object-detection benchmarks	Supports candidate design concepts; does not validate the exported SSD checkpoint.
Tan *et al.* [9]	Scalable efficient detector	COCO object detection	Supports BiFPN/compound-scaling context; COCO results are not a fire-data baseline.
Kong *et al.* [16]	Temporal fire evidence	Surveillance video	Motivates temporal processing; does not validate the three-frame rule.
Qiao and Zulkernine [14]	Cooperative perception	Connected-vehicle LiDAR fusion	Explains exchange levels; different sensor, geometry and task.
Qu *et al.* [2]	Collaborative UAV edge intelligence	Architectural resource/failure analysis	Motivates resource cooperation; unlike this decision-level two-node workflow.
Kang *et al.* [15]	Hierarchical aerial task offloading	Simulated learning-based resource allocation	Motivates hierarchical allocation; not a prototype baseline.
ROS 2 documentation [17], [18]	DDS QoS and middleware	Specified middleware semantics	Defines mechanisms; cannot establish SwarmNet latency or completion.

The literature establishes strong components: efficient single-stage detectors, edge runtimes, temporal evidence, cooperative perception and typed robotic communication. The research gap addressed here is their integration into a traceable heterogeneous prototype in which model choice, artifact identity, resource cost, temporal filtering, secondary verification and mission logic are assessed under explicit evidence boundaries. The goal is not to claim a universally novel swarm concept. It is to determine what this particular division of responsibility supports, what it costs, and which claims remain untested.

CHAPTER 3

System Requirements and Tiered Architecture

This chapter converts the research questions into system requirements and defines the boundary between the stationary perception workflow and the two-vehicle mission software. It presents intended behaviour and verified implementation status; measurements of performance are reserved for Chapters 6 and 7.

3.1 Functional and non-functional requirements

The requirements were derived from the research questions, the available hardware and the need to preserve auditable evidence. A requirement can be implemented without being experimentally validated. Table 3.1 therefore identifies both its architectural purpose and its current evidence status.

Table 3.1: System requirements and final evidence status.

ID	Requirement	Related question	Status boundary
FR1	Sentinel shall perform local initial visible-flame detection and expose the artifact identity and timing needed for analysis.	RQ1, RQ2	Implemented and exercised on the stationary Sentinel.
FR2	A three-consecutive-frame temporal filter shall gate verification requests.	RQ3	Implemented; its comparative effect requires the accepted ablation evidence.
FR3	Each verification request, Scout verdict and final decision shall be correlated by experiment, trial and event identity.	RQ3	Implemented and exercised in stationary ROS 2 tests.
FR4	Scout shall perform independent full-frame visual verification and return at most one application verdict for an event.	RQ1, RQ3	Implemented and exercised under bounded laboratory conditions.
FR5	Sentinel shall emit one terminal decision: `CONFIRMED`, `REJECTED`, `TIMEOUT` or `ERROR`.	RQ3	Implemented; physical alarm actuation is outside scope.
FR6	The mission layer shall model confirmed, rejected and degraded branches for both vehicles without enabling unverified physical flight.	RQ4	Development simulation branches executed; the formal repeated campaign was not executed.
NFR1	Raw evidence shall be append-only and include configuration, timing, model identity and logging-drop counters.	All	Implemented in the enhanced protocol and experiment bundles.
NFR2	Runtime claims shall refer to an exact device, model hash, precision, input shape and service configuration.	RQ1, RQ2	Mandatory analysis rule; historical 416-pixel and deployed 640-pixel Sentinel engines are distinct.
NFR3	A missing, duplicate, unrelated or late verdict shall not generate a second accepted decision.	RQ3, RQ4	Enforced by the coordination state machine; bounded fault evidence is reported later.
NFR4	Evaluation shall keep offline, device, stationary, simulation and external evidence categories separate.	All	Thesis-wide reporting constraint.

No universal numeric latency, power or accuracy requirement was imposed after the system had been measured. Instead, each experiment freezes an acceptance criterion before its result is interpreted. This avoids converting an observed value into a retrospective design target. Safety requirements are categorical: physical integrated AI tests are stationary and propeller-free, and the mission adapter must fail closed unless an exact simulator-only precondition is satisfied.

3.2 Sentinel and Scout platforms

The Sentinel is a customized Holybro X500 V2 platform with a Pixhawk 6C flight controller, NVIDIA Jetson Nano 4 GB computer and Intel RealSense D435i camera. Its role is broad-area hazard detection, temporal filtering, creation of target metadata and publication of the final decision. NVIDIA's published processor and memory specification provides resource context [12]; the D400 family datasheet provides camera-family context [21]. Actual camera mode, runtime and latency are taken from experiment configuration and logs rather than inferred from those specifications.

The Scout is a ModalAI PX4 Development Kit with a VOXL 2 computer based on the QRB5165. Its role in the stationary workflow is to receive a correlated alert, capture its own current image and perform full-frame secondary verification. The official

VOXL 2 feature matrix describes its processor, memory and accelerators [13]. The retained runtime evidence, rather than the feature matrix, determines which software backend was actually used.

Table 3.2: Platform roles and architecture-relevant components. Specifications are nominal; measured configurations are reported with the experiments.

Attribute	Sentinel	Scout
Vehicle basis	Customized Holybro X500 V2	ModalAI PX4 Development Kit
Flight controller	Pixhawk 6C	PX4-compatible ModalAI platform
Edge computer	Jetson Nano, 4 GB	VOXL 2 / QRB5165, 8 GB nominal memory
Camera path	Intel RealSense D435i	Modal Pipe Architecture camera stream
Perception role	Initial hazard detection and temporal filtering	Independent full-frame hazard verification
Accepted deployment family	YOLOv8n / TensorRT	YOLOv8s / TensorFlow Lite
Communication role	Alert publisher, verdict subscriber, decision publisher	Alert subscriber, verdict publisher
Validated physical mode	Stationary, propellers removed	Stationary, propellers removed

Physical configuration and scenario fit

The two platforms are not interchangeable carriers for the same software. The vehicles were assembled and configured for this project, with most physical customization concentrated on the Sentinel. Its open X500 frame accommodated the added Jetson Nano, D435i, wiring and power-conversion hardware, whereas the Scout retained the more integrated ModalAI compute, camera and flight stack. This difference supports the intended division between persistent initial inspection and dispatched secondary verification, but it does not by itself establish flight endurance, payload margin or maneuverability.

The Sentinel companion computer is supplied from the airframe power distribution through an intermediate DC–DC regulator. A three-conductor serial connection links the Jetson-side interface to the Pixhawk, and the D435i is mechanically attached to and connected directly to the Jetson. The accepted repository preserves the logical serial endpoint and baud rate, but not a complete connector-level pin map, regulator part number, calibrated voltage/current series or measured camera pose. The thesis therefore reports the integration topology while treating electrical efficiency, energy use and installed geometry as unmeasured quantities.

Table 3.3 separates resource constraints that were measured from those that remain design context. This separation is important: the term *resource-constrained* is supported

directly for compute, memory, thermal observation, runtime compatibility and wireless communication, but not by a calibrated electrical-power or airborne-endurance experiment.

Table 3.3: Platform constraints, evidence status and architectural consequence.

Constraint	Evidence in this study	Consequence for the architecture
Compute and memory	Exact deployed runtimes, latency distributions, host CPU, available memory and swap were recorded on both edge computers.	Model, tensor layout and runtime are selected per platform rather than from predictive accuracy alone.
Thermal operation	Host thermal-zone values were sampled during bounded stationary runs; cooling configuration and ambient conditions were not a controlled thermal experiment.	Sustained operation is evaluated directly, but no causal throttling or airborne cooling claim is made.
Electrical power and energy	The Sentinel power topology is documented qualitatively; calibrated voltage, current, regulator efficiency and energy per decision were not measured.	No energy-efficiency ranking or flight- endurance improvement is claimed.
Payload and geometry	Installed components are identified, but complete mass, rotor spacing, dimensions, centre of gravity and camera pose were not admitted as measured evidence.	Gazebo airframes remain provisional simulation proxies rather than independently identified physical models.
Wireless communication	Compact ROS 2 request–reply events were measured over SwarmNet in stationary laboratory conditions.	Raw video remains local, reducing link load, while access-point coverage and interference remain deployment risks.
Sensing	RGB camera paths were exercised on both vehicles; the final physical campaign did not validate depth, inertial, thermal, gas or particulate sensing for hazard decisions.	A second RGB decision improves traceability but is not independent physical confirmation of combustion.

The available sensor set is consequently broader than the evaluated sensing path. The D435i family provides RGB, stereo depth and inertial capabilities [21], but the accepted Sentinel detector used only the colour stream. Pixhawk telemetry was observed during the final endurance run, yet its estimator and vehicle sensors were not inputs to the fire decision. The Scout used its ModalAI camera stream through MPA; its PX4, IMU and QVIO services were present as platform load but were not validated as parts of an AI-controlled physical mission. Table 3.4 makes this availability–use distinction explicit.

Table 3.4: Available platform capabilities versus their evaluated use.

Component	Evaluated use	Boundary
D435i colour camera	Sentinel RGB fire inference	Stationary live camera path exercised; no industrial-site calibration.
D435i depth and IMU	None in the accepted hazard-decision path	Available device capabilities, not evaluated project results.
Pixhawk 6C sensors and estimator	Receive-only telemetry/service coexistence	No integrated arming, navigation or physical mission.
ModalAI camera through MPA	Scout current-frame RGB/YUV verification	Stationary bridge and decoding exercised, including unavailable/stale-frame faults.
Scout PX4, IMU and QVIO services	Concurrent platform services during the endurance run	Their presence contributes load; navigation accuracy was not evaluated.

The labels “recall-oriented” and “precision-oriented” describe the intended roles of the two stages. They are not assumed properties of YOLOv8n and YOLOv8s. Oper-

ating thresholds and measured errors determine whether the final configurations actually realize those priorities.

3.3 Responsibility allocation and data flow

Figure 3.1 shows the stationary tiered architecture. The Sentinel captures a frame, performs local inference and applies the temporal filter. When the configured count is reached, it creates one alert containing the event identity, detector confidence, temporal count, source frame timing, model hash and normalized source-image box. ROS 2/DDS carries this compact record over the SwarmNet wireless network.

The Scout treats the alert as a trigger. It does not crop its image around the Sentinel box and does not navigate to the transferred coordinates. It performs an independent full-frame inference on its current camera frame and publishes a correlated `CONFIRMED`, `REJECTED` or `ERROR` verdict. The Sentinel accepts only a verdict for the active event and publishes one terminal decision. A timeout is generated locally if a matching verdict does not arrive within the configured interval.

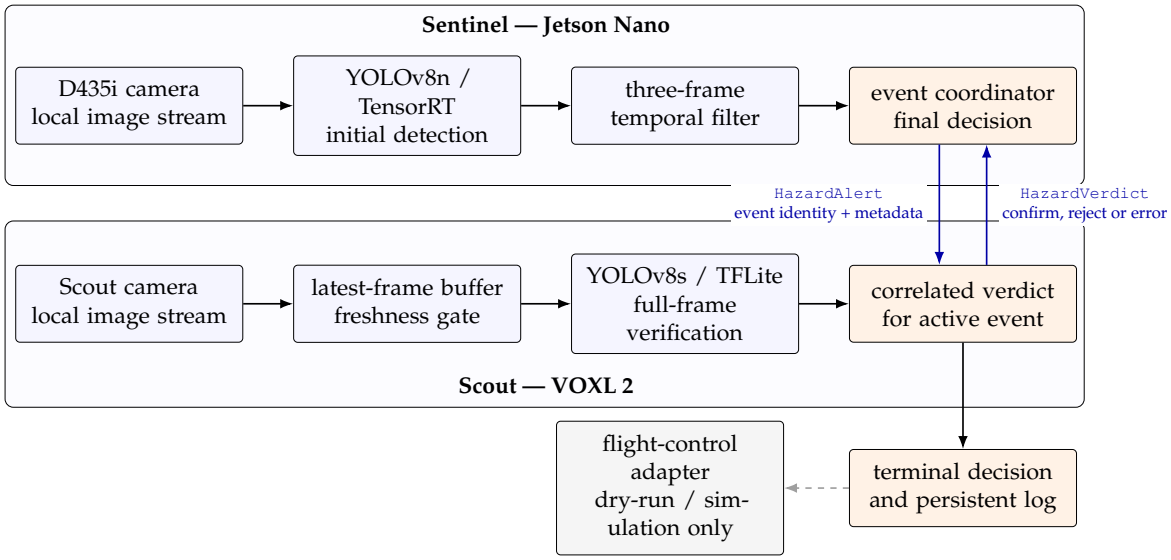


Figure 3.1: Tiered edge architecture. Solid paths belong to the evaluated stationary perception and ROS 2 workflow. The dashed flight-control path is implemented for constrained dry-run and simulation use but was not validated as integrated physical flight.

This division keeps raw video local to each device and limits the network exchange to event metadata and decisions. It also means that verification is possible only when the Scout's current field of view contains relevant visual evidence. The tiered workflow provides no implicit camera-to-camera correspondence. That constraint is part of the architecture rather than a later experimental failure.

Onboard execution and ground-station boundary

The physical perception architecture is fully onboard in the computational sense. Sentinel camera capture, TensorRT inference, temporal filtering, alert publication and final-decision handling execute on the Jetson Nano; Scout camera acquisition, decoding and TFLite verification execute on VOXL 2. The Sentinel also hosts the SwarmNet wireless access point. The laboratory laptop was used for configuration, bounded test orchestration, monitoring and evidence retrieval, but it was not the inference server or the required final- decision processor in the evaluated two-device path. Loss of an external Internet or cloud service therefore does not, by design, remove local perception.

This property must not be described as a *full aerial experiment*. The accepted physical evidence is stationary and propeller-free; aerial state transitions were exercised only in simulation. A precise description is a *fully onboard edge-processing architecture with stationary physical validation and simulated aerial mission execution*.

Keeping images local reduces radio traffic and avoids making hazard decisions dependent on continuous raw-video transport. It also distributes failure and maintenance across two dissimilar embedded environments. The Sentinel access point and coordinator concentrate connectivity and finalization responsibility, so their loss can isolate the Scout even though inference is decentralized. Within a warehouse-like scenario, local processing is attractive where external connectivity is weak or restricted, but metallic structures, aisles, occlusion, multipath propagation and changing illumination can degrade both radio and RGB performance. These are design implications, not measured industrial-site results.

3.4 Communication interfaces and coordinate conventions

The enhanced interface uses three typed ROS 2 messages. Their roles are summarized in Table 3.5. All messages carry a ROS header and the identifiers needed to join records across the workflow.

Table 3.5: Enhanced decision interfaces.

Topic and type	Direction	Principal semantics
`/swarm/hazard_alert` / `HazardAlert`	Sentinel to Scout	Experiment, trial and event IDs; source device; hazard type; confidence; temporal count; source/alert monotonic times; model hash; normalized source-image box and frame dimensions.
`/swarm/hazard_verdict` / `HazardVerdict`	Scout to Sentinel	Matching IDs; verdict; hazard type and confidence; receipt/source/verdict times; Scout model hash; reason.
`/swarm/hazard_decision` / `HazardDecision`	Sentinel to logger or consumer	Matching IDs; terminal decision; both confidences; detection, alert, verdict and decision times; reason.

The alert fields `target_u`, `target_v`, `bbox_width` and `bbox_height` are normalized in the Sentinel source-camera frame. The origin is the top-left; u increases to the right and v increases downwards. These values are not latitude, longitude, local vehicle coordinates or Scout image coordinates. Earlier legacy `TargetHazard` messages used ambiguous location naming. The enhanced message makes the image-space meaning explicit and is the normative interface for the final evaluation.

ROS header time supports cross-device record alignment when clock synchronization is known. Each `*_monotonic_ns` field is valid for durations on its originating device. Monotonic values from different devices cannot be subtracted directly. Application round-trip time can be calculated on the initiating Sentinel without assuming that the two monotonic clock epochs are equal.

Figure 3.2 makes the request, response and timeout paths explicit. Event identifiers are opaque and cannot be reused within an experiment. A duplicate, late or unrelated verdict cannot close a different active event.

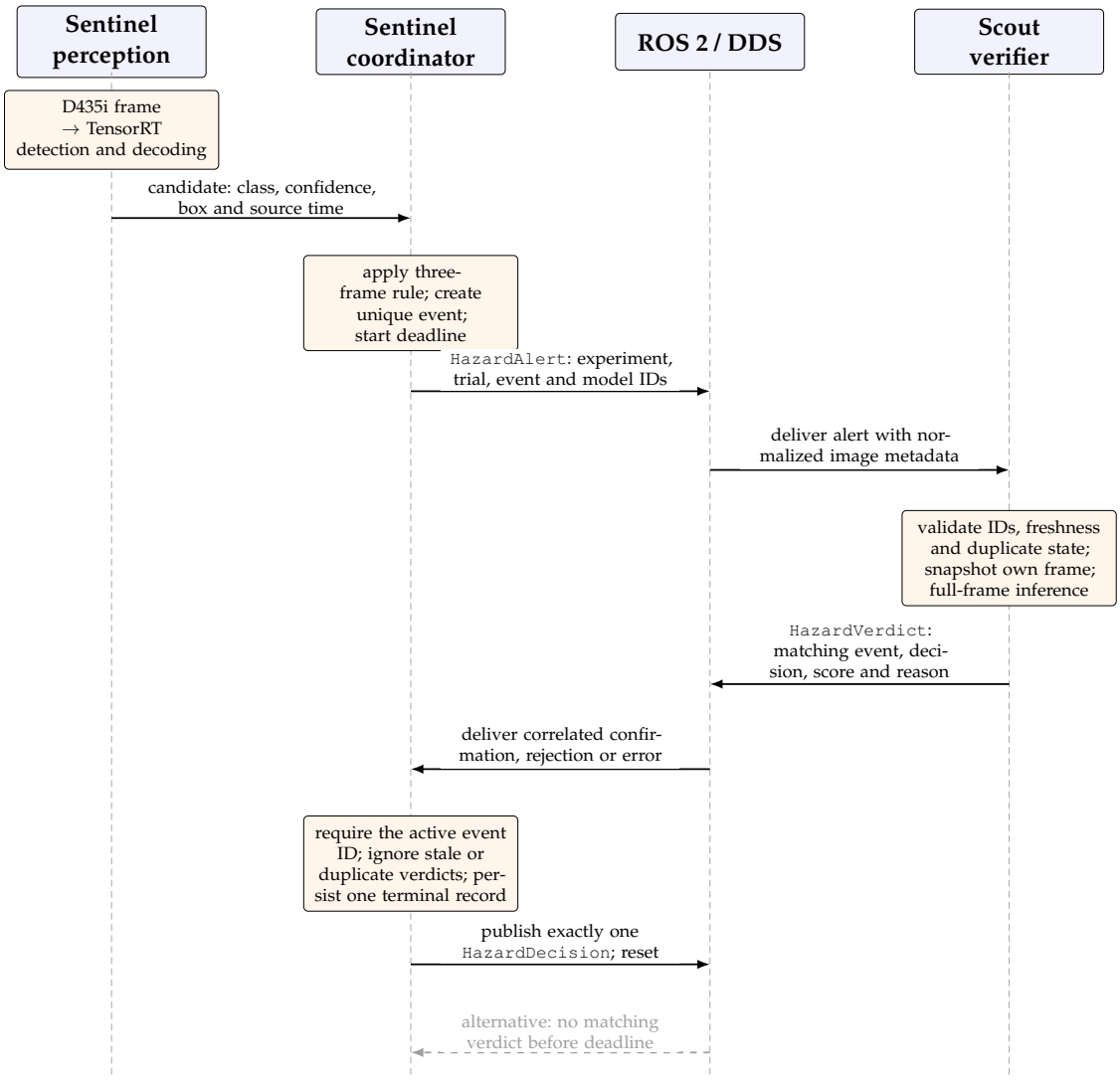


Figure 3.2: Detailed correlated Sentinel–Scout decision sequence. The Scout validates the request and performs independent full-frame inference on its own fresh image. The coordinator accepts only a matching active-event verdict and records exactly one terminal outcome; the dashed path is the deadline-expiry alternative. Image-space target metadata is retained for traceability and is not used to steer the Scout in the evaluated stationary workflow.

3.5 Coordination states and mission boundary

The perception coordinator and the flight mission controller use related but distinct state models. In the stationary workflow, the coordinator begins in `SEARCHING`. A qualifying candidate moves it to `TRACKING`; an alert moves it to `AWAITING_SCOUT`. A matching verdict produces `CONFIRMED`, `REJECTED` or `ERROR`; expiration produces `TIMEOUT`. Exactly one terminal decision is recorded before the coordinator resets to `SEARCHING`.

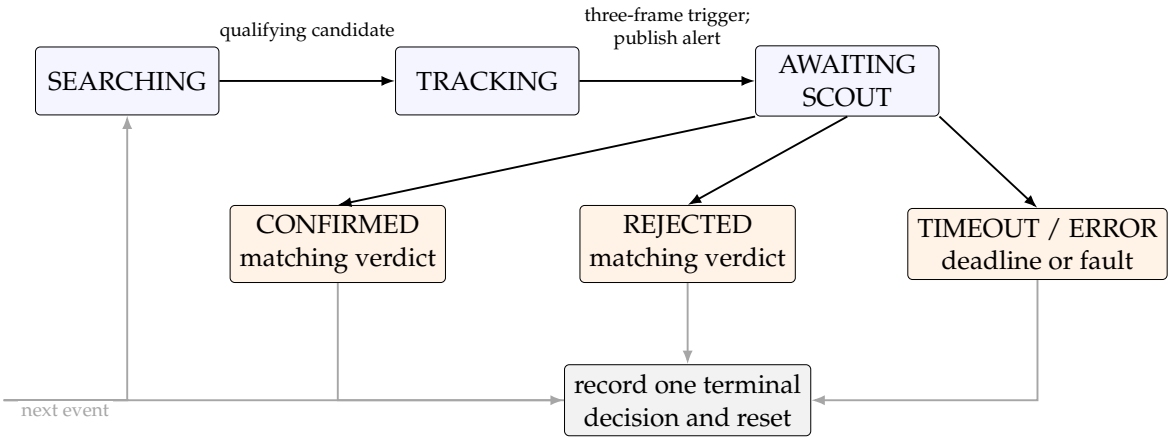


Figure 3.3: Stationary perception and coordination states. Terminal decisions are recorded once before the coordinator returns to search.

The mission layer adds vehicle states such as `PREFLIGHT`, `TAKEOFF`, `PATROL`, `SUSPECT_HOLD`, `SCOUT_DISPATCH`, `VERIFYING`, `CONFIRMED_HOLD`, `REJECTED_RESUME`, `DEGRADED`, `RTL` and `LANDED`. Its target positions are scenario-defined local-ENU coordinates. They are not derived from the Sentinel image box. Two development PX4/Gazebo runs exercised the nominal confirmed and rejected branches on provisional vehicle proxies. A later accepted visual run executed the confirmed branch from preflight through patrol, hold, Scout dispatch, verification, alarm publication, return and landing while retaining the corresponding state-transition log. These runs establish executable control paths under their recorded simulation conditions, not a reliability rate or physical-flight validation.

The formal repeated simulation campaign was not executed before the final evidence freeze. Measured physical parameters and the complete preregistered fault set did not close every readiness gate. Consequently, RQ4 receives only a bounded answer from the retained development runs. The visual record is supporting evidence for one run and does not replace the formal repeated campaign or its machine-readable telemetry.

3.6 Resource, logging and safety constraints

The Sentinel has less nominal memory and an older GPU architecture than the Scout platform, but hardware specifications alone do not determine the model assignment. Runtime availability and graph compatibility also matter. The Sentinel therefore uses a TensorRT deployment path, while the accepted Scout configuration uses TensorFlow Lite. Candidate screening remains broader than these historical deployments so that the thesis can distinguish a deployed artifact from a model selected under the later controlled comparison.

Logging is moved off the inference path through a bounded queue and a dedicated writer. This reduces synchronous file-I/O interference while making overload visible through drop counters. Raw JSON Lines records are append-only. Corrections, exclusions and derived metrics are stored separately so that an analysis cannot silently replace the observation from which it was produced.

The communication layer uses explicit correlation and a timeout rather than equating DDS reliability with successful application completion. A time-limited rejection cooldown can prevent immediate repeated requests for the same normalized image region. Persistent evidence or a sufficient confidence increase can override the cooldown, and expiry makes the region eligible again. This policy must be evaluated; it is not assumed to improve detection.

The physical safety boundary is strict. Integrated perception experiments use stationary aircraft with removed propellers. The architecture does not convert an image coordinate into a flight setpoint. The flight adapter is used only under explicit dry-run or simulator safeguards. Physical arming, autonomous movement and industrial operation require separate validation outside the evidence presented here.

3.7 Design rationale and research-question traceability

The architecture addresses RQ1 by allowing model and runtime choices to be evaluated per device rather than assuming homogeneous hardware. It addresses RQ2 by defining the concurrent camera, inference, communication and logging services whose sustained operation must be measured. It addresses RQ3 through explicit single-frame, temporal-filter and verification stages with measurable timestamps and terminal outcomes. The separate mission controller supplies the software boundary for RQ4, while keeping simulator coordinates and physical perception evidence distinct. RQ5 remains principally a data and evaluation question; the architecture contributes the frozen model and preprocessing contracts needed to test external transfer.

The expected benefit is fewer unsupported terminal confirmations than a single-frame Sentinel decision, at the cost of additional delay, energy use and failure opportunities.

That statement is a hypothesis. The architecture makes the comparison observable, but only the accepted experiments in Chapters 6 and 7 can determine its magnitude under defined conditions.

CHAPTER 4

Dataset Preparation and Model Development

This chapter describes the evidence pipeline that converts heterogeneous third-party imagery into a controlled detector comparison. It first separates the historical project assets from the fire-only research dataset, then documents annotation and leakage controls, frozen data partitions, training, model selection, export and conversion-fidelity checks. Predictive results are reported in Chapter 7; the purpose here is to make clear which inputs, decisions and artifacts produced those results.

4.1 Dataset provenance and historical development

Historical task definition

The project did not begin with the final one-class experiment. The historical Sentinel and Scout deployments used models and datasets that included fire and smoke-related classes. Those artifacts are retained as part of the engineering history, but they are not treated as the source of the controlled six-model comparison. Live smoke was removed from the formal validation scope because a safe and repeatable physical smoke source was not available. The controlled task therefore uses one localized target class, *fire*, while smoke-only imagery is excluded from the positive class.

The recovered base export identifies itself as Drone-Fire-Smoke version 1. Its local metadata records 41,880 images, 640-pixel stretch preprocessing and flip, brightness and exposure augmentation. The public dataset is distributed through Roboflow Universe under CC BY 4.0 [22]. I did not collect these images, and the available export

does not provide reliable original video or capture-session identifiers. Consequently, the image count cannot be interpreted as the number of independent scenes.

Historical development also introduced environmental negative images to expose the detector to fire-like visual confounders. Filename and content mapping recovered images from the Weather Image Recognition dataset, which contains multiple weather categories and declares a CC0 licence [23], and from the Weather Classification dataset, which also declares CC0 [24]. The former records the Weather Phenomenon Database as its source [25]. A further 711 images were mapped to the Stanford Background Dataset, while 21 historical files had no surviving source match. The Stanford and unresolved groups were excluded from the research freeze because their reuse conditions or provenance were not sufficiently established. This exclusion is a licensing and traceability decision, rather than an assertion that their image content was invalid.

Fire-only curation decision

Label inspection divided the 41,880-image base export into 20,584 images with at least one fire annotation, 21,190 smoke-only or other images, and 106 empty or background images. Fire-positive images retained only class-0 fire boxes. Smoke-only images were excluded rather than silently relabelled as negative, because their source annotations were not designed to establish the absence of visible fire. The approved environmental sources contributed 2,583 candidate background images. Exact hashing removed 53 duplicate negative copies, leaving 2,530; with the 106 base backgrounds, the provisional dataset contained 20,584 positive and 2,636 background images.

The distinction between a negative example and erroneous data is important. Fog, cloud, glare, sunrise, sandstorm and similar scenes are legitimate fire-free observations. Retaining them permits image-level false-alert behaviour to be measured. Their inclusion does not prove that these conditions represent the future industrial domain, and the positive and negative images come from different source collections, so source-specific cues may remain.

From provisional data to the frozen research manifest

The provisional 23,220-image manifest passed structural checks, but further analysis showed that three inferred source groups contained 16,364 records, or 70.5% of the provisional collection. Allowing those groups to dominate separate partitions would create a strong source/domain confound. The larger manifest was therefore preserved as an intermediate artifact and was not used for the final comparison.

Figure 4.1 summarizes the controlled lineage. I first resolved ten flagged annotations into immutable source-family manifest version 10, which contained 23,214 images

and 29,095 fire boxes. I then excluded the three dominant mega-groups, retained 6,850 records from 2,628 inferred groups, and deterministically capped every retained group at 50 records using seed `20260729`. This produced a 5,185-record candidate with more balanced positive and background support. Whole inferred groups, rather than individual images, were assigned to train, validation and test targets.

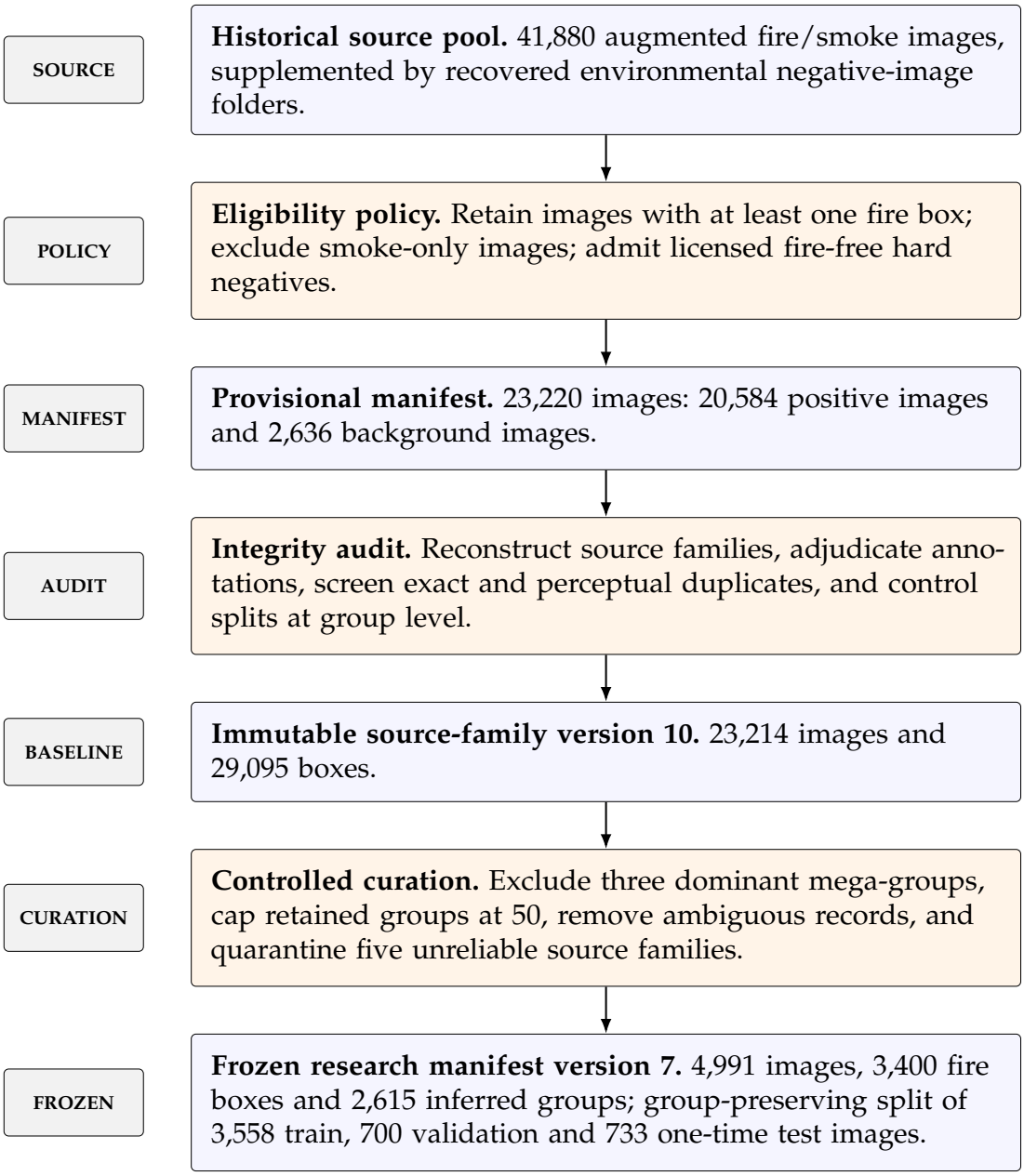


Figure 4.1: Dataset lineage from the historical fire/smoke export to the frozen fire-only research split. Every transformation created a new manifest; the source images, source annotations and previous manifests were preserved.

A split- and box-size-stratified review then led to the exclusion of six ambiguous positive records and the removal of 28 non-fire or redundant boxes from nine records. A new sample exposed systematic errors in augmented sibling images, so five source

families were quarantined as complete units, removing 188 further records. The resulting version-7 freeze contains 4,991 records, 3,400 retained fire boxes and 2,615 inferred source groups. This 78.5% reduction relative to version 10 is an independence and quality control; it does not mean that 78.5% of the source imagery was intrinsically false.

Table 4.1 records the sources that remain in the freeze. The URLs and recorded licence strings are stored per record. The original dataset pages remain the attribution authority; the manifest is the local traceability record.

Table 4.1: Recorded provenance of the frozen version-7 dataset. Counts refer to retained records after grouping, capping and adjudication.

Recorded source	Records	Recorded reuse status
Drone-Fire-Smoke version 1	2,461	CC BY 4.0; attribution required
Weather Image Recognition / WEAPD derivative	1,366	CC0 as recorded on the Kaggle page
Weather Classification	1,164	CC0 as recorded on the Kaggle page
Total	4,991	Third-party imagery; not collected for this thesis

4.2 Annotation policy, data health and leakage control

Class and annotation representation

The controlled experiment has one target class, *fire*. It is represented as class index 0 in YOLO text labels and category identifier 1 in COCO and TFRecord exports. A positive record must contain at least one retained fire box; a background record must contain none. The same manifest was materialized into framework-specific representations so that the candidate frameworks did not receive different semantic partitions.

For each YOLO row with normalized centre (x_c, y_c) and normalized width and height (w, h), the audit reconstructed the edges

$$x_{\min} = x_c - \frac{w}{2}, \quad x_{\max} = x_c + \frac{w}{2}, \quad y_{\min} = y_c - \frac{h}{2}, \quad y_{\max} = y_c + \frac{h}{2}. \tag{4.2.1}$$

Valid rows require finite values, class index zero, $x_c, y_c \in [0, 1]$, $w, h \in (0, 1]$, and all reconstructed edges within the image to a numerical tolerance of 10^{-9}. The audit also checked record/image/label existence, positive image dimensions, duplicate rows and consistency between record type and retained box count.

The earlier 23,220-record audit found no malformed row or geometric error, but the deterministic 45-image visual queue identified 35 acceptable, four questionable and

six apparently incorrect cases. Those proportions are not an estimate of population label error: the queue was deliberately stratified by split and box size, and one reviewer first judged contact-sheet renders. Full-resolution adjudication removed six records and 16 boxes when producing version 10. The subsequent curated lineage repeated semantic review, removed the further ambiguous records and boxes described above, and quarantined the five affected source families. All 50 records in the final version-7 split/size-stratified queue were accepted. This final check still cannot prove that all remaining boxes are semantically correct.

Image and group integrity

The final audit decoded all 4,991 referenced images. It reported no corrupt or unreadable file, no defined annotation-format problem, no invalid class or box geometry, and no inferred group identifier spanning more than one split. Seventeen exact-duplicate sets remain, but every set is contained within a single split. A difference-hash screen retained 456 cross-split candidates because low-detail sky, fog, glare and flame images can collide under a compact perceptual hash. A stabilized 108-pair, high-risk manual queue contained no remaining visually same-scene pair.

Keeping inferred groups intact is intended to reduce leakage from augmented or neighbouring observations. Closely related samples across training and evaluation partitions can inflate apparent generalization [4]. The cited study concerns a different imaging domain; it supports the dependence mechanism rather than providing a numerical baseline here. Because the source export lacks authoritative session IDs, filename parsing, hashes and manual review may over-group unrelated images or under-group related ones. The split is therefore described as *leakage-controlled*, not as proven capture-session independent.

Object-size boundary

The training split contains 2,505 fire boxes. The validation split contains 392 boxes and the one-time test split contains 503. At stored image resolution, none of the validation boxes is below the COCO 32^2-pixel small-object boundary, while the test split contains only three such boxes. Standard COCO-small average precision is therefore unavailable for validation and too sparsely supported for a general test-set conclusion [26]. Descriptive recall is additionally stratified by the ground-truth box area as a fraction of image area: below 1%, 1–5%, 5–20% and at least 20%. These relative bins do not recreate the image-space size of the small lighter flame used in the stationary laboratory demonstration.

4.3 Frozen research splits and external data

Partition roles and reproducibility anchors

The final split was assigned at inferred-group level toward a deterministic 70/15/15 target. Table 4.2 gives the resulting composition. The training partition fits model parameters. The validation partition selects checkpoints, confidence thresholds and the validation interpretation. The test partition was sealed until those choices and their hashes were frozen.

Table 4.2: Frozen fire-only version-7 composition. A background image contains no retained fire box.

Split	Positive images	Background images	Total images
Training	1,731	1,827	3,558
Validation	319	381	700
Test	354	379	733
Total	2,404	2,587	4,991

The authoritative records JSONL has SHA-256 `478803efb7f8f4f6c3fe74d586 58958bccfa2d0f58ec50a1a5c16535e393a7dd`. The COCO annotations are separately hashed for training, validation and test. The validation and test annotation hashes are respectively `1a93f800d63bf355bb586039da245cd612cbdd7052 102cbfd6da1409312ab7c6` and `651e2782db96cbbdf2ba718494f7b57447 f96aea6d7719fac350132f1b72a06d`. These anchors allow generated YOLO, COCO and TFRecord materializations to be checked without treating generated copies as new datasets.

The one-time test split was opened on 9 August 2026 only after the six validation outputs, checkpoint identities, confidence thresholds, evaluator hash and selection interpretation were frozen. All six candidates were then evaluated once at their validation-selected thresholds. No test-driven threshold tuning, checkpoint reselection or retraining followed. Chapter 7 reports the predictive outcomes and the rank reversal observed between validation and test; the present chapter records only the procedural boundary.

External-evaluation eligibility

External UAV and industrial-like sources are reserved for RQ5. They did not enter training, validation or the one-time internal test. At the final evidence freeze, no external inference result was admitted. FASDD-UAV has a publisher-reported CC BY-SA 4.0 licence and 25,097 UAV samples, but the exact archive was not acquired, hashed, decoded, audited for overlap and frozen locally. These publisher values are not substituted for verified local counts.

The MmodalFire archive is retained and hash-verified, and all 28 videos were decoded to their ends without a decoder failure. However, only eight of 28 video ranges passed the strict consolidated timestamp-alignment rule; the remaining ranges did not complete event/condition review and overlap checks. Optical character recognition is used only as review assistance. Until eligible non-overlapping manifests are frozen, these sources cannot contribute model metrics. Any later outcomes should be described as external UAV-domain and industrial-domain proxy evaluations, not as operational industrial-site validation.

4.4 Candidate detectors and training protocol

Controlled candidate matrix

The experiment compares six pretrained single-stage detectors. Four YOLOv8 configurations form a 2×2 capacity/resolution comparison: nano and small variants at 416 and 640 pixels. SSD MobileNet V2 FPNLite and EfficientDet-Lite0 provide mobile-oriented non-YOLO comparators [6], [7], [8], [9]. The screening seed is `20260729` for every candidate. Input resolution is part of the candidate configuration; comparisons between 416, 640 and 320 pixels are therefore deployment-oriented comparisons rather than pure architecture ablations.

Table 4.3: Six controlled detector configurations and their recorded training completion. Predictive and runtime results are presented in Chapter 7.

ID	Model / framework	Input	Batch	Completion	Training lineage
N416	YOLOv8n / PyTorch	416 px	16	48 epochs	Early stopping under the 100-epoch, patience-20 policy
N640	YOLOv8n / PyTorch	640 px	4	87 epochs	Optimizer-state resume from epoch 51
S416	YOLOv8s / PyTorch	416 px	8	33 epochs	Early stopping under the same policy
S640	YOLOv8s / PyTorch	640 px	4	33 epochs	Early stopping under the same policy
SSD320	SSD MobileNet V2 FPN-Lite / TensorFlow OD API	320 px	8	44,500 steps	One-class pipeline; corrected bounded validation signature
EDL0	EfficientDet-Lite0 / TFLite Model Maker	320 px	4	100 epochs	Full-state recovery through checkpoints 50 and 90

All candidates use the same version-7 record allocation. Pretrained initialization, environment locks, dataset hashes, checkpoints and failure lineage are retained per run. The four YOLO configurations ran in an isolated Windows Python 3.12 environment with PyTorch 2.12, Ultralytics 8.4.45 and an NVIDIA RTX 3050 Laptop GPU. Their starting weights were hash-verified. The training request specified 100 maximum epochs, early-stopping patience 20, automatic mixed precision and the largest stable recorded batch for each configuration.

SSD training used TensorFlow 2.15.1 and a pinned TensorFlow Object Detection API revision under WSL Python 3.11. The pipeline used a 320-pixel input, batch size 8, 445 steps per epoch, 100 maximum epochs and a cosine-decay momentum schedule. An initial common validation attempt created excessive TensorFlow retracing as image shapes changed, reaching 7.5 GB WSL resident memory and severe swap activity. That attempt was preserved. The predictor was corrected with one dynamic-spatial-dimension signature; the bounded retry used two concrete traces and completed normally. This was a runtime/evaluator correction, not a change to the model or ground truth.

EfficientDet-Lite0 used TFLite Model Maker 0.4.3 and TensorFlow 2.8.1 in a separate WSL Python 3.9 environment. The legacy loader initially failed on genuine zero-box background examples. Instead of deleting 2,587 background records or generating artificial boxes, I introduced a source-controlled empty-safe mask with the same semantics for non-empty examples. A one-epoch gate confirmed non-zero training and validation box loss before the full run. The full-state lineage, including optimizer iterations, was preserved through interruption and resume; checkpoint 90 supplied the final ten epochs, and the epoch-100 run ended normally.

The six runs constitute a single-seed screen. A three-seed finalist stage was originally planned, but it was not executed before the selection freeze. Consequently, between-seed variance cannot be estimated and no claim of statistical superiority is based on the training runs. Failed, interrupted and superseded attempts remain part of the engineering record and are excluded from accepted predictive comparisons unless a protocol explicitly adopts a recovered run.

4.5 Validation-based selection and held-out evaluation

Each native framework generated COCO-format predictions down to confidence 0.001 on the same 700 validation images. A common evaluator computed COCO AP over IoU thresholds 0.50:0.05:0.95 and AP50. For each candidate, the reporting confidence threshold was selected from 0.01 to 0.99 by maximizing object-level F1 at IoU 0.50. A numerical tie was resolved by higher recall and then by the lower threshold. The selected threshold was frozen before the test split was opened.

For a confidence threshold t, greedy class-aware matching at IoU 0.50 gives true positives $TP(t)$, false positives $FP(t)$ and false negatives $FN(t)$. The selection quantities are

$$P(t) = \frac{TP(t)}{TP(t) + FP(t)}, \tag{4.5.1}$$

$$R(t) = \frac{TP(t)}{TP(t) + FN(t)}, \tag{4.5.2}$$

$$F_1(t) = \frac{2P(t)R(t)}{P(t) + R(t)}. \tag{4.5.3}$$

An object detector has no unique count of true-negative boxes. Negative behaviour is therefore reported as the fraction of background images with at least one prediction at t, together with false-positive counts, rather than as a fabricated box-level specificity.

Model selection considered AP50–95, validation-selected F1, recall, precision, background rejection, observed host latency, artifact size and demonstrated device feasibility as separate dimensions. No post-hoc weighted score was introduced. A candidate is Pareto-efficient when another candidate cannot match or improve every selected dimension and strictly improve at least one. Sentinel interpretation emphasizes recall subject to its device constraints; Scout interpretation emphasizes precision and rejection of background alerts. These role preferences do not convert laptop screening into evidence of deployment.

The selection freeze was written before the test opening and records all six thresholds and checkpoint lineages. The test runner verified the freeze and test-annotation hashes, refused silent overwrites and retained per-image predictions and timing. A Windows-to-WSL path-normalization defect stopped the first invocation after the first model's inference. The raw predictions were preserved, the artifact-level path handling was corrected, and evaluation continued without rerunning or changing those predictions. This recovery is part of the evidence lineage rather than an unreported clean run.

4.6 Export, optimization and deployment fidelity

Fidelity before performance

A converted model is not assumed to be equivalent to its native checkpoint. The deployment registry requires comparison on the same 64 hashed validation calibration images before a converted artifact can enter device timing. Native and converted predictions use declared preprocessing and decoding. Within each image and class, detections are matched at box IoU at least 0.50. The frozen conversion rule requires a detection match rate of at least 0.95 and a mean absolute matched-confidence differ-

ence no greater than 0.02. A failure on either condition blocks performance ranking for that converted artifact.

The four YOLO checkpoints were exported as fixed batch-1, fixed-input FP32 ONNX graphs at opset 12 without simplification. ONNX 1.22 accepted all graph structures, but graph validity is distinct from prediction fidelity. Table 4.4 records the current prepared form and disposition. The detailed counts belong to Chapter 7; here, `FIDELITY_FAILED` means that the frozen conversion gate was not met.

Table 4.4: Prepared candidate artifacts and conversion-fidelity disposition. These are laptop preparation results, not edge-device benchmarks.

Candidate	Prepared form	Fidelity disposition	Permitted next step
YOLOv8n-416	FP32 ONNX, opset 12	Pass	TensorRT build/validation on Sentinel; ONNX Runtime attempt on Scout
YOLOv8n-640	FP32 ONNX, opset 12	Failed	Independent TensorRT candidate may be built; current ONNX is rejected
YOLOv8s-416	FP32 ONNX, opset 12	Pass	TensorRT build/validation on Sentinel; ONNX Runtime attempt on Scout
YOLOv8s-640	FP32 ONNX, opset 12	Failed	Independent TensorRT candidate may be built; current ONNX is rejected
SSD MobileNet V2 FPNLite-320	Float32 TFLite	Failed	Audit another native/conversion path or classify as unsupported
EfficientDet-Lite0-320	Native TFLite	Not applicable	Attempt only where a compatible TFLite runtime is available

The SSD conversion produced the expected number of outputs after its tensor roles and resize semantics were corrected, but only 87.14% of detections matched the native predictions, below the fixed 95% requirement. Its small matched-confidence error did not override the unmatched detections. The EfficientDet artifact is the native TFLite output of its training toolchain, so a separate source-to-converted comparison is not applicable; device runtime support and resource cost still require measurement.

Historical deployed models versus screened candidates

The six frozen checkpoints are research-screening artifacts. They must not be silently substituted for the models already exercised in the two-device system. The final Sentinel workspace contains a distinct YOLOv8n TensorRT engine with input shape $[1, 3, 640, 640]$, output shape $[1, 6, 8400]$ and SHA-256 `16f9018ac299c6f52561 0a877d798392ec26d53fccb84b589ff6533d76589bf2`. Earlier 416-pixel measurements refer to a different engine with output shape $[1, 6, 3549]$ and cannot be applied to this deployment.

The historical Scout deployment uses a distinct TFLite artifact with input $[1, 640, 640, 3]$, output $[1, 7, 8400]$ and SHA-256 `d533e320081be17ac91fb77c20 a9de09f4bffada0d7f88c88326d57cd893bc35`. The tensor interface is float32 and the seven output channels are consistent with four box values and three class

scores. It belongs to the historical multi-class development line, rather than to the one-class six-model freeze.

The strengthened exact-device campaign for the six new candidates was not executed before the final evidence freeze. Its protocol requires 20 warm-up and 200 measured inferences over the same 64-image calibration manifest, one-second resource sampling, artifact and image hash verification, and separate preprocessing, runtime, postprocessing and complete-pipeline timing. None of the twelve candidate/platform pairs has a frozen outcome. The existing historical Sentinel and Scout measurements establish the demonstrated pair; they do not fill the six-model device matrix.

4.7 Chapter evidence boundary

This chapter establishes a reproducible data and model-development lineage: the version-7 dataset is frozen, all six single-seed candidates completed, validation-based thresholds were frozen, and the test split was evaluated once without post-test tuning. The chapter does not claim that the dataset is capture-session independent, that all annotations are error free, that any model family is universally superior, or that the six candidates have been validated on both edge devices. External transfer and the strengthened exact-device matrix remain explicit evidence gaps. These boundaries determine which numerical results can be interpreted in Chapters 7 and 8.

CHAPTER 5

System Implementation and Integration

Chapter 3 defined the division of responsibility between the Sentinel and Scout, and Chapter 4 separated the historical deployed models from the controlled six-model study. This chapter describes how the stationary two-device system and the simulation mission layer were implemented. It concentrates on software paths, interfaces, state control, logging and recovery. Experimental protocols and numerical outcomes are reserved for Chapters 6 and 7.

5.1 Implementation scope and status

The repository contains two related implementation layers. The first is the stationary physical perception workflow: a TensorRT node on the Sentinel, a TensorFlow Lite verification node on the Scout, and a correlated ROS 2/DDS request–verdict exchange. The second is a PX4/Gazebo mission implementation that exercises vehicle-state transitions using simulated vehicles. The two layers share event semantics but do not share evidence status. In particular, the simulation does not convert an image-space detection box into a geographic target, and it cannot substitute for physical flight.

Table 5.1 records the status used throughout this chapter. “Implemented” means that source code and configuration exist; “exercised” additionally requires retained runtime evidence. Neither term implies that the component passed every later experimental protocol.

Table 5.1: Final implementation status for the retained study evidence.

Component	Status	Evidence boundary
Sentinel camera and TensorRT node	Implemented and exercised	The cleanly rebuilt stationary deployment executed the final 640-pixel TensorRT engine in the retained ten-minute and thirty-minute campaigns.
Scout camera and TFLite verifier	Implemented and exercised	Live camera decoding and independent full-frame verification were exercised in bounded stationary trials; no coordinate-guided navigation was used.
Temporal filter and coordinator	Implemented and exercised	Three-frame gating, correlated verdicts, timeouts, cooldown and one terminal decision are enforced in source code; comparative effects belong to later chapters.
ROS 2/DDS exchange on SwarmNet	Implemented and exercised	Typed messages crossed the stationary wireless link during unloaded and live-inference campaigns; airborne operation and controlled radio interference were not evaluated.
MAVLink/physical-flight path	Safety locked	Telemetry configuration was retained, while the stationary nodes reject enabled flight commands and the local dry-run adapter blocks arm and motion calls.
Dual-PX4/Gazebo mission layer	Development branches exercised	Confirmed and rejected development paths were executed on provisional proxies; the preregistered repeated formal campaign remains gated.

Figure 5.1 relates the principal processes to the interfaces defined in Chapter 3. Each device retains its own image stream. Only the event record and terminal decision cross the perception boundary.

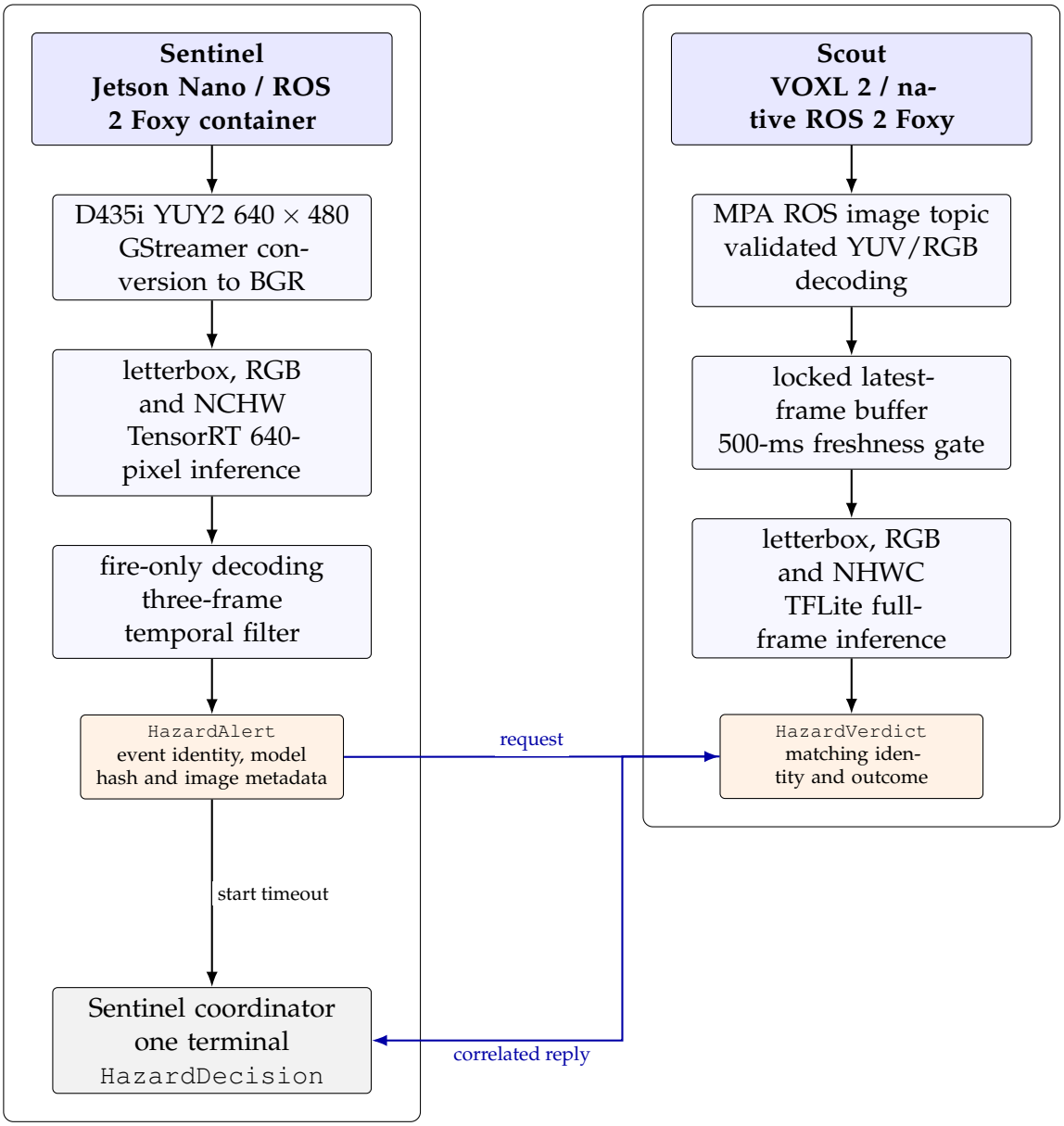


Figure 5.1: Implemented stationary software pipeline. The Sentinel bounding box is transferred for event traceability; the Scout verifier uses its own latest full frame. Flight commands remain disabled in both physical nodes.

5.2 Physical, power and data integration

The Sentinel was built as a customized X500 V2 system rather than treated as an opaque commercial appliance. The Pixhawk 6C provides the flight-controller boundary, while the Jetson Nano was added as a companion computer and the D435i as the perception sensor. The Jetson receives power from the airframe power distribution through a DC–DC regulator; the camera connects to the Jetson; and a three-conductor serial link provides the Jetson–Pixhawk telemetry path. Figure 5.2 records this functional topology together with the more integrated Scout path.

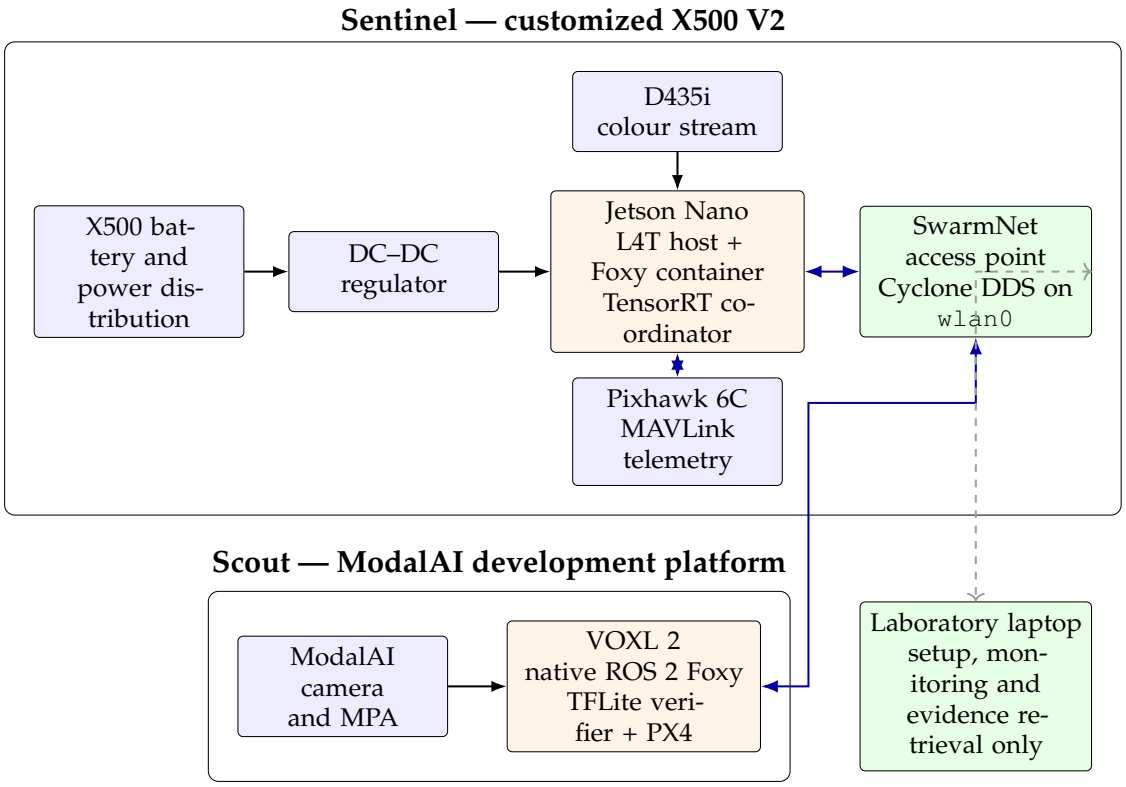


Figure 5.2: Physical power, data and software integration boundary. Solid paths are required by the onboard implementation. The dashed laptop path is used for configuration, monitoring and evidence retrieval and is not the hazard-inference or final-decision path. Electrical ratings and connector-level pin mapping were not admitted as calibrated measurements.

The retained MAVProxy configuration identifies the Jetson serial endpoint as `/dev/ttyTHS1` at 57,600 baud. It does not independently identify which physical connector pins were populated, so the manuscript does not infer a TX/RX/ground mapping from the three-conductor count alone. Similarly, the accepted evidence does not preserve the regulator model, its efficiency or a calibrated input/output time series. The power topology is therefore an implementation fact, while regulator adequacy and energy cost remain unmeasured engineering quantities.

Camera and compute placement also influence payload, balance, field of view, cooling and vibration. The installed D435i pose, complete-aircraft mass, dimensions, rotor spacing and centre of gravity were not measured under the registered geometry protocol before the evidence freeze. The simulation uses provisional platform proxies for this reason. These omissions do not invalidate stationary perception measurements, but they prevent the stationary results from establishing airborne endurance, aerodynamic performance or a calibrated camera-to-body transform.

The Scout retains the ModalAI development platform's integrated VOXL 2, camera and PX4 services. It was not used as a remote accelerator for the Sentinel: it captures and verifies its own frame. This separation preserves the tiered role boundary but requires two independently functioning camera/runtime stacks.

5.3 Heterogeneous operating environments

The two devices do not run an identical operating-system image. Compatibility is established at the ROS 2 interface and DDS wire level rather than by forcing the same host distribution onto dissimilar vendor platforms. Table 5.2 summarizes the retained software inventory.

Table 5.2: Retained onboard software environments and compatibility role.

Layer	Sentinel	Scout
Host platform	Jetson Linux R32.7.6 / JetPack 4.6.6, Ubuntu 18.04.6 and kernel `4.9.337-tegra`	ModalAI VOXL 2 host; the exact base distribution/version was not preserved in the accepted thesis inventory
ROS 2 placement	Foxy in the L4T-compatible `dustynv/ros:foxy-ros-base-l4t-r32.7.1` container	Native Foxy at `/opt/ros/foxy`
Build/deployment	Docker 20.10.21, NVIDIA runtime, host networking and a bind-mounted ROS workspace	Native `colcon` workspace and ModalAI platform services
Camera/runtime	D435i, GStreamer, CUDA and TensorRT	Modal Pipe Architecture bridge, validated YUV/RGB decoding and TensorFlow Lite
Middleware compatibility	Common enhanced message package, Cyclone DDS, selected `wlan0`, multicast, matching domain per experiment and compatible QoS	

Containerization was chosen because the Jetson Nano depends on its NVIDIA L4T host stack, whereas the project required the same ROS 2 Foxy message and middleware generation exposed natively by the Scout. The container retains access to the

NVIDIA runtime, camera devices, host wireless interface and mounted workspace without replacing the Jetson host distribution. This gives a repeatable ROS user space but introduces device-pass-through, host-network and container-image dependencies. It is therefore a compatibility mechanism, not evidence that the hosts are homogeneous or that container overhead is zero.

Using Foxy on both nodes simplifies custom-message generation and reduces cross-distribution uncertainty. Identical Ubuntu releases are not required for DDS communication. What must agree is the message schema, ROS domain, middleware discovery, network interface and compatible QoS. The failed first transport attempt, in which Cyclone DDS selected the Sentinel container's Ethernet interface rather than SwarmNet, demonstrates why matching ROS labels alone are insufficient.

5.4 Sentinel camera and inference pipeline

The Sentinel node is a ROS 2 Foxy Python process executed in the Jetson Nano software environment. Its retained configuration selects the Intel RealSense D435i colour endpoint at `/dev/video2`, a 640×480 YUY2 stream at 30 frames/s, and an explicit GStreamer capture string. The pipeline converts YUY2 to BGR and terminates in an `appsink` configured with one buffer, frame dropping and disabled clock synchronization. These options bound queued camera work: if inference is slower than the source stream, the application uses a recent frame instead of processing an increasingly old backlog.

The camera path is explicit because two generic alternatives failed during bring-up. Direct OpenCV/V4L2 access did not establish the intended conversion path, and the pre-existing QGroundControl streaming service could hold `/dev/video2` exclusively. The accepted preflight procedure temporarily paused that service, opened the explicit GStreamer pipeline, acquired the bounded probe frames, and restored the service. This was an operator-controlled resource arbitration procedure, not automatic camera self-recovery. A failed read during normal node operation is logged as `camera_read_failed` and that inference iteration is skipped.

Preprocessing and TensorRT execution

The preprocessing function derives its target geometry from the engine rather than assuming that a model name implies a resolution. It applies the configured letterbox transform, pads with value 114, converts BGR to RGB, normalizes to $[0, 1]$, and produces a contiguous batch-1 NCHW tensor. The transform object is retained so decoded boxes can be mapped back to the source frame.

At initialization, the TensorRT adapter deserializes the engine and enumerates every binding. It rejects dynamic or non-positive shapes, requires exactly one input and one output, and checks for a batch-1 NCHW RGB input. Host buffers are page locked and

device buffers are allocated from the binding-derived element counts. Each inference verifies the tensor data type and element count before asynchronous host-to-device transfer, engine execution and device-to-host transfer on one CUDA stream. The stream is synchronized before a copied output array is returned. A failed execution is raised to the node and logged as an `inference_error`; it is not converted into a zero-confidence detection.

The accepted final workspace engine is the 640-pixel artifact identified in Section 4.6. Its input and output shapes are $[1, 3, 640, 640]$ and $[1, 6, 8400]$. The implementation must not be described with the timing of the earlier 416-pixel engine. The node stores the engine SHA-256, binding metadata, camera configuration, thresholds, resize mode and disabled flight-command flag in its session manifest before processing frames.

Detection and dispatch

Decoded predictions are filtered to the approved live class, *fire*. The node records the number of retained detections and uses the highest-confidence box as the temporal-filter observation. While a Scout request is pending, the Sentinel continues to record local observations but does not create a second event. When the filter triggers, the source-frame box is normalized and stored with the source dimensions. A new opaque event identifier is attached to the experiment and trial identifiers before publication on `/swarm/hazard_alert`.

The message does not contain a geographic position. Fields named `target_u`, `target_v`, `bbox_width` and `bbox_height` refer only to the Sentinel camera image, as defined in Section 3.4. The Sentinel therefore implements target designation at the event-record level without claiming cross-camera localization.

5.5 Scout camera and verification pipeline

The Scout node runs in the native ROS 2 Foxy environment on VOXL 2. It subscribes to the Modal Pipe Architecture image bridge with the sensor-data QoS profile and maintains a thread-safe latest-frame buffer. The configured topic is `/hires_small_color`. Camera preflight showed that the concrete ROS payload can be YUV 4:2:2 even when the platform camera configuration uses other format terminology. The decoder therefore validates the message dimensions, row step and payload length before accepting YUV 4:2:0, YUV 4:2:2, RGB8 or BGR8 layouts. Unsupported or inconsistent data produces a `camera_decode_error` record.

MPA is useful because it exposes the Scout's vendor-supported camera pipeline without requiring the verification node to implement a new low-level camera driver. The `voxl_mpa_to_ros2_node` bridge converts that platform stream into a ROS 2 image topic consumed by the verifier. This reduced camera integration effort and

kept acquisition on the Scout, but it did not remove format risk: live metadata and ROS encoding terminology did not initially make the concrete byte layout unambiguous. Explicit payload-geometry checks and YUV-to-BGR conversion were therefore required. The bridge and camera server also become runtime dependencies, as demonstrated by the retained unavailable- camera and stale-frame fault experiments.

The latest-frame buffer stores a copied BGR image, a local monotonic timestamp and a sequence number under a lock. On a verification request it returns a snapshot only if a frame exists, time has not regressed, and the frame is no older than the configured 500 ms limit. Missing, stale or clock-regressed states produce an explicit `ERROR` verdict. They are not interpreted as evidence that fire is absent.

Bounded verification work

The alert callback validates the message structure and requires the expected experiment and trial identifiers. Event identifiers are retained in a thread-safe set, so duplicate requests are logged and ignored. Accepted work is placed in a single-element non-blocking queue served by a dedicated worker. If that queue is occupied, the node immediately returns a correlated `ERROR` verdict with reason `verification_worker_busy`. This policy makes overload visible and prevents the DDS callback from waiting for a slow inference.

For each accepted request, the worker snapshots the latest local camera frame, applies the same letterbox/RGB/normalization function with NHWC layout, and invokes the deployed TFLite interpreter with four threads. The adapter requires one float32 input and one output and rejects an unexpected tensor interface. The historical deployment accepts $[1, 640, 640, 3]$ and returns $[1, 7, 8400]$. As explained in Chapter 4, this is a distinct multi-class deployment artifact rather than one of the later fire-only screening candidates.

The worker performs independent full-frame fire decoding at the configured threshold. A retained fire detection yields `CONFIRMED`; otherwise the normal verdict is `REJECTED`. The Sentinel box is stored in the alert record but never selects a Scout crop. The implementation records queue delay, frame age, inference duration, raw maximum fire score, decoded detections and, when enabled, the diagnostic-frame hash. A second set prevents more than one verdict from being published for an event.

The Scout bring-up history is retained because it changed the interpretation of the live system. An output-coordinate assumption initially produced invalid decoded behaviour, and an event-synchronized diagnostic established the correct configuration for the retained artifact. Earlier pre-correction trials remain excluded from accepted performance summaries. The correction altered decoding configuration; it did not retrospectively change the recorded model or camera evidence.

5.6 Temporal filter and correlated decision handling

The three-frame rule is implemented by a small stateful filter. For threshold τ and required count $K = 3$, its update is

$$k_t = \begin{cases} k_{t-1} + 1, & c_t \geq \tau, \\ 0, & c_t < \tau, \end{cases} \qquad z_t = \mathbb{1}[k_t \geq K]. \tag{5.6.1}$$

A non-finite confidence is rejected. A low-confidence frame resets the count, and a successful dispatch resets the filter. The rule counts inference observations rather than elapsed time, so three frames do not represent a fixed physical duration when the achieved inference rate changes.

Listing 5.1 summarizes the implemented control path. It complements the architectural state diagram in Figure 3.3 by showing where validation and terminal outcomes occur.

Listing 5.1: Simplified implemented detection–verification control path.

```
on_sentinel_frame(frame):
    detections = infer_and_decode(frame, live_class="fire")
    best = highest_confidence(detections)
    if event_is_pending():
        log_tracking_observation(best)
        return
    if temporal_filter.update(confidence(best)) and best exists:
        alert = create_correlated_alert(best)
        publish(alert)

on_scout_alert(alert):
    require_valid_session_and_new_event(alert)
    if worker_queue_full():
        publish(ERROR, reason="verification_worker_busy")
    else:
        enqueue(alert)

scout_worker(alert):
    frame = latest_fresh_local_frame()
    if frame unavailable:
        publish(ERROR, reason=frame_status)
    else:
        result = independent_full_frame_inference(frame)
        publish(CONFIRMED if result has fire else REJECTED)

on_sentinel_verdict(verdict):
    require_matching_active_event(verdict)
    publish_exactly_one_terminal_decision(verdict)
```

The Sentinel coordinator accepts a verdict only while it is awaiting the Scout and only for the active event. Finalized identifiers are retained so a duplicate cannot generate another decision. The timeout timer runs independently of the inference timer; expiration creates `TIMEOUT` with zero Scout confidence. A rejection creates an expiring, image-space cooldown around the rejected box centre. This replaced an earlier permanent spatial blacklist that could have suppressed a later genuine flame at a similar image coordinate. The cooldown is a request-rate control, not a world-coordinate exclusion zone.

Every accepted outcome becomes one `HazardDecision`. A confirmed event also emits an `ALARM` console record. No GPIO siren, remote dashboard, emergency-service call or automatic physical response was implemented. This is the boundary between the demonstrated software decision and any operational alarm system.

5.7 Evidence logging and fault recovery

Each physical node creates a session directory whose path contains the experiment, trial, device and node-run identifiers. In formal mode, experiment and trial identifiers are mandatory and a pre-existing directory causes the run to fail rather than overwrite evidence. The session manifest is created with exclusive file mode and records the model path and hash, tensor interface, camera settings, thresholds, runtime options, safety state and test scope.

Runtime events are serialized as JSON Lines by a dedicated writer thread. The producer submits a copy of each record to a queue with capacity 4096 and never waits for disk I/O. Queue overflow increments a locked dropped-record counter, which is itself included in inference records. The writer flushes at the configured interval, and orderly shutdown drains the stop marker, flushes the file and calls `fsync`. This design reduces synchronous logging interference; it cannot guarantee that evidence survives sudden power loss or a damaged filesystem.

Records contain both UTC wall time and local monotonic nanoseconds. Monotonic differences are used for durations on one device. Sentinel and Scout monotonic epochs are unrelated and must not be subtracted. Wall time supports chronology, but the retained experiments do not establish sub-millisecond clock synchronization. The round-trip design in Chapter 6 therefore uses the Sentinel clock for both publication and callback endpoints.

Expected failures are converted into explicit events or error verdicts:

- Sentinel camera and inference failures skip the affected iteration and preserve their reason;
- Scout decoding, stale-frame and worker failures return correlated `ERROR` outcomes rather than ordinary rejection;

- duplicate and mismatched events are retained as ignored or suppressed records; and
- timeout is a terminal outcome distinct from rejection.

Platform recovery is a separate engineering layer. The Scout camera server was restarted under operator control after a bounded stream failure. The Sentinel camera service required controlled pause and restoration during camera access. Later, the Sentinel microSD developed extensive filesystem inconsistency. A complete sector image and configuration archives were frozen before bounded repair attempts, and the damaged card was rejected for further experiments. The clean-rebuild runbook requires storage, camera, engine, ROS 2, SwarmNet and safety acceptance gates before any new device result is admitted. This recovery chronology belongs in Appendix C; it is not evidence of runtime fault tolerance by the deployed perception node.

5.8 SwarmNet and ROS 2/DDS integration

SwarmNet is hosted by the Sentinel access point and gives the Sentinel, Scout and laboratory laptop a local wireless path. ROS 2 Foxy is containerized on the Sentinel and installed natively on the Scout. The enhanced interface package defines the three messages summarized in Table 3.5; the node package implements the publishers, subscribers, camera paths and backends.

The first injected DDS attempt exposed a multi-interface discovery failure. Cyclone DDS selected the Sentinel container's Ethernet interface while the Scout used `wlan0`, so no intended subscriber was discovered. The corrected configuration explicitly selects Cyclone DDS on both devices, binds to `wlan0`, permits multicast and uses an isolated ROS domain for each test. The public configuration contains no access-point password or remote access credential.

The physical workflow uses three topic paths:

- `/swarm/hazard_alert`: Sentinel request with model identity, persistence count, source timing and image-space box;
- `/swarm/hazard_verdict`: Scout response with matching event identity, model identity, frame timing, confidence and reason; and
- `/swarm/hazard_decision`: Sentinel terminal record joining the two stages.

Publisher and subscription queues use depth 10 in the enhanced nodes. The Scout image subscription uses the ROS sensor-data profile, while the compact event topics use the default reliable profile created by the depth argument. The implementation does not expose a custom deadline, lifespan or durability policy. Chapter 6 treats

application-level event counts and round-trip timing as the measured quantities; it does not infer packet-level loss from the absence of missing application replies.

5.9 MAVLink and physical-flight boundary

The recovered Sentinel configuration includes a MAVProxy service that expects the Pixhawk serial endpoint on `/dev/ttyTHS1` at 57,600 baud and forwards MAVLink to a laboratory-network destination. These values are recovered configuration facts and require revalidation after the clean rebuild. MAVLink is designed for constrained vehicle telemetry links [20], but the presence of a service definition does not prove correct estimator, navigation or actuator behaviour.

Both stationary perception nodes declare `enable_flight_commands` and abort initialization if it is true. The separate local mission dry run creates a deterministic lawnmower preview in an arbitrary local ENU frame and exercises hold, resume and confirmation logic through a safety-locked adapter. That adapter refuses arm, takeoff, waypoint and velocity calls. The dry run contains no live MAVLink, PX4, GPS, estimator, collision-avoidance or motor path. It verifies a software boundary rather than vehicle motion.

5.10 Dual-vehicle simulation implementation

The simulation layer uses two namespaced PX4/Gazebo instances with distinct vehicle identities, PX4 instances, MAVLink system identifiers and ROS 2 command and telemetry topics. Scenario manifests provide local-ENU patrol, hold, verification, home and geofence positions. These positions are scenario inputs; they are not calculated from the Sentinel camera bounding box.

The coordinator implements the mission states introduced in Section 3.5. In a confirmed branch, the Sentinel holds, the Scout is dispatched to the scenario verification point, and both vehicles proceed through the configured safe return and landing path. In a rejected branch, the Sentinel resumes its patrol and the Scout returns to its standby path before landing. Development runs exercised both branches using distinct X500 V2-derived Sentinel and ModalAI/Starling-derived Scout proxies. Their results establish executable command paths under those recorded simulator conditions, not formal success probabilities.

Figure 5.3 shows six registered points from the accepted confirmed-hazard visual run. The authoritative instance and spawn manifests identify the orange vehicle as the Sentinel X500 V2-derived proxy and the blue vehicle as the ModalAI/Starling-derived Scout proxy. This mapping supersedes ambiguous labels attached to earlier development screenshots. The red fire marker and its local-ENU location are scenario

truth: they trigger and assess the mission logic but do not represent camera-model inference or vision-derived localization.

The continuous recording from the final v17 run was accepted for thesis and defence presentation after its role mapping and sequence were checked; this qualitative review did not replace the synchronized manifest, event log or PX4/ROS 2 records. The later v6 visual derivation retained the identical accepted video hash and corrected only the thesis contact sheet: the safe-standoff panel now uses the registered capture frame and the alarm panel reproduces exact fields from the retained event log.

Fault support is fail closed. The mission-node-restart case, for example, terminates the first coordinator with a distinct restart-required status while leaving both simulator instances running. A separate recovery process then re-discovers both identities, enters degraded/RTL handling, publishes safe setpoints and verifies landing without sending a new arm command. Source and mocked orchestration tests exist, but a mocked handoff is not a Gazebo result.

The accepted visual pilot closes one presentation gate, but the repeated 310-run matrix remains incomplete; therefore no reliability rate or general fault-recovery claim is made in this chapter.

5.11 Integration boundary

The implementation realizes the tiered architecture defined in Chapter 3 without silently extending its scope. The Sentinel performs local fire detection and temporal gating; the Scout independently verifies its own current view; ROS 2 correlates the request, verdict and final decision; and append-only records preserve the evidence path. The physical nodes remain stationary and reject flight enablement. The simulated mission uses scenario coordinates and remains separate from image-space perception.

The implementation is therefore sufficient for the bounded experiments defined next. Four validation gaps nevertheless remain explicit: the six-model exact-device matrix, the common-stimulus tiered-workflow ablation, the formal repeated simulation campaign and external transfer evaluation were not completed. Chapters 6 and 7 retain those distinctions when defining and interpreting the evidence.

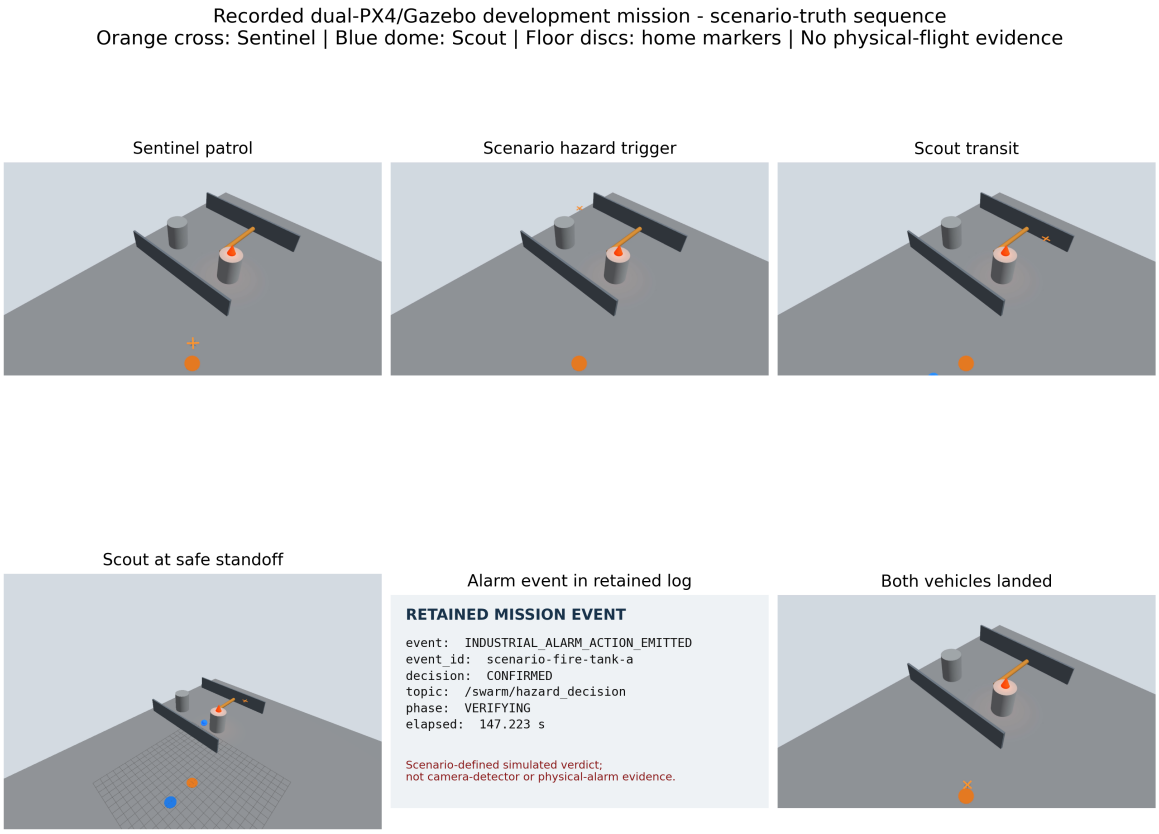


Figure 5.3: Accepted Gazebo confirmed-hazard development sequence. The orange cross-shaped Sentinel patrols and holds at the scenario trigger; the blue dome-shaped Scout dispatches to a 4.5 m commanded horizontal standoff point; verification produces the simulated alarm event; and both proxies return and land. The fifth panel reports the exact alarm fields retained in the mission event log rather than substituting an unrelated camera frame. The coloured floor discs are home markers, not vehicles. The visual frames and log panel are linked to the retained event log and continuous video. Airframe dynamics remain provisional simulation proxies, and the hazard trigger, acquisition, and verdict are scenario truth rather than detector outputs.

CHAPTER 6

Experimental Methodology and Data Analysis

Chapters 4 and 5 defined the frozen data, candidate models, deployed software and integration boundaries. This chapter specifies how those artifacts are evaluated. Its purpose is to make the unit of analysis, timing boundary, decision rule and execution status explicit before numerical outcomes are reported in Chapter 7. The distinction is important because the repository contains both completed experiments and strengthened protocols prepared after limitations were found in earlier work.

The five research questions require different evidence. Offline images can compare predictive behaviour, but cannot establish sustained operation on an edge device. A camera-free device replay can measure a processing path, but cannot include acquisition latency. A stationary lighter trial can establish that the integrated request–verdict path executed under one bounded condition, but cannot estimate autonomous-flight reliability or industrial transfer. The methodology therefore preserves each evidence category rather than combining unlike observations into one performance claim.

6.1 Study design and evidence hierarchy

The study uses a staged engineering-evaluation design. Each stage answers a different part of the research questions and admits evidence only when its configuration, raw records and acceptance decision are retained. The stages are:

1. **offline predictive evaluation**, using the frozen validation and test manifests to compare six detector configurations;

2. **exact-device measurement**, using a fixed image replay to measure executable model–runtime–platform combinations;
3. **stationary integrated evaluation**, using synchronized event identifiers and manually marked visible-flame intervals on the physical Sentinel and Scout;
4. **application communication evaluation**, using correlated ROS 2 requests and replies on the SwarmNet network;
5. **mission simulation**, using two namespaced PX4/Gazebo vehicles and scenario-defined local-ENU truth; and
6. **external proxy evaluation**, admitted only after licensing, annotation alignment and overlap checks pass.

Evidence is ordered by traceability rather than by how favourable it appears. Raw machine-readable records and immutable manifests are the primary observations. Deterministic analysis outputs derived from those records are calculated evidence. Tables and figures in Chapter 7 transcribe accepted derived evidence, while engineering interpretations are reserved for Chapter 8. Screenshots and operator notes provide context, but they do not replace a missing log, configuration or ground-truth interval.

Table 6.1 records the execution boundary at this chapter release. "Executed" means that the retained protocol produced an accepted evidence bundle. "Development only" means that an executable path was demonstrated but the planned repeated campaign was not completed. "Pending" means that the protocol exists but cannot contribute a result.

Table 6.1: Experimental evidence classes and execution status at the Chapter 6 release.

Evidence class	Status	Admissible scope
Six-model validation and frozen test	Executed	Predictive comparison on the curated internal data under the specified object-matching and operating point rules.
Historical deployed-device replay	Executed	Camera-free processing measurements for the exact historical model, runtime, shape and device; not the strengthened six-model device matrix.
Twelve model/platform device cells	Pending	The strengthened protocol is frozen, but no cell is admitted until conversion, fidelity and device gates pass.
Concurrent-service endurance	Executed with limitations	A stationary 30-minute complete-stack run combined both live perception paths, ROS 2, logging, Scout PX4 services and Sentinel Pixhawk/MAVProxy telemetry; the Scout latency and strict Jetson swap criteria were not met.
Stationary integrated flame trials	Executed, bounded	Proof of the correlated physical perception workflow under marked laboratory exposure; no flight, localization or industrial-site inference.
ROS 2 request–reply trials	Executed	Separate 300-event camera-free campaigns and a 1,000-event campaign under concurrent live inference load; all timing uses one Sentinel monotonic clock.
Dual-vehicle mission scenarios	Development only	Executable nominal and selected fault paths with simulator-local truth; no formal reliability estimate.
External proxy evaluation	Pending	No transfer result is admitted until archive, licensing, alignment and contamination gates close.

The mapping in Table 6.2 prevents an unanswered question from receiving an inferred positive answer. It also identifies the experimental unit used by each protocol. Images, source-family groups, application events, physical exposures and simulation runs are not interchangeable independent samples.

Table 6.2: Research-question mapping to protocols, experimental units and final evidence disposition.

RQ	Principal protocol	Unit of analysis	Status
RQ1	Frozen offline validation/test and exact-device screening	Object, image, source-family group; model–platform cell	Offline comparison executed; strengthened device matrix not executed.
RQ2	Full-service endurance with camera, inference, communication and logging active	Device run and time-indexed resource sample	Thirty-minute accepted complete-stack evidence exists; the run passed duration and process-survival gates but failed the Scout latency and strict swap gates.
RQ3	Common-stimulus stage ablation and synchronized stationary trials	Labelled frame sequence or marked physical exposure; correlated event	Integrated proof-of-function executed; comparative effect not evaluated.
RQ4	Preregistered two-vehicle simulation campaign	Retained run with a unique seed and attempt identifier	Development paths executed; formal 310-run campaign not executed.
RQ5	Frozen-model inference on eligible external proxy datasets	Eligible image or independently defined video event	Admission work recorded; inference evaluation not executed.

6.2 Offline predictive evaluation

The internal predictive comparison uses the curated fire-only manifest from Chapter 4. Its validation split contains 700 images, 392 annotated fire boxes and 381 background images distributed across 307 reconstructed source-family groups. The one-time test split contains 733 images and 503 fire boxes. All six detector configurations use the same manifests, label policy and prediction floor of 0.001. The validation data are used to choose an operating threshold; the test data are evaluated once with that frozen threshold and cannot be used for later tuning.

Object matching and detection metrics

Predictions are sorted by confidence and matched greedily to unmatched ground-truth boxes in the same image. A pair is a true positive when its intersection over union is at least 0.50. For predicted box B_p and ground-truth box B_g, intersection over union is

$$\mathrm{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}. \tag{6.2.1}$$

An unmatched prediction is a false positive and an unmatched ground-truth box is a false negative. Precision, recall and F1 are then

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}. \tag{6.2.2}$$

The denominator is stated whenever a metric is reported. Object precision is not described as an image-level false-alarm probability. For the background subset, the reported background-image false-positive proportion is

$$p_{\mathrm{BG,FP}} = \frac{\#\{\text{background images with at least one retained prediction}\}}{\#\{\text{eligible background images}\}}. \tag{6.2.3}$$

This proportion measures whether an image contains any retained false detection. It is not a false-positive rate per hour or per video frame.

Average precision follows the COCO convention across IoU thresholds from 0.50 to 0.95 in steps of 0.05, with AP50 reported separately [26]. These ranking metrics are threshold-integrated and are therefore distinguished from the single operating-point counts used to compare deployed decisions.

Operating-point selection and scale analysis

For each model, thresholds from 0.01 through 0.99 in increments of 0.01 are evaluated on validation data. The selected threshold maximizes object-level F1 at IoU 0.50. A tie is resolved first in favour of higher recall and then in favour of the lower threshold. This deterministic rule prevents a later manual choice based on the test set. The selected threshold and model hash are written to the frozen evaluation record before the one-time test pass.

Recall is also stratified by ground-truth box area relative to image area: less than 1%, 1–5%, 5–20% and at least 20%. The bins describe apparent target scale in the curated images; they are not physical fire-size categories because camera distance and calibration are unavailable. Laptop inference timing uses batch size one after ten excluded warm-up iterations. It is a software-screening measure and is not transferred to either edge platform.

Uncertainty intervals are calculated by a grouped nonparametric percentile bootstrap. The resampling unit is the reconstructed source-family group, not the individual image. Two thousand bootstrap samples use seed 20260729 and resample the 307 validation groups with replacement. The 2.5th and 97.5th percentiles form the descriptive 95% interval. This follows the bootstrap principle of resampling observed units [27] while retaining within-group dependence [28]. The intervals do not correct annotation error, source-selection bias or imperfect group reconstruction, and overlapping intervals are not treated as a formal significance test.

6.3 Exact-device screening and resource measurement

Two device-evidence programmes must remain separate. The first consists of completed historical measurements of the deployed Sentinel and Scout configurations. Those runs use their recorded runtimes, tensor interfaces, model hashes and input shapes. Their camera-free replay begins with a fixed 480×640 black BGR image, applies the deployed preprocessing, executes the runtime and decodes the output. Earlier baseline runs used five warm-up and 50 measured iterations; later retained deployed measurements used their own documented 100-iteration configuration. Each result is reported only with its exact protocol.

The second programme is the strengthened cross-platform screen specified after the six-model offline study. It defines 12 candidate cells: each of six models on each of two edge platforms. A frozen, hashed sequence of 64 validation-calibration images is replayed in manifest order and cycled as needed. Each admitted cell requires 20 warm-up iterations followed by 200 measured iterations at batch size one. Resource telemetry is sampled at 1 Hz. The timed pipeline includes preprocessing, runtime execution and decoding; camera acquisition, network transport and event coordination are excluded.

Each cell records source model hash, converted-artifact hash, runtime and version, precision, input and output shapes, preprocessing, decoder, threshold, device identity, power mode, cooling state and concurrent services. Conversion success alone is insufficient. A fidelity check must establish that the converted runtime behaves within the frozen acceptance rules on the calibration set before latency or resource values are ranked. The final status vocabulary is `DEPLOYED`, `TOO_SLOW`, `OUT_OF_MEMORY`, `CONVERSION_FAILED`, `RUNTIME_UNSUPPORTED` or `FIDELITY_FAILED`. A failed cell remains part of the comparison rather than disappearing from its denominator.

None of the 12 strengthened cells was admitted to the final evidence set. The registered campaign was not executed before the evidence freeze. Chapter 7 may therefore report historical exact-deployment measurements, but it cannot present them as the completed strengthened matrix.

Timing boundaries

Figure 6.1 defines the timing quantities used across the thesis. Runtime inference measures only the framework call. Device-pipeline time includes preprocessing, runtime and decoding. The historical replay and strengthened device screen are camera-free, so neither includes exposure, driver buffering or image acquisition. Application round-trip time begins when the Sentinel publishes an alert and ends when the matching verdict callback is received on the Sentinel. Alert-to-decision time continues until the terminal decision is recorded.

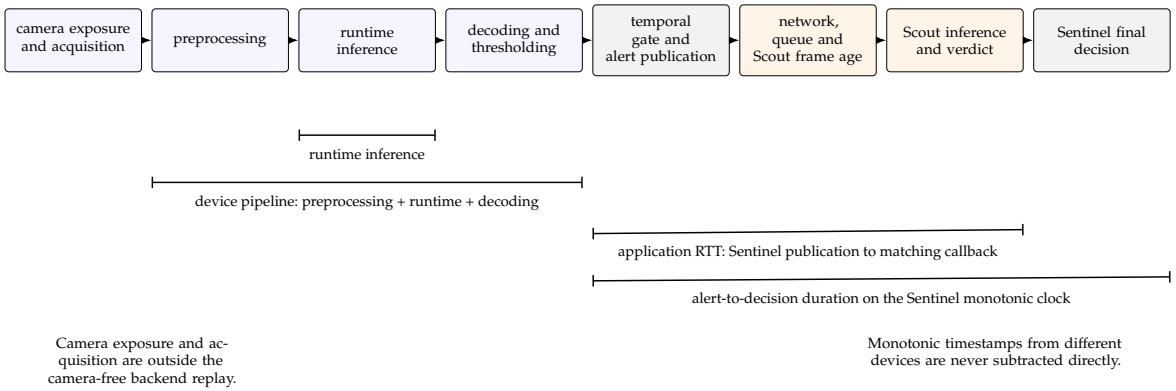


Figure 6.1: Timing boundaries used by the evaluation. Brackets identify different measured quantities; they must not be compared as if they covered the same processing path.

For an ordered sample $x_1, \ldots, x_n$, each latency or resource series is summarized by sample size, arithmetic mean, median, minimum, maximum, range and sample standard deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{6.3.1}$$

Percentiles are added where the retained sample size supports them. Throughput is computed from the complete measured interval rather than as the reciprocal of a rounded mean latency. Warm-up observations are retained separately but excluded from the reported steady measurement. No outlier is removed solely because it is slow; exclusions require a recorded protocol violation or invalid run state.

The two Sentinel engines require particular care. Historical Test 1 used a 416-pixel engine with tensors $[1, 3, 416, 416] \rightarrow [1, 6, 3549]$. The final ROS workspace uses a distinct 640-pixel engine with tensors $[1, 3, 640, 640] \rightarrow [1, 6, 8400]$. Latency or throughput from the 416-pixel artifact is never assigned to the deployed 640-pixel engine without a new controlled measurement of the matching hash.

6.4 Full-service endurance

RQ2 asks whether a selected configuration can operate sustainably with its required concurrent services. A short backend loop cannot answer that question. The registered endurance protocol therefore ran each device for at least 30 minutes with the services required by its role active: camera input, deployed preprocessing and inference, ROS 2 communication, asynchronous evidence logging and the platform services needed for the test. The model hash, input shape, runtime, power mode, cooling arrangement, ambient conditions and service list must be frozen before the run.

Resource samples were requested at 1 Hz where the platform exposes the signal. The retained fields include CPU load, GPU or accelerator load, resident and available

memory, temperature, power when a calibrated or platform-supported reading is available, processing time, achieved throughput, camera-read failures, inference errors, dropped log records and communication outcomes. Unavailable measurements are marked unavailable; they are not estimated from nominal hardware specifications.

Acceptance required completion of the full interval without process death, unbounded memory growth, thermal shutdown, silent evidence loss or a safety state violation. A time series is inspected for warm-up behaviour, sustained drift and discrete failures. Temperature and processing time are plotted against elapsed time, while resource summaries retain their units and sample counts. A trend may be described, but a causal thermal-throttling claim requires corresponding frequency or platform throttle evidence.

A later complete-stack run operated the live D435i/TensorRT Sentinel path, the live Scout camera/TFLite path, ROS 2 transport, asynchronous logging, Scout PX4 services, Sentinel Pixhawk/MAVProxy telemetry and approximately 1 Hz host-resource sampling for more than 30 minutes. The preregistered 360-alert probe used a mismatched trial identifier and was correctly rejected by the Scout. This setup failure is retained. A corrected 36-alert probe exercised current-frame Scout inference during the final 180.245 s without treating the synthetic alerts as camera-detection accuracy observations. Chapter 7 therefore reports duration, service coexistence and resource behaviour across the full interval, but request–verdict latency only for the separate corrected three-minute probe.

6.5 Temporal-filter and tiered-workflow evaluation

RQ3 requires a common-stimulus comparison of three decision stages:

1. a single Sentinel frame at the frozen confidence threshold;
2. the same Sentinel sequence with the three-consecutive-frame temporal filter; and
3. the retained Sentinel alert followed by a correlated Scout verdict.

A valid comparative unit must contain a complete ordered frame sequence or a synchronized physical exposure with event-level ground truth. The same frames, thresholds and eligibility rules are applied to all stages. The analysis then counts ground-truth events retained, unsupported alerts removed and additional decision delay introduced by filtering and verification. Event sensitivity is the proportion of eligible hazard events that reach the relevant stage. False-alert burden must use a declared exposure denominator, such as per eligible negative event, per labelled negative sequence or per observed hour. These denominators cannot be reconstructed from a handful of selected alert records.

For the implemented filter, an observation qualifies when its highest retained fire confidence reaches threshold τ. The run length resets after a non-qualifying observation, and an alert is created when the run length reaches three. Filter delay can be measured in processed observations. Conversion to seconds requires the actual observation timestamps because the inference rate is not assumed constant.

The accepted physical trials use stationary, propeller-free vehicles and a handheld lighter positioned approximately 20–25 cm from cameras mounted at approximately 19 cm for the Sentinel and 9.5 cm for the Scout. Ignition and extinction intervals were marked separately from node startup. Correlated event identifiers join the Sentinel alert, Scout verdict and Sentinel decision. The analysis script verifies those joins and hashes the input logs before writing a derived JSON result; raw records are not edited.

Only post-correction trials are eligible. Early Scout trials used an incorrect image-decoding or output-decoding assumption and are retained as diagnostic history. The later 0.05 Scout threshold was selected after diagnostic evidence, so the physical trials using it are bounded engineering observations rather than a preregistered accuracy estimate. The strong visible-flame exposure, the small-flame exposure and six deliberately forced unlit requests answer different questions and are reported separately. Forced unlit requests check the correlated rejection path; they do not estimate an environmental false-positive rate.

The older unsynchronized lighter recordings remain historical proof-of-function only. Ignition was not synchronized to script or log start, so script-start-to-detection time is invalid. Unlit intervals cannot be counted as false negatives, and trial-level success rates cannot be inferred without known exposure ground truth. Because the retained physical evidence is small, heterogeneous and dependent, no confidence interval or significance test is attached to it. The complete comparative tiered-workflow ablation was not executed.

6.6 Communication and fault experiments

The completed communication protocol consists of three sequential, camera-free ROS 2 trials with 100 request–response events each. The trials use isolated domain identifiers 43, 44 and 45, Cyclone DDS bound to `wlan0`, reliable volatile event topics, queue depth 100 and a 1-second application timeout. The Sentinel publishes one request only after the previous event has completed or timed out, which avoids uncontrolled queue growth but also means that the observations form a serial sequence on one network setup.

Response completeness is

$$C = \frac{N_{\text{matching replies}}}{N_{\text{published requests}}}. \tag{6.6.1}$$

This is an application-level measure. It is not packet loss: DDS may retransmit, and the experiment does not capture every link-layer packet. Application RTT for event (i) is calculated entirely on the Sentinel monotonic clock,

$$RTT_i = \frac{t^S_{\text{callback},i} - t^S_{\text{publish},i}}{10^6} \text{ ms.} \tag{6.6.2}$$

The superscript (S) indicates the Sentinel clock. Scout and Sentinel monotonic epochs are never subtracted.

All 300 intended observations, including slow tails, remain in the summary. The analysis reports the descriptive statistics defined in Equation 6.3.1, counts above 10, 20, 50 and 100 ms, and identifies Tukey tail observations for inspection without deleting or winsorizing them. No confidence interval or significance claim is made because the events are serial, share one radio environment and do not represent independent network deployments.

A separate loaded protocol publishes 1,000 sequential correlated events on an isolated ROS 2 domain while both camera/inference nodes, asynchronous loggers, resource samplers and an independent periodic tiered-workflow probe remain active. The same Sentinel-clock RTT definition is used. Completeness receives a two-sided Wilson interval because the outcome is a bounded delivery proportion; latency remains descriptive because the serial events share one run and one radio environment. The Sentinel Pixhawk is excluded from this loaded protocol, so it does not become complete-stack evidence.

A separate 30-sample-per-device clock-offset probe uses midpoint estimates based on a request and response timestamp. The minimum-RTT sample is retained as the least-queued offset indication. This supports chronology checks only; it does not justify retroactively subtracting unsynchronized cross-host wall times for fine-grained latency.

Fault evidence is categorized by layer. The physical coordinator tests retain duplicate, mismatched, stale and timed-out verdict behaviour. Simulation tests exercise mission-level cases such as lost alert, Scout unavailable, delayed verdict, sensor or inference error, low battery, invalid geofence and duplicate/stale verdicts. A software unit test or mocked process handoff is not reported as a physical-network or Gazebo fault result.

Four stationary hardware fault protocols use five unique event identifiers per condition. Removing the Scout ROS camera interface tests fail-closed `ERROR/no_frame`; proving a live preflight and then stopping only the bridge tests `ERROR/stale_frame`. Removing the Scout verification process tests the Sentinel's unchanged 10 s timeout. Finally, injecting two `CONFIRMED` verdicts after each event has already finalized as `TIMEOUT` tests idempotent rejection of late and duplicate

outcomes. These are deliberate interface or dispatch-boundary faults, not estimates of natural fault frequency or wireless-delay distributions.

6.7 Simulation and external-evaluation protocols

Two-vehicle mission simulation

The simulation unit is one retained run with a unique scenario, seed and attempt identifier. Each run starts two namespaced PX4 v1.17 SITL instances and the corresponding ROS 2 Jazzy mission processes. Vehicle identity, local-ENU scenario coordinates, geofence, battery conditions, fault injection, expected state path and terminal acceptance rule are fixed in the manifest. Simulator truth is used only to assess the commanded mission path; it is not presented as vision-based localization.

Development runs exercised nominal confirmed and rejected branches on provisional vehicle dynamics, together with selected bounded faults. A later accepted visual confirmed-hazard run used distinct X500 V2-derived Sentinel and ModalAI/Starling-derived Scout proxies and retained synchronized mission events, video and hashes. These records show that the software path can execute in that development environment. The formal campaign is a separate preregistered programme of 310 runs: 50 nominal confirmed, 50 nominal rejected, 50 repeated-confounder runs and 20 repetitions for each of eight fault scenarios. Seeds and run identifiers are unique and deterministic. Automatic retry is prohibited; a rerun receives a new attempt identifier so that failed attempts remain visible.

Formal acceptance requires the scenario preflight, expected state sequence, terminal vehicle state, safety invariant checks, complete logs and artifact hashes to pass. Selected runs also require visual captures, but video does not replace telemetry. The accepted visual run closes that requirement for one confirmed-hazard pilot. A final qualitative review additionally accepted the continuous recording from the v17 run for presentation after checking role labels, continuity and framing; this disposition adds no mission trial to the denominator. In the final evidence set, the repeated campaign was not executed because measured airframe parameters, the full fault set and the retained-run matrix have not all passed. Consequently, development traces cannot be converted into a mission success percentage.

External proxy evaluation

External evaluation freezes the internal models, preprocessing and operating thresholds. A dataset is eligible only after five gates close: a compatible licence is recorded; the acquired archive and extracted files are hashed; files decode successfully or every failure is enumerated; labels or event intervals can be mapped to the fire-only policy;

and exact, perceptual and source-family overlap with training, validation and test data is checked. The eligible manifest is frozen before inference.

The FASDD publisher reports 25,097 UAV samples under CC BY-SA 4.0, but the archive has not been acquired and admitted. Its sample count is therefore dataset context rather than an evaluated denominator. The MmodalFire archive and licence checks passed for 28 videos. Decoding produced 2,974,209 frames against 2,974,228 container-reported frames, with 19 one-frame warnings. These are readiness observations, not detector results. Condition and event alignment is incomplete: the stricter timestamp consolidation validated eight of 28 ranges and left 20 for review. No external inference is admitted while those gates remain open.

If admitted, video analysis defines independent events or condition intervals before computing metrics; millions of neighbouring frames are not treated as independent trials. Results are stratified by eligible dataset, acquisition condition and apparent target scale. Any finding is described as transfer to a specific UAV or industrial-like proxy dataset. It does not establish operation at an industrial site.

6.8 Statistics, reproducibility and exclusions

Every reported table must identify the artifact and configuration, sample size, unit, threshold, timing boundary and evidence source. Raw observations remain append-only. Derived files record their input paths and SHA-256 hashes, analysis command or script version, configuration hash, creation time and acceptance outcome. A correction creates a new derived artifact; it does not silently edit the source log.

Data cleaning is rule-based. A record is excluded only when a predefined eligibility condition fails or a documented protocol violation makes the quantity uninterpretable. Examples include pre-correction Scout decoding, missing formal identifiers, a mismatched model hash, an incomplete camera preflight or a run outside its marked ground-truth interval. Runtime failures, timeouts, conversion failures and out-of-memory outcomes are substantive outcomes and remain in the experiment register. Missing values are reported as missing rather than imputed from neighbouring samples.

Descriptive statistics are the default because several experiments are bounded engineering studies rather than random samples from a defined population. Confidence intervals are used only where the resampling unit and dependence structure are defensible, as in the grouped bootstrap for internal predictive metrics. Statistical significance is not claimed merely because two point estimates differ or two intervals fail to overlap. For repeated simulation, a binomial success proportion and interval may be reported only after the formal campaign is admitted and the denominator includes every retained attempt.

The evidence cutoff for this final controlled revision is 30 September 2026, after admission and author visual acceptance of the v17 mission delivery. Late evidence entered only through coordinated revisions of the affected methodology status, result table, research-question answer, abstract and conclusion. This rule prevents a late number from being inserted without updating the assumptions and limitations that give it meaning.

Evidence boundary. The offline six-model evaluation, historical deployed-device measurements, loaded ROS 2 round-trip trial, 30-minute complete-stack endurance run, accepted visual mission and four repeated hardware fault cases are executed evidence. The strengthened 12-cell device screen, common-stimulus ablation, formal 310-run simulation campaign and external transfer evaluation were not executed. Chapter 7 preserves these boundaries and does not infer a full positive answer for RQ1, RQ3, RQ4 or RQ5.

CHAPTER 7

Experimental Results

This chapter reports the accepted observations produced by the protocols in Chapter 6. Results are ordered from offline model behaviour to device execution, physical integration, communication, simulation and external-data readiness. Each numerical result retains its experimental unit and configuration. In particular, laptop inference calls, camera-free device pipelines and live alert-to-decision paths are not combined into a single latency measure.

Table 7.1 states the final evidence boundary. The distinction prevents a prepared protocol or a successful development run from being reported as a completed experiment.

Table 7.1: Results admitted to the final evidence set. A protocol that was not executed or a development-only result remains visible as a limitation rather than being converted into a positive finding.

Evidence programme	Status	Result admitted in this chapter
Six-model internal evaluation	Executed	Validation-selected operating points and one-time frozen-threshold test results.
Conversion fidelity	Partly executed	Four YOLO ONNX comparisons, SSD TFLite comparison and native EfficientDet artifact readiness.
Strengthened 12-cell device screen	Not executed	No model–platform ranking; the preparation gate remained blocked.
Historical device measurements	Executed	Exact recorded pipeline and integrated-node measurements, each with its own configuration boundary.
Concurrent-service endurance	Executed with limitations	More than 30 minutes of live camera, inference, ROS 2, logging and flight-service coexistence; duration and process-survival gates passed, but Scout p95 latency and strict Jetson swap gates failed.
Stationary physical workflow	Executed, bounded	Condition-specific confirmation, rejection and latency observations with propellers removed.
ROS 2 request–reply trials	Executed	Two distinct 300-event camera-free campaigns and one 1,000-event campaign under concurrent inference load.
Hardware fault repetitions	Bounded subset executed	Five-repetition camera- interface, stale-frame, unavailable-process and late/duplicate-verdict outcomes; network interruption and rejection-memory tests were not executed.
Mission simulation	Development evidence only	Two nominal branches, four fault pilots and one accepted end-to-end visual confirmed-hazard run; no formal success probability.
External proxy evaluation	Not evaluated	Readiness observations only; no external predictive-performance result.

7.1 Dataset audit and six-model predictive results

The common evaluator processed all six candidates on the frozen validation split of 700 images and 392 annotated fire boxes. The split contained 381 background images and 307 reconstructed source-family groups. Table 7.2 reports the threshold selected by maximum object-level F1 at IoU 0.50 and the metrics observed at that operating point. The final column is the mean single-image inference-call time from the laptop screen; it is not an edge-device or end-to-end latency.

YOLOv8n-416 produced the largest validation AP50–95 (0.236), AP50 (0.566), precision (0.681) and F1 (0.612). YOLOv8n-416 and YOLOv8n-640 tied for the largest recall (0.556). EfficientDet produced no above-threshold prediction on the 381 validation background images, but it also had the lowest validation AP50–95 and recall. The zero background count is limited to this retained subset and threshold.

After the checkpoints, thresholds and evaluator were frozen, the one-time test pass used 733 images, comprising 354 positive images, 379 background images and 503 annotated fire boxes across 306 reconstructed source-family groups. No test-driven threshold adjustment or checkpoint reselection followed.

EfficientDet ranked first on test AP50–95, precision and F1 despite ranking last on validation AP50–95. It tied YOLOv8s-640 for the largest test recall. F1 decreased from validation to test for every candidate, and the grouped intervals were broad and overlapping. Figure 7.1 shows the AP50–95 rank change without altering the frozen selection rule.

Table 7.2: Common validation results on 700 images and 392 fire boxes. B-FPI is the proportion of 381 background images containing at least one retained prediction. Each timing row contains 700 post-warm-up inference calls.

Candidate	AP50–95	AP50	Threshold	Precision	Recall	F1	B-FPI	Mean (ms)
YOLOv8n-416	0.236	0.566	0.45	0.681	0.556	0.612	0.0026	9.89
YOLOv8n-640	0.220	0.542	0.36	0.553	0.556	0.555	0.0105	7.91
YOLOv8s-416	0.203	0.497	0.30	0.549	0.533	0.541	0.0079	11.14
YOLOv8s-640	0.213	0.532	0.22	0.576	0.548	0.562	0.0052	10.99
SSD MobileNet-320	0.185	0.491	0.81	0.667	0.500	0.571	0.0052	27.44
EfficientDet-320	0.161	0.461	0.41	0.582	0.434	0.497	0.0000	1263.86

Table 7.3: One-time test results at validation-frozen thresholds. The grouped 95% F1 intervals resample the 306 retained test source-family groups.

Candidate	Thr.	AP50–95	AP50	Precision	Recall	F1	95% F1 interval	TP/FP/FN	B-FPI
YOLOv8n-416	0.45	0.119	0.303	0.541	0.300	0.386	[0.236, 0.544]	151/128/352	0.0554
YOLOv8n-640	0.36	0.101	0.260	0.409	0.274	0.329	[0.175, 0.450]	138/199/365	0.0396
YOLOv8s-416	0.30	0.130	0.319	0.508	0.320	0.393	[0.236, 0.537]	161/156/342	0.0792
YOLOv8s-640	0.22	0.133	0.359	0.545	0.328	0.409	[0.245, 0.545]	165/138/338	0.0343
SSD MobileNet-320	0.81	0.114	0.303	0.546	0.237	0.330	[0.162, 0.479]	119/99/384	0.0554
EfficientDet-320	0.41	0.168	0.403	0.679	0.328	0.442	[0.274, 0.596]	165/78/338	0.0343

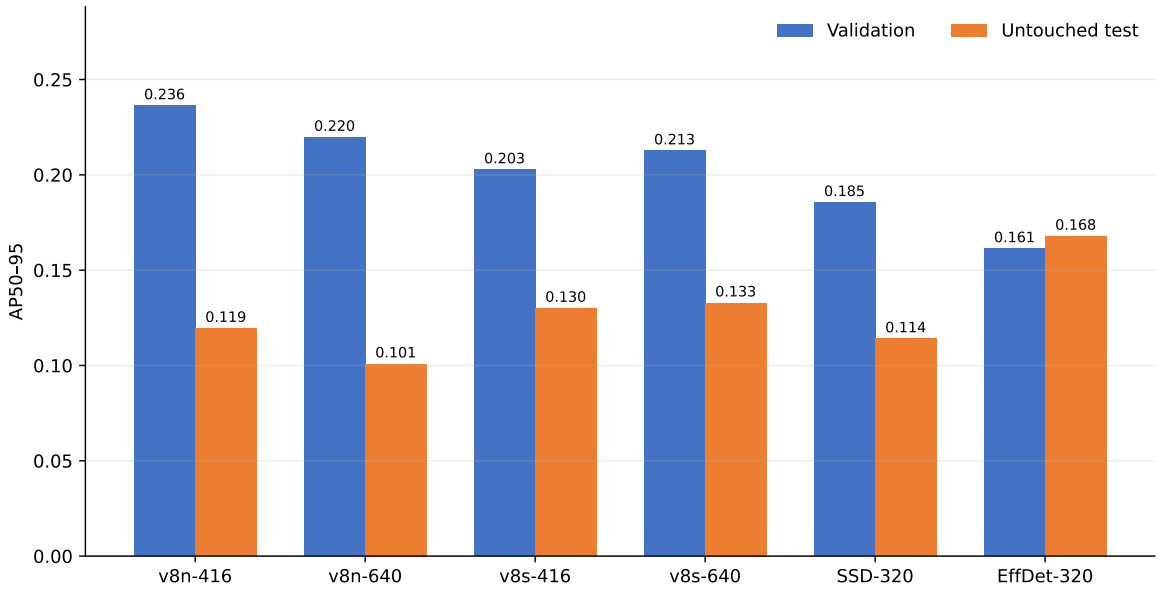


Figure 7.1: AP50–95 on the 700-image validation split and the one-time 733-image test split. Test inference used checkpoints fixed before the test split was opened; AP is integrated over confidence rather than evaluated at the selected operating threshold.

Only three test boxes met the COCO small-object definition at stored resolution. A separate relative-area audit found 54 boxes occupying less than 1% of their image. Those counts are insufficient for a general small-fire claim and do not validate the approximately 2 cm lighter condition used later.

The two-objective validation screen placed YOLOv8n-416 and YOLOv8n-640 on the exploratory AP–latency Pareto front (Figure 7.2). The four YOLO measurements and SSD used GPU-capable framework paths, whereas the EfficientDet export used CPU/XNNPACK. The plot therefore describes available runtime paths rather than an architecture-only speed comparison.

7.2 Conversion fidelity and deployment outcomes

All four fixed-batch YOLO checkpoints exported to ONNX and passed the ONNX checker. Conversion fidelity was evaluated on the same 64-image calibration manifest used to preserve the native reference predictions. Under the frozen match-rate and confidence-error gates, the two 416-pixel ONNX candidates passed, while both 640-pixel ONNX candidates failed. YOLOv8n-640 missed the 0.95 match-rate gate by 0.0004; the criterion was not relaxed after inspection.

Table 7.4: Prepared deployment artifacts and laptop conversion-fidelity outcomes. Ready indicates eligibility for a later runtime attempt, not successful execution on either edge device.

Candidate	Prepared form	Match rate	Conf. MAE	Disposition
YOLOv8n-416	ONNX FP32	0.9603	0.000056	Fidelity pass; ready for runtime attempt.
YOLOv8n-640	ONNX FP32	0.9496	0.000255	`FIDELITY_FAILED`.
YOLOv8s-416	ONNX FP32	0.9598	0.000223	Fidelity pass; ready for runtime attempt.
YOLOv8s-640	ONNX FP32	0.9169	0.000212	`FIDELITY_FAILED`.
SSD MobileNet-320	TFLite FP32	0.8714	0.003910	`FIDELITY_FAILED`.
EfficientDet-320	Native TFLite	N/A	N/A	Native artifact ready for device attempt.

The SSD conversion allocated and produced the same number of retained outputs as the native evaluator, but only 5,577 of 6,400 detections matched at IoU 0.50, yielding a 0.8714 match rate. Its small matched-confidence error did not override the failed detection-match criterion. EfficientDet was already evaluated in its native TFLite form, so cross-format fidelity was not applicable. Its artifact was classified ready for a device attempt rather than deployed.

The strengthened screen required 12 model–platform outcomes. In the final evidence set, the executor and all artifact-integrity checks had passed, but both devices were not ready for the campaign. Consequently, none of the 12 cells contributes latency, memory, thermal or deployment-ranking evidence in this chapter. The historical

640-pixel Sentinel TensorRT and Scout TFLite deployments remain separate evidence and are not invalidated by the later ONNX experiments.

7.3 Exact-device runtime and endurance

The retained 3 August camera-free benchmark used the exact final deployed artifacts, ten excluded warm-up iterations and 100 measured iterations per device. The measured pipeline included preprocessing, runtime execution and decoding. Table 7.5 shows the descriptive results. The platforms, models, runtimes and tensor layouts differ, so the ratio is a system-role observation rather than a controlled model comparison.

Table 7.5: Exact deployed backend-pipeline results, excluding camera capture, ROS 2 communication and decision handling (100 iterations per device).

Platform	Runtime	Input tensor	Mean (ms)	Median (ms)	P95 (ms)
Sentinel	TensorRT FP16	$1 \times 3 \times 640 \times 640$	58.151	57.868	59.680
Scout	TensorFlow Lite	$1 \times 640 \times 640 \times 3$	999.431	991.630	1105.076

The reciprocal mean rates were approximately 17.20 and 1.00 pipelines per second. The Sentinel row belongs to the final engine with input $[1, 3, 640, 640]$, output $[1, 6, 8400]$ and its recorded engine hash. It is not the historical 416-pixel engine with output $[1, 6, 3549]$; no value from the latter is assigned to the final deployment.

Figure 7.3 presents the complete retained distributions together with the later communication campaign. Each panel uses its own horizontal scale, avoiding the loss of detail that would result from placing millisecond and approximately one-second distributions on one linear axis.

A separate July before-and-after software experiment measured the effect of replacing Python anchor iteration with vectorized NumPy decoding. Under its own fixed black-frame 5/50 protocol, mean decode time changed from 148.920 to 1.290 ms on the Sentinel path and from 83.134 to 0.992 ms on the Scout path. Mean complete-pipeline time changed from 204.324 to 57.049 ms and from 1070.811 to 979.667 ms, respectively. These are separate serial code-revision runs with incomplete control of background load and starting temperature. They establish the measured software bottleneck reduction in that historical configuration, not a confidence interval or a replacement for the 3 August exact-deployment benchmark.

Trial `camera10m-threads1-20260928-r1` operated both live camera and deployed inference paths, ROS 2 transport, asynchronous logging and 1 Hz host-

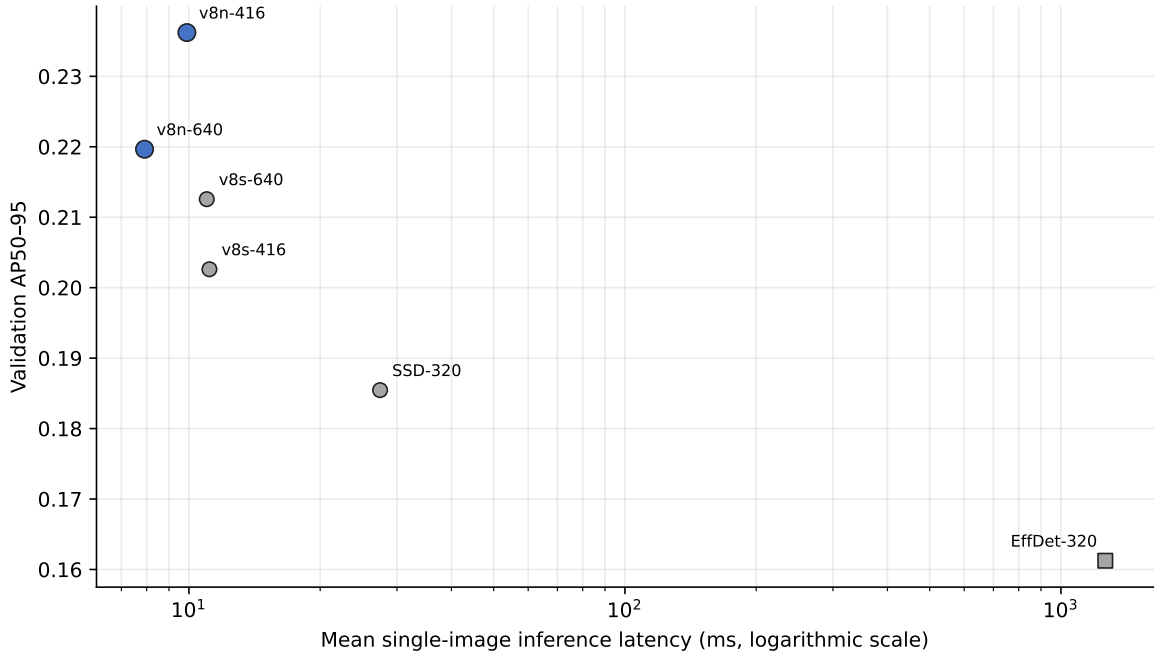


Figure 7.2: Validation AP50–95 versus mean single-image laptop inference-call latency. Blue points identify the two-objective exploratory Pareto front. The logarithmic latency axis and runtime distinction are retained because the six candidates did not use an identical accelerator path.

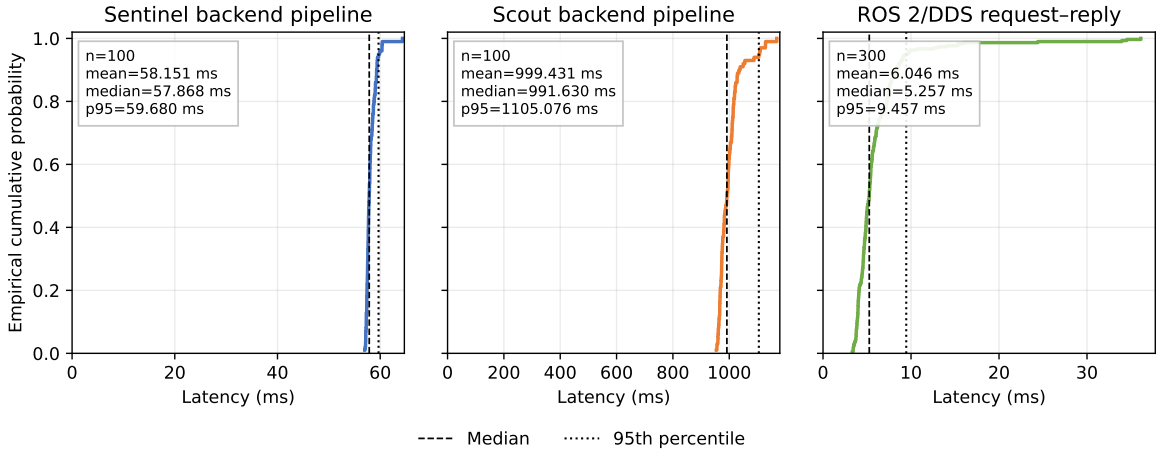


Figure 7.3: Empirical cumulative distributions for the exact deployed camera-free backend pipelines and the 3 August correlated ROS 2/DDS request–reply campaign. Dashed and dotted lines identify the median and 95th percentile within each panel.

resource sampling. The periodic probe received 108 of 108 correlated current-frame Scout verdicts; both nodes exited normally, no camera error or logger drop was recorded, and swap did not grow. Sentinel inference covered 9,411 frames with 61.239 ms p95 latency. Scout inference covered 108 requests with 2,504.387 ms p95 runtime. Alert-to-verdict latency was 2,529.161 ms mean and 2,616.852 ms p95, so the run failed the registered Scout p95 target of at most 2 s. Minimum host-wide available memory was 1,581.8 MiB on Sentinel and 4,879.5 MiB on Scout.

The observation is useful but incomplete: the Sentinel Pixhawk was unpowered, synthetic alerts bypassed Sentinel detection, and the interval was ten minutes rather than the registered thirty. Therefore no synchronized 30-minute complete-stack sustainability result is inferred from this ten-minute trial alone; the later registered run is reported below.

The later registered run `fullstack-20260929` closed the duration and flight-service gaps. The Sentinel resource record lasted 1,939.709 s with 1,552 samples; the Scout record lasted 1,934.836 s with 1,833 samples. Both wrappers exited normally. The Sentinel processed 30,426 frames at 60.035 ms mean and 61.170 ms p95 inference time. Mean host CPU utilization was 32.65% on Sentinel and 42.44% on Scout; minimum available memory was 1,589.9 and 4,873.8 MiB, respectively, and the largest valid reported temperatures were 51.0 and 48.2 °C. No logger drop was recorded. MAVProxy was active and reported its link as operational before and after the run; a pre-existing fence-breach message remained visible, and no vehicle command was sent.

The strict acceptance result was conditional rather than an unqualified pass. Jetson swap increased by 0.922 MiB, violating the registered no-growth rule. The original 360-alert probe carried the wrong trial identifier and produced 0/360 accepted requests because the Scout correctly rejected the mismatches. That failed setup is retained. The corrected probe returned 36/36 correlated verdicts during a separate 180.245 s interval near the end of the run. Its mean and p95 alert-to-verdict times were 2,526.056 and 2,617.746 ms, so the registered 2 s Scout target also failed. These 36 events demonstrate live service coexistence but are not a 30-minute continuous tiered-workflow-load result.

The result partially answers RQ2: the complete stationary service configuration survived the registered duration with stable available-memory and temperature observations, but it did not satisfy every acceptance criterion. In particular, the Scout latency target and strict Sentinel swap rule remain measured resource constraints rather than discarded failures.

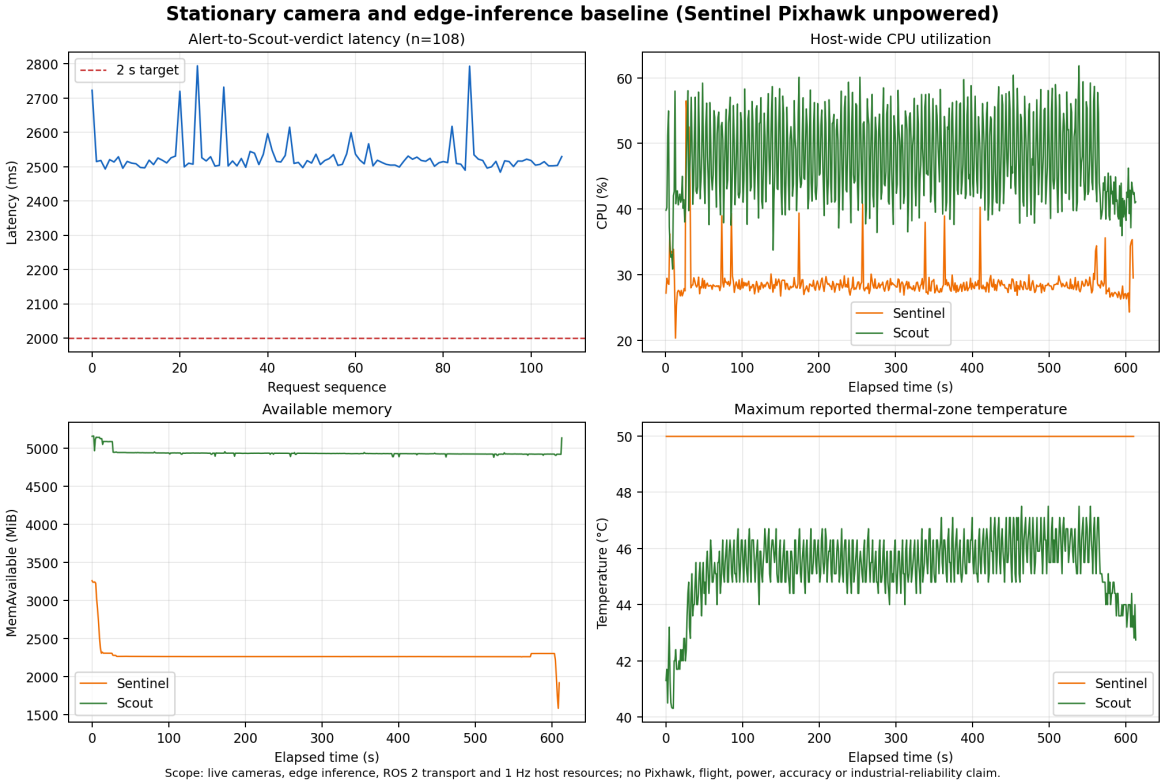


Figure 7.4: Ten-minute stationary camera and edge-inference baseline. The latency panel exposes the failed 2 s Scout target; the remaining panels show host-wide CPU, available memory and maximum reported thermal-zone temperature. The Sentinel Pixhawk was unpowered, so the figure is not complete-stack endurance evidence.

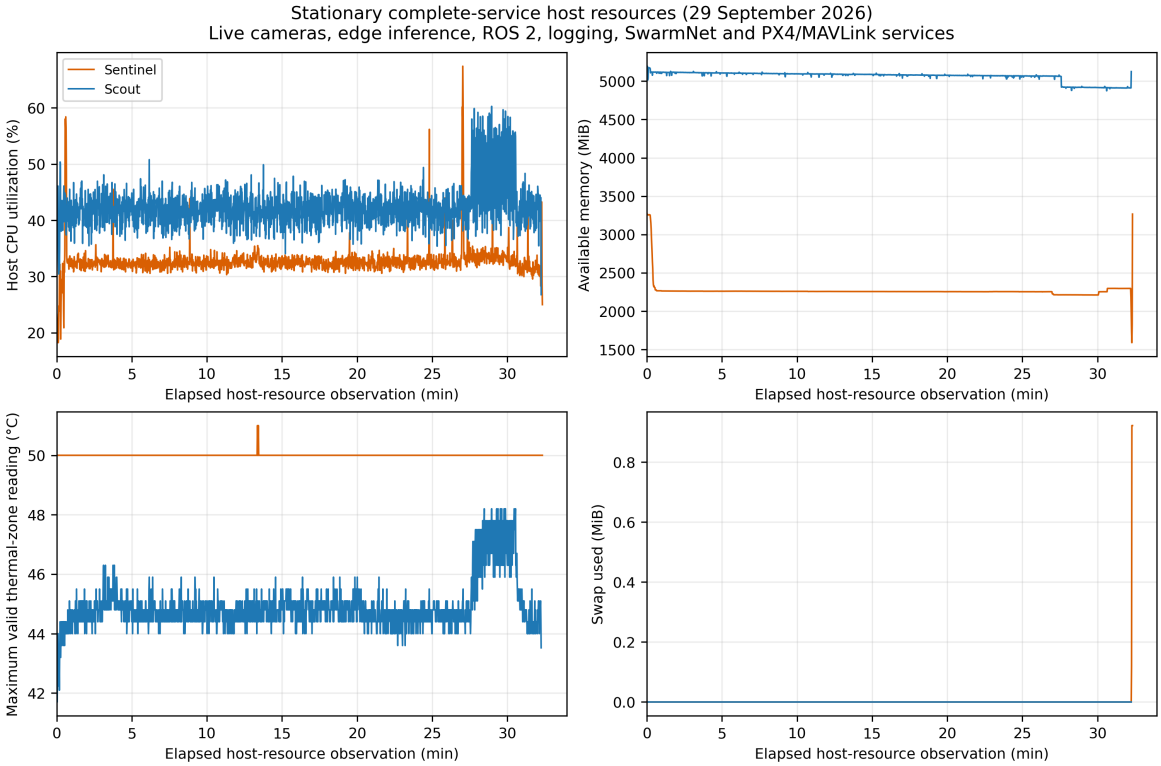


Figure 7.5: Stationary complete-stack resource series for the Sentinel and Scout. Coverage exceeds 30 minutes on both platforms. CPU, available memory, swap and maximum valid thermal-zone temperature are host observations at their retained sample times; the small Sentinel swap increase is not suppressed.

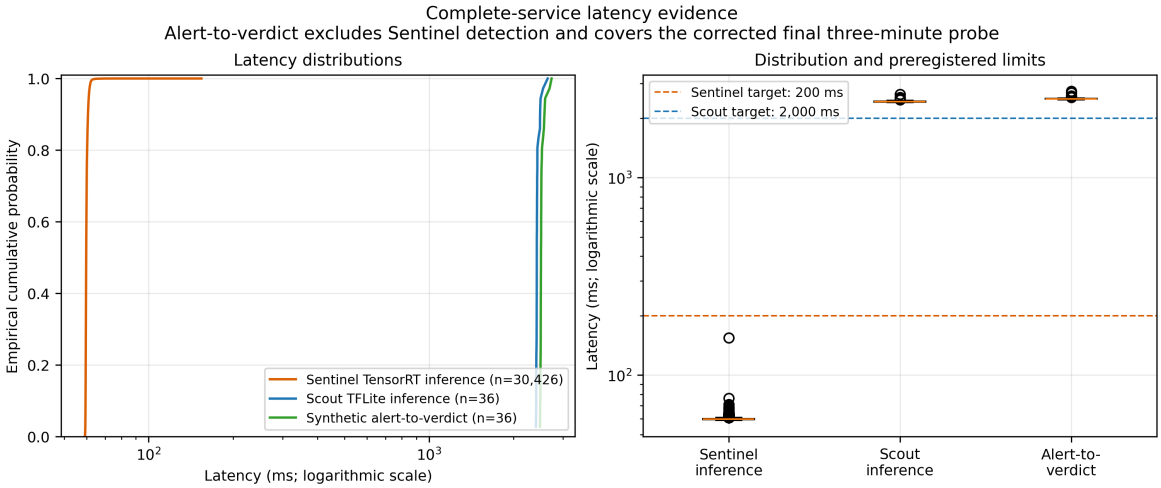


Figure 7.6: The Sentinel distribution contains 30,426 inference frames. The Scout and alert-to-verdict distributions contain the 36 corrected probe events from the final 180.245 s and therefore describe that bounded probe, not the full endurance interval. The Scout p95 exceeds the registered 2 s criterion.

7.4 Stationary workflow, filtering and verification

The accepted physical trials used stationary vehicles with propellers removed. The final Sentinel ran the 640-pixel TensorRT engine and three-consecutive-frame temporal filter. The Scout used the retained 640-pixel TFLite model with a 0.05 threshold selected after diagnostic inspection. Event identifiers matched across alert publication, Scout receipt, Scout inference, verdict publication and final Sentinel decision.

Table 7.6: Accepted stationary event observations. Fractions are conditional counts within the recorded scene and interval; they are not population recall, specificity or industrial reliability estimates.

Condition	Requests	Confirmed	Rejected	Admissible observation
Strong visible flame	13	13	0	Complete correlated path accepted every request inside the 28.740 s marked interval.
Approx. 2 cm unstable flame	10	1	9	The same path was sensitive to the smaller, unstable condition.
Forced unlit live frame	6	0	6	Correlated requests were rejected in the retained unchanged unlit scene.

The strong-flame run contained 14 total requests when one pre-interval rejected request was included. Scout inference had mean 1256.648 ms, median 1244.841 ms and range 1214.606–1416.544 ms. Sentinel alert-to-final-decision time had mean 1438.437 ms, median 1416.765 ms and range 1405.759–1591.737 ms. During the bounded Sentinel node run, 2,220 inference events had mean duration 60.606 ms and maximum 174.018 ms; no logger drop was recorded.

The small-flame run contained 11 total requests, including one pre-interval rejection. Mean Scout inference and alert-to-final-decision times were 1278.495 and 1463.527 ms. The maximum raw Scout score was 0.088566 and only one visible-interval request exceeded the 0.05 decision threshold. The Sentinel again retained 2,220 inference events, with mean duration 60.537 ms, and both loggers reported zero dropped records.

The forced unlit observations comprised one request and a five-request replication. All six received correlated rejected verdicts. In the five-event replication, the maximum raw fire score had mean 0.000890 and range 0.000441–0.002052; Scout inference had mean 1315.770 ms. These requests were injected after Sentinel detection and therefore do not measure Sentinel false alerts or environmental specificity.

The temporal filter's reset and third-frame trigger behaviour passed unit tests. However, the accepted trials retained post-filter requests rather than every frame-level

candidate and counter transition. The number of single-frame alerts removed, event retention and filter-added time cannot be reconstructed without inventing unlogged observations. The planned quantitative three-stage ablation is consequently not estimable from this campaign.

7.5 Communication and recovery

Two independent camera-free request–reply campaigns each contained three sequential 100-event trials. All 600 requests received one correlated Scout verdict, and neither campaign recorded an event-identity mismatch or responder duplicate. They are reported separately because their distributions and raw records differ.

Table 7.7: Separate bounded ROS 2/DDS application round-trip campaigns. The measurements use the Sentinel monotonic clock and are not one-way latency or packet-loss estimates.

Campaign	Replies	Mean	Median	P95	P99	Maximum
			milliseconds			
29 July	300/300	8.152	5.878	12.758	32.407	400.453
3 August	300/300	6.046	5.257	9.457	24.501	36.120
28 September loaded	1000/1000	4.897	4.393	7.984	—	44.011

The 29 July campaign retained one 400.453 ms event below the one-second timeout; the next three largest values were below 36 ms. This single event raised its pooled mean and standard deviation while leaving the median near the later campaign. The 3 August result therefore does not erase the earlier tail. Both 300-event campaigns excluded camera acquisition, model inference, image traffic, movement, controlled interference and long-duration load.

The 28 September campaign ran while the live D435i/TensorRT Sentinel node, the live Scout camera/TFLite verifier, both asynchronous loggers, resource samplers and a separate domain-85 tiered-workflow probe were active. All 1,000 events returned one correlated response, with zero missing, mismatched, duplicate or unexpected events. The observed delivery fraction was 1.0 (two-sided Wilson 95% interval 0.99617–1.0). The independent load witness concurrently received 36 of 36 fresh camera-based Scout verdicts. This remains stationary application-level evidence; the Pixhawk was unpowered, no link-layer packet capture was taken and one-way latency is not inferred.

A clock-offset probe further showed why cross-host timestamps were not used as one-way delay. The minimum-round-trip estimates placed the Sentinel and Scout wall clocks approximately 178.535 and 201.010 ms ahead of the laptop, respectively, with a derived Scout–Sentinel difference of approximately 22.475 ms and a conservative

combined uncertainty of ± 5.088 ms. The round-trip results avoid that subtraction by using only one Sentinel monotonic clock.

Four repeated fault cases produced accepted evidence. Table 7.8 reports their specified outcomes; each case used five unique events and retained its deliberate fault boundary.

Table 7.8: Stationary hardware fault-injection results. Fractions describe the registered repetitions and are not natural failure-rate estimates.

Fault	Expected outcome	Result	Boundary
ROS camera interface absent	`ERROR/no_frame`	5/5; zero inference	Camera service itself remained active.
Stream stopped after live preflight	`ERROR/stale_frame`	5/5; zero post-stop inference	Three normal camera/model verdicts first established readiness.
Scout verifier absent	Sentinel `TIMEOUT` at 10 s	5/5; zero Scout verdicts	Synthetic detections entered at the dispatch boundary.
Two verdicts after each timeout	Reject finalized duplicates	10/10 rejected; zero extra decisions	Deterministic late/duplicate injection, not network delay.

Mean fault-response latency was 47.389 ms for the unavailable camera interface and 35.341 ms for the stale stream. Scout-process absence produced timeout decisions at 10,023.481 ms mean. Rejection of the ten post-timeout verdicts took 2.631 ms mean and 2.840 ms p95 at the callback. Network interruption and repeated rejection-memory cases were not completed; consequently these results support specific fail-closed and idempotent paths, not a general recovery-rate or mean-time-to-recover claim.

7.6 Simulation outcomes

The dual-PX4 development environment executed both nominal decision branches through namespaced ROS 2 Jazzy and PX4 SITL interfaces. The confirmed branch completed Sentinel patrol and hold, Scout dispatch, verification, confirmed hold, return and landing. The rejected branch returned the Scout to standby, resumed Sentinel patrol and ended with both vehicles landed. Table 7.9 also records the retained fault pilots.

Table 7.9: Development simulation outcomes using provisional Gazebo airframe proxies and scenario-defined local-ENU targets. Each row is one retained pilot, not a formal repeated campaign.

Scenario	Outcome	Terminal state	Retained observation
Confirmed hazard v1	Pass	Both `LANDED`	Confirmed branch completed through the real SITL command interfaces.
Rejected confounder v1	Pass	Both `LANDED`	Rejected branch resumed Sentinel patrol; Scout required six mode/arm attempts after temporary health rejections.
Lost alert v2	Pass	Both `LANDED`	No Scout command attempt; typed `DEGRADED`, `RTL`, `LANDED` sequence retained.
Scout unavailable v2	Pass	Both `LANDED`	No Scout command attempt; distinct failure reason and safe sequence retained.
Delayed verdict v1	Pass	Both `LANDED`	Timeout moved both previously flown vehicles through the safe sequence.
Sensor/inference error v1	Pass	Both `LANDED`	Error retained an unverified-hazard record and initiated the safe sequence.

The first lost-alert and Scout-unavailable pilots exposed premature assignment of the Scout `TAKEOFF` state despite no actuation. The controller was corrected and the v2 logs showed a direct `PREFLIGHT` to `DEGRADED` transition. No wrong-system command or namespace cross-talk was observed in the retained pilots. The two nominal and four fault passes do not estimate mission reliability. A later accepted visual confirmed-hazard run used system ID 2 for the orange X500 V2-derived Sentinel and system ID 3 for the blue ModalAI/Starling-derived Scout. It reached `PATROL` at 39.821 s, received the scenario hazard at 105.215 s, began Scout transit at 133.424 s, reached the 4.5 m commanded horizontal standoff at 144.124 s, published the scenario-defined confirmed verdict and alarm at 147.223 s, and ended with both vehicles landed and disarmed at 214.253 s. Figure 5.3 provides the visual sequence; its complete continuous video, event log and manifest are hash-linked. The Sentinel resumed independent patrol while the Scout verified the target. The red fire marker, trigger, target-acquisition event and verdict were simulator scenario truth, not outputs of a detector. The accepted v17 run used a recorded 60 s landing bound after the retained v16 attempt missed the earlier 30 s bound during recording; this threshold change and both attempts remain traceable.

The continuous recording from the final v17 run was subsequently inspected in full. The qualitative review established that the retained presentation was understandable and role-consistent for thesis and defence use; it is not an additional simulator run and does not change the quantitative or reliability interpretation. The v6 derived delivery preserves that video's hash while replacing the misleading alarm still with a hash-linked event-log panel and using the registered safe-standoff capture frame.

The preregistered 310-run campaign was not executed; several formal injectors and the final airframe-parameter freeze remain open. The accepted visual run is therefore one traceable development observation, not a mission-success rate.

7.7 External generalization

No external predictive-performance result was admitted. FASDD-UAV licensing was resolved to CC BY-SA 4.0 and its publisher reported 25,097 UAV samples, but the archive had not been acquired, hashed, decoded, mapped or screened for overlap. No FASDD-UAV file entered training or evaluation.

The MmodalFire archive and license passed integrity checks. All 28 retained videos decoded without a decoder failure, yielding 2,974,209 frames compared with 2,974,228 container-reported frames; the nineteen one-frame differences remain metadata warnings. The stricter alignment consolidator validated 8 of 28 video ranges and left 20 for independent review. Event/condition alignment, overlap checks and a final eligible manifest remained open.

These readiness observations show why RQ5 has no evaluated result rather than a negative result. No frozen model was run on an eligible external manifest, so the chapter reports neither an external accuracy estimate nor a transfer conclusion.

Evidence boundary. In the final evidence set, RQ2 has a 30-minute complete-stack result with two failed acceptance criteria; RQ3 has no common-stimulus temporal/tiered-workflow ablation; RQ4 is supported by development simulation and one accepted visual mission only; and RQ5 has no admitted external inference result.

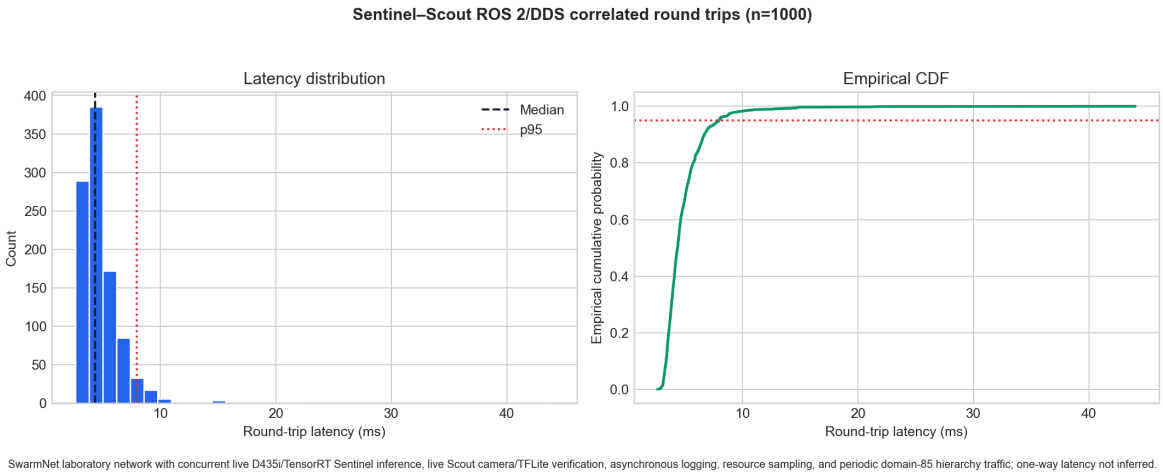


Figure 7.7: Distribution and empirical cumulative distribution of the 1,000 Sentinel-clock ROS 2/DDS round trips under concurrent live camera and inference load. The median and p95 markers are descriptive for this stationary SwarmNet run; one-way latency and link-layer loss are not inferred.

CHAPTER 8

Discussion and Limitations

Chapter 7 showed that the evidence is uneven across the five research questions. The internal model comparison, historical device measurements and stationary two-node workflow produced quantitative observations. A 30-minute complete-stack run established bounded concurrent service survival while exposing unmet Scout latency and strict swap criteria. Comparative temporal-filter effects, formal mission reliability and external transfer were not established. This chapter interprets that pattern rather than treating every implemented component as an experimentally validated contribution.

The central hypothesis proposed that a resource-aware tiered edge architecture can divide hazard-detection responsibilities between heterogeneous drones. The results support the narrower claim that the implemented Sentinel and Scout can execute that division under bounded stationary conditions and that the division exposes measurable predictive, runtime and latency trade-offs. They do not establish that the tiered workflow improves accuracy in general, that the selected pair is optimal, or that the system is ready for autonomous industrial use.

8.1 Answers to the research questions

Table 8.1 gives the final evidence-bounded answer to each research question. "Bounded answer" means that the retained evidence answers part of the question under stated conditions while another evidence layer remains unavailable. "Not evaluated" means that no eligible experiment supports a performance answer.

Table 8.1: Evidence-bounded answers to the five research questions.

RQ	Status	Answer supported by the accepted evidence
RQ1	Bounded answer	The six candidates exhibited different predictive and laptop-runtime trade-offs, and the historical Sentinel and Scout pair executed on their devices. The strengthened 12-cell screen was not completed, so the experiment does not select a supported candidate for every platform.
RQ2	Bounded answer	A stationary 30-minute complete-stack run completed without process death, OOM, thermal shutdown or logger loss while both flight-service paths were active. It missed the Scout p95 target and showed 0.922 MiB Sentinel swap growth, so not every acceptance criterion was satisfied.
RQ3	Bounded answer	The Scout produced both correlated confirmations and rejections, and the added decision time was measured. The effects of the temporal filter and tiered workflow on event retention and false-alert burden are not estimable from the retained frame-level evidence.
RQ4	Bounded answer	Two nominal branches, four selected fault paths and one hash-linked visual confirmed-hazard mission completed in development simulation. The formal repeated campaign and physical two-vehicle mission were not completed, so correct behaviour is demonstrated only for the retained pilots.
RQ5	Not evaluated	Neither external source reached a frozen eligible inference manifest. No transfer-performance conclusion is supported.

RQ1: predictive performance and platform support

The internal comparison rejects the idea of a universal detector winner. YOLOv8n-416 led validation AP50–95 and F1, while EfficientDet led test AP50–95, precision and F1. All candidates lost F1 between validation and test, and their grouped intervals overlapped. The ranking reversal is compatible with sensitivity to split composition, source domains, target scales and model inductive biases. It does not justify replacing the validation-frozen selection with the model that happened to rank first on the test split.

Runtime also changed the ordering. EfficientDet's stronger test result used an available CPU/XNNPACK path whose laptop inference call was approximately two orders of magnitude slower than the GPU-capable YOLO paths. Conversely, YOLOv8n-640 had the lowest observed YOLO laptop time but lower validation AP than YOLOv8n-416. These comparisons establish a multi-objective selection problem; they do not show which model will preserve its ranking after conversion and execution on either edge device.

The historical exact-device evidence shows that the final Sentinel TensorRT and Scout TFLite artifacts were executable in their assigned roles. It cannot complete the six-candidate selection question because only that deployed pair was measured under the retained 100-iteration device protocol. The answer to RQ1 is therefore descriptive for offline predictive trade-offs and historical deployment, but incomplete for candidate-by-platform support.

RQ2: sustainable operation with concurrent services

The ten-minute run first showed that both live cameras and deployed inference paths could coexist with ROS 2, logging and resource sampling. The later complete-stack

run extended that observation beyond 30 minutes and added Scout PX4 services plus Sentinel Pixhawk/MAVProxy telemetry. Both wrappers closed normally, no logger drop or OOM occurred, minimum available memory remained 1,589.9 MiB on Sentinel and 4,873.8 MiB on Scout, and maximum valid reported temperatures were 51.0 and 48.2 °C.

RQ2 therefore has a bounded rather than universal answer. The evidence supports stationary service coexistence for the measured interval, not unlimited or airborne sustainability. Moreover, the corrected tiered-workflow probe covered only the final 180.245 s. Its 2.618 s p95 alert-to-verdict time exceeded the registered 2 s target, and Sentinel swap increased by 0.922 MiB. Preserving these failures is important: resource-constrained operation was feasible under the observed load, but the registered configuration was not compliant with every performance criterion.

RQ3: temporal filtering and secondary verification

The stationary evidence demonstrates a functional second decision stage. The Scout confirmed all 13 requests in the stronger visible-flame interval, confirmed one of ten requests in the smaller unstable-flame interval, and rejected all six forced unlit requests. A fixed acknowledgement would not have produced this condition-sensitive pattern. The event identity chain also shows that the returned verdict was connected to the correct Sentinel request.

The evidence does not quantify improvement. The forced unlit requests bypassed Sentinel detection, so they cannot measure false alerts prevented by the tiered workflow. The logs did not retain every pre-filter candidate and filter-counter transition, so they cannot reconstruct how many single-frame candidates the three-frame rule removed or how many labelled hazards it retained. A single-frame, filtered and verified comparison on a shared labelled sequence is still required.

The measured cost of verification is material. Mean Scout inference exceeded 1.25 s in both accepted positive trials, and mean Sentinel alert-to-final-decision time was approximately 1.44–1.46 s. The architecture therefore adds an independent RGB decision at the cost of delayed resolution. Whether that trade-off is appropriate depends on the operational cost of an unsupported alert relative to the cost of waiting for confirmation. The present study measures the latency but does not define a safety-critical deadline for an industrial plant.

The repeated fault cases strengthen the coordination result without turning it into a reliability estimate. Camera-interface loss and a post-preflight stale stream failed closed in all five registered repetitions; absence of the Scout process generated one timeout per event; and all ten verdicts injected after timeout were rejected without an additional final decision. These outcomes demonstrate specific defensive paths. They do not measure natural fault rates, wireless recovery or rejection-memory effectiveness.

RQ4: two-vehicle mission behaviour and faults

The simulation pilots demonstrate executable nominal branches and selected safe-state transitions through real PX4 SITL and namespaced ROS 2 interfaces. They also exposed a genuine state-assignment defect: two pre-fix fault pilots assigned `TAKEOFF` before Scout actuation. The corrected v2 runs moved directly from `PREFLIGHT` to `DEGRADED`. Retaining both versions is stronger engineering evidence than reporting only the repaired runs because it shows the failure, correction and regression outcome.

Six passing pilots do not support a reliability estimate. The rejected nominal scenario required six mode/arm attempts after temporary health failures, which shows that a final pass can contain operational variability. The incomplete 310-run campaign, provisional airframe proxies and scenario-defined target coordinates prevent a broader claim about mission correctness. RQ4 is therefore answered for the existence of the implemented branches, not for their probability of success or their physical-flight behaviour.

RQ5: external transfer

RQ5 has no evaluated performance answer. FASDD-UAV had not been acquired and locally audited. MmodalFire passed archive and decode checks, but only 8 of 28 video ranges passed the stricter timestamp consolidation and the event labels and overlap gates were still open. Running the detector before those gates closed would have produced numbers without a defensible denominator or independence claim. Reporting the question as not evaluated preserves the value of the future evaluation and avoids using readiness work as surrogate accuracy evidence.

8.2 Predictive performance and resource trade-offs

The comparison illustrates why predictive performance, runtime portability and device feasibility must remain separate selection dimensions. An offline checkpoint can rank well and still lack a faithful conversion. The two 416-pixel YOLO ONNX artifacts passed the frozen fidelity gate, while both 640-pixel ONNX exports and the SSD TFLite export failed. Successful export and runtime allocation were therefore insufficient indicators of equivalent behaviour.

The YOLOv8n-640 ONNX result is a useful borderline example. Its match rate of 0.9496 was close to the 0.95 rule, and its matched confidence error was small. Rounding the result upward or relaxing the gate would nonetheless change the experiment after observing the outcome. Preserving the failure supports a repeatable comparison and leaves room for a separately registered alternative conversion.

The historical deployed pair reflects system roles rather than a model-family optimum. The Sentinel 640-pixel TensorRT pipeline averaged 58.151 ms in the camera-free benchmark, while the Scout TFLite pipeline averaged 999.431 ms. The devices, models, runtimes and output interfaces all differed. The ratio supports the observation that the deployed Sentinel path was faster than the deployed Scout path; it does not isolate whether hardware, model size, runtime, delegate use or implementation caused the difference.

The decoder experiment reinforces this point. Vectorization reduced the historical Sentinel complete-pipeline mean from 204.324 to 57.049 ms while backend inference changed only slightly. Before optimization, Python post-processing hid much of the benefit of TensorRT. On the Scout, inference already dominated, so the same decoder change yielded a smaller pipeline reduction. Resource-aware design therefore requires profiling the complete software path on each platform rather than inferring the bottleneck from model architecture alone.

No energy-efficiency ranking is supported. Laptop resource samples combined host-wide and device-wide quantities over unequal durations, while the physical complete-stack record did not include calibrated power measurements. The 30-minute temperature and resource series supports bounded thermal and memory observations, but not a causal throttling or energy-efficiency conclusion. The strengthened device screen remains necessary before all candidate models can be compared on common platform-specific objectives.

8.3 Contribution and cost of the tiered workflow

The primary architectural contribution is the explicit division of responsibilities. The Sentinel performs initial hazard detection, persistence filtering and request publication. The Scout acquires a current frame, makes a second model-based judgement and returns a correlated verdict. The Sentinel then records the final decision. Correlation fields, typed outcomes and asynchronous logging turn two unrelated detector calls into an auditable decision workflow.

This division provides an additional rejection opportunity, but the Scout is not independent physical ground truth. Both nodes use RGB imagery and learned fire detectors, so their errors may be correlated through appearance, lighting, domain shift or shared training sources. A thermal, particulate or gas sensor would provide a different physical measurement; it was not part of the tested verification path.

The strong- and small-flame observations show why a second decision must not be assumed beneficial in every condition. The stronger exposure produced 13 confirmations from 13 marked requests, while the smaller unstable exposure produced one confirmation from ten. Possible explanations include apparent flame area, motion, viewpoint, focus, exposure, threshold calibration and training-domain mismatch.

The trial does not isolate their causal effects. It does establish that the Scout can reject requests that the Sentinel has already promoted, including requests associated with a real but difficult visual target.

The communication time was small relative to Scout inference in the bounded campaigns. Typical camera-free round-trip medians were approximately 5–6 ms, and the loaded 1,000-event median was 4.393 ms, whereas Scout inference was approximately 1.25–1.32 s in the accepted live conditions. The earlier 400.453 ms communication tail remains operationally relevant: timeouts based only on a median would ignore observed variability. Under loaded, moving or interfered conditions, the communication share could change, so the present ratio is not generalized beyond the retained stationary setup.

The tiered workflow also creates new failure surfaces: request loss, unavailable Scout, stale or duplicate verdicts, incompatible clock interpretations, timeout selection, and disagreement handling. The architecture addresses several of these with event identity and explicit terminal states, but it does not remove them. Its contribution is a testable and traceable coordination boundary, not automatic reliability from adding a second vehicle.

8.4 Onboard execution and warehouse-scenario implications

The implemented perception and decision path is onboard even though the physical experiment is not aerial. The Sentinel performs capture, TensorRT inference, filtering, coordination and finalization on its Jetson Nano, while the Scout performs camera acquisition and TFLite verification on VOXL 2. The Sentinel-hosted SwarmNet access point carries compact ROS 2 records between them. The laboratory laptop supports setup, observation and evidence transfer; it is not the required inference or final-decision computer. This distinction is central to the title: the work evaluates distributed edge computation on UAV hardware, but physical flight remains outside the accepted evidence.

Onboard processing has four principal advantages for the intended scenario. First, it avoids continuous raw-video transmission and reduces the radio load to alerts, verdicts and final decisions. Second, perception does not require an external Internet or cloud service. Third, imagery can remain on the vehicle, which may simplify privacy and site-data handling. Fourth, different runtimes can be optimized for the two heterogeneous computers. The observed SwarmNet round-trip times show that compact decision messages were inexpensive relative to Scout inference in the stationary tests; they do not establish the same relationship under flight or industrial radio interference.

The same design creates costs. Compute, camera and radio load draw from the vehicle energy budget; the added Sentinel computer, regulator, camera and mounting contribute payload; and two embedded environments must be deployed, monitored and recovered. No calibrated electrical-power or flight-endurance measurement quantifies those costs in this study. The Sentinel is also both the access-point host and the final-decision coordinator. Loss of that node or its wireless service can remove the coordination path even if the Scout can still execute local inference. The architecture is therefore distributed in computation but not free of concentrated dependencies.

A warehouse-like deployment adds conditions not reproduced by the laboratory. Metallic structures and machinery can produce occlusion and multipath radio propagation; aisles constrain line of sight and safe trajectories; illumination can change abruptly; and smoke can obscure the RGB appearance on which both detectors depend. Indoor GPS denial would require a validated local estimator, shared frame convention, camera calibration, obstacle sensing and safe-standoff planner. The simulator exercised scenario-defined local coordinates, not this physical localization chain. Local onboard processing is consequently a reasonable architectural response to constrained connectivity, but the present evidence does not prove coverage, collision safety or hazard-detection reliability in a warehouse.

8.5 Failures and alternative explanations

The evidence contains failures at data, model, conversion, software, integration and mission levels. Treating them as part of the result clarifies which changes improved the system and which questions remain uncertain.

The validation-to-test rank change may arise from domain composition rather than training instability alone. The source-family split reduces direct leakage, but reconstructed identities cannot capture every related frame or scene. Annotation judgement, retained negatives and the scarcity of COCO-defined small boxes can also change metrics. One seed per configuration does not separate model choice from training variance.

Conversion failures have several plausible sources: exporter graph changes, preprocessing differences, coordinate conventions, suppression details and runtime kernels. The Scout live bring-up supplied a concrete instance. The camera stream required explicit UYVY decoding, and the model output used normalized coordinates while the shared decoder initially expected model-pixel coordinates. Those defects produced repeated zero-confidence outcomes until the interfaces were inspected and corrected. The early records are therefore configuration failures, not evidence that the detector could not see flame.

The 0.05 Scout threshold was selected after observing diagnostic illuminated and unlit scores. It made the later proof-of-function trials possible, but it was not preregistered

or calibrated on a representative Scout validation set. This choice can explain part of the observed confirmation pattern and prevents the live counts from becoming an unbiased accuracy estimate.

Hardware state also affected execution. The Scout battery depleted and the device rebooted, clearing volatile deployment files. Three of five repeated negative JPEG payloads were lost after their hashes had been logged, although the structured event records remained complete. The incident supports the need for persistent staging and transfer checks. It does not invalidate the retained scores, but it limits later visual inspection of those three frames.

The simulation pre-fix state error and temporary arming rejections show two different failure classes. The first was a controller defect corrected in code. The second came from PX4 health readiness and required no bypass. Combining them under a single mission-failure count would conceal whether the remedy is a software change, a preflight wait policy or a simulator configuration update.

8.6 Threats to validity

Construct validity

The detector operationalizes visible flame appearance in RGB images. It does not measure heat, combustion products or particulate concentration. A fire bounding box is therefore not proof of a physical fire. Physical smoke was not evaluated in the final two-device campaign and is outside the validated hazard scope.

Hazard verification is also limited to a second RGB model judgement. The final alarm was a logical decision and retained log output, not an independently certified emergency alarm. Image-space coordinates were transferred as event metadata, but the Scout used full-frame inference and did not use those coordinates to crop, navigate or localize a target.

Internal validity

The physical trials were stationary, with propellers removed and a handheld lighter approximately 20–25 cm from the cameras. Camera height, flame stability, focus, exposure, viewpoint and battery state were not fully controlled. The accepted positive intervals were marked independently, but requests within one interval were temporally dependent. Historical unsynchronized lighter trials cannot supply ignition-to-detection latency or frame-level false-negative counts.

Framework-specific preprocessing and decoding could not be made identical for all six candidates. The fidelity programme measures some of that risk but does not remove native-framework differences. Laptop runs also occurred sequentially and

may include operating-system or display workload. These conditions restrict causal statements about architecture alone.

External validity

The curated internal images and one laboratory room do not represent the range of industrial sites, weather, lighting, occlusion, smoke, vibration, altitude, radio interference or fire scale. The lighter stimulus is especially different from an industrial hazard in fuel, geometry, heat and background context. The term industrial in the title identifies the application problem, not the validated deployment environment.

The development simulator used provisional airframe proxies and scenario-defined local-ENU targets. Its successful state transitions do not establish real estimator behaviour, aerodynamics, collision clearance or radio performance. Physical two-vehicle flight generalization remains outside the evidence.

Statistical conclusion validity

All six training configurations used one controlled seed. The grouped bootstrap resampled inferred source families and therefore reflects only within-dataset resampling variability under that grouping. It does not correct selection bias, label error, unknown dependence or between-seed variance. Overlapping intervals are not treated as a formal equivalence or difference test.

The live request counts are small and dependent within exposures. Fractions such as 13/13 and 1/10 are direct condition-specific observations rather than binomial estimates of population recall. The 300-event communication campaigns use serial messages from consecutive sessions, and 300/300 replies do not prove zero DDS packet loss because retransmission can occur below the application layer.

Table 8.2: Principal validity threats, their effect on interpretation and the validation needed to reduce them.

Threat	Consequence for this thesis	Required follow-up
RGB construct only	Verification does not establish physical combustion or smoke presence.	Add calibrated thermal, particulate or gas sensing under an approved safety protocol.
Small, stationary lighter trials	Counts do not estimate industrial or flight reliability.	Preregister repeated conditions, synchronized labels, varied target scale, distance, viewpoint and hard negatives.
One seed per model	Ranking may include training-run variance.	Repeat selected candidates over multiple frozen seeds.
Inferred source grouping	Residual relatedness and domain bias may remain.	Strengthen provenance and collect independently sourced evaluation data.
Incomplete device matrix	Platform-specific model support is unknown.	Complete fidelity-valid 20/200 cells on both devices.
Single complete-stack endurance run	Repeatability, calibrated power and airborne thermal behaviour remain unknown.	Repeat the registered run across controlled ambient and power conditions, then add staged flight only under an approved safety protocol.
Missing stage ablation	Filter and tiered-workflow effect sizes are unknown.	Replay the same labelled sequence through all three decision stages.
Development simulation only	No mission success probability or physical flight claim is supported. Qualitative review of the visual delivery checks presentation clarity, not simulator fidelity.	Freeze airframe parameters and execute the formal retained-run matrix before staged physical tests.
External manifests blocked	Transfer behaviour was not evaluated.	Close license, alignment, overlap and eligibility gates before inference.

8.7 Remaining validation and deployment boundaries

The current contribution ends at a stationary two-node perception workflow and development simulation. The Sentinel Pixhawk participated only through receive-only telemetry in the accepted complete-stack endurance run; no integrated arming, take-off, navigation, collision avoidance, return or landing was demonstrated on the real vehicles. The word swarm denotes coordinated heterogeneous roles in a two-node architecture; it does not imply a validated autonomous formation.

Localization is likewise incomplete. The Sentinel message carries image-space coordinates and event context, while the simulator uses scenario-defined local truth. A physical system would need camera calibration, a transform from image and vehicle frames to a shared world frame, estimator uncertainty and a validated safe standoff trajectory. Without those elements, the architecture supports target designation at the messaging level but not geographic localization or autonomous Scout dispatch.

Operational deployment would require more than additional accuracy data. It would need hazard-specific safety requirements, alarm deadlines, human oversight, fail-safe behaviour, data-governance rules, radio planning, environmental qualification and compliance with the applicable aviation and site procedures. Those requirements were not evaluated and are not inferred from the laboratory or simulator results.

The evidence supports the central hypothesis to a limited degree: roles can be divided across the tested heterogeneous edge nodes, and the implemented workflow can produce correlated confirmations and rejections. Evidence is insufficient to conclude that the division is sustainable beyond the measured stationary interval, improves

accuracy relative to a common baseline, transfers to eligible external domains or operates correctly as a physical two-vehicle industrial mission. Chapter 9 therefore states a conditional conclusion and prioritizes the missing validations rather than presenting the prototype as a completed deployment.

Evidence boundary. The discussion answers RQ1, RQ2, RQ3 and RQ4 only within their completed evidence layers. RQ2 has a bounded answer and RQ5 was not evaluated. No result establishes physical smoke detection, geographic localization, autonomous industrial flight or universal reliability.

CHAPTER 9

Conclusions and Future Work

This thesis investigated whether a tiered edge architecture can divide visual hazard-detection responsibilities between two heterogeneous, resource-constrained drone platforms. The Sentinel was assigned initial hazard detection, temporal filtering and target designation; the Scout was assigned secondary visual hazard verification. The work combined dataset curation, six-model screening, deployment conversion, device-specific integration, ROS 2/DDS communication, stationary two-node trials and development simulation. The evidence supports a functioning division of responsibilities, but only within the completed offline, stationary and simulated conditions.

9.1 Contributions and principal findings

The first contribution is an explicit tiered edge architecture whose roles, messages and decision states are traceable. A Sentinel alert carries a unique event identity and image-space context; a Scout verdict returns the corresponding identity, outcome and score; and the Sentinel records the final decision. This interface converts two detector executions into a correlated workflow and makes confirmation, rejection, timeout and degraded behaviour observable. The contribution is architectural and implementation-oriented: it does not by itself prove that a two-stage system is more accurate or reliable than a single detector.

The second contribution is an evidence-controlled model-development process. The curated fire-only study froze source-family-aware splits, screened six configurations, selected operating points from validation data and reserved the test split for one-time

evaluation. YOLOv8n-416 led the validation comparison with AP50–95 of 0.236 and F1 of 0.612, while EfficientDet led the one-time test comparison with AP50–95 of 0.168 and F1 of 0.442. The reversal, overlapping grouped-bootstrap intervals and single training seed preclude a universal ranking. Conversion fidelity added another selection constraint: both 416-pixel YOLO exports passed the frozen match-rate gate, whereas the two 640-pixel YOLO exports and the SSD export did not. Predictive ranking, runtime availability and deployment fidelity must therefore be evaluated separately.

The third contribution is the integration and profiling of the historical deployed pair. The final Sentinel TensorRT path used a 640-pixel engine and averaged 58.151 ms in the retained camera-free 100-iteration benchmark. The Scout TFLite path averaged 999.431 ms under its corresponding protocol. These measurements establish executable platform-specific paths, not a controlled model or hardware comparison. The older 416-pixel Sentinel benchmark remains a separate historical result and is not attributed to the final deployment. Decoder profiling further showed that software around the neural network can dominate end-to-end cost: vectorization reduced the historical Sentinel complete-pipeline mean from 204.324 to 57.049 ms, while the Scout remained inference-bound.

This deployed decision path was onboard in the computational sense: each vehicle acquired and processed its own camera stream, and the Sentinel hosted coordination and final-decision handling. The laboratory laptop supported configuration, observation and evidence retrieval but was not required for inference. This property strengthens the edge-computing contribution without changing the experimental boundary: the accepted physical trials remained stationary, and aerial mission execution remained simulated.

The fourth contribution is a bounded stationary demonstration of secondary verification. The Scout confirmed 13 of 13 correlated requests in a marked stronger-flame interval, confirmed one of ten in a smaller unstable-flame interval and rejected six of six forced unlit requests. Mean alert-to-final-decision time was approximately 1.44–1.46 s in the accepted positive conditions, with Scout inference accounting for most of the delay. These are condition-specific observations, not estimates of industrial accuracy. The forced unlit requests bypassed Sentinel detection, and the logs did not retain the complete pre-filter candidate sequence needed to measure the effect of the temporal filter.

The fifth contribution is a documented coordination and simulation boundary. Two 300-event ROS 2/DDS campaigns returned all application-level replies, with median round-trip times of 5.878 and 5.257 ms; the earlier campaign also retained a 400.453 ms tail. A separate 1,000-event campaign under concurrent live inference load delivered every reply with 7.984 ms p95 RTT. Four five-repetition hardware fault cases demonstrated fail-closed camera-interface, stale-stream and unavailable-Scout

paths plus idempotent rejection of ten post-timeout verdicts. Development simulation completed two nominal and four selected-fault pilots and exposed a mission-state defect that was corrected and rerun. One later hash-linked visual run completed the confirmed-hazard sequence from patrol through Scout dispatch, verification, alarm publication and landing with distinct Sentinel and Scout proxies. Its final continuous delivery was reviewed and accepted for thesis and defence presentation, while the synchronized logs remained authoritative for event timing. These bounded cases do not provide a general reliability estimate, and the incomplete formal campaign does not support a claim of physical mission correctness.

9.2 Assessment of the central hypothesis

The central hypothesis is supported to a limited and conditional degree. The tested heterogeneous edge nodes can execute different perception roles, exchange correlated events and produce both confirmations and rejections in a stationary two-node workflow. The results also demonstrate that the division has a measurable cost: secondary verification adds roughly a second of Scout inference and increases final-decision latency. Typical communication time was small relative to that inference cost under the retained laboratory network, although the observed tail prevents a general low-latency guarantee.

The evidence does not establish four stronger interpretations of the hypothesis. First, the complete stationary service configuration survived a registered 30-minute run with both perception paths, communication, logging and flight services active, but it did not satisfy every criterion: the Scout p95 alert-to-verdict time was 2.618 s and Sentinel swap increased by 0.922 MiB. This supports bounded coexistence, not unlimited or airborne sustainability. Second, no common-sequence ablation measured whether the temporal filter and Scout stage reduce false alerts while retaining labelled hazards. Third, the simulation pilots and stationary physical trials do not demonstrate autonomous two-vehicle flight, geographic localization or industrial-site operation. Fourth, transfer performance was not evaluated because neither external data source reached a frozen eligible inference manifest.

The answer to the research problem is consequently narrower than the intended application. The project establishes a traceable prototype and a reproducible evaluation framework for heterogeneous role division. It does not establish a validated industrial hazard-response system. In particular, the final physical evidence concerns visible flame appearance in RGB imagery; physical smoke, heat, gas and particulate sensing were not evaluated.

9.3 Prioritized future work

Future work should first close the evidence gaps that prevent answers to the existing research questions. Architectural extensions should follow only after the current prototype has passed those checks.

Complete the registered validation programme

The first priority is the strengthened candidate-by-device screen. Each model admitted to a device must first pass the frozen conversion-fidelity gate, then complete the registered warm-up and measured iterations with artifact, configuration and runtime hashes. The completed 30-minute full-service study should then be repeated across controlled ambient and power conditions, adding calibrated power measurement and a sustained tiered-workflow load. One-second resource records should cover memory, temperature, CPU/GPU load, power, dropped frames, queue depth and service failures. These tests are required to answer RQ1 fully and to strengthen the conditional answer to RQ2.

The second priority is a common-stimulus stage ablation. A frozen labelled sequence should be replayed through single-frame Sentinel detection, the three-consecutive-frame temporal filter and Scout verification. The analysis should report event retention, false-alert burden, false-negative burden and decision latency at every stage, with the experimental unit and uncertainty method defined in advance. This directly answers the unresolved part of RQ3 without inferring filter performance from incomplete live logs.

The third priority is to complete the registered simulation campaign using frozen airframe parameters, scenario seeds, mission timeouts and pass/fail rules. Failures and retries must remain in the retained denominator. After the software and simulation gates pass, physical work should proceed in stages: bench integration with both flight controllers, restrained or propeller-free tests, controlled single-vehicle flight and finally supervised two-vehicle missions under approved safety procedures. This sequence is necessary before RQ4 can extend beyond development simulation.

The fourth priority is external evaluation. FASDD-UAV requires acquisition, licence and provenance checks. MmodalFire requires closure of timestamp, event-label, overlap and eligibility gates. Only a frozen eligible manifest should enter inference. Results should be reported per source domain and target scale, with overlap exclusions and unavailable conditions visible, so that RQ5 measures transfer rather than hidden data relatedness.

Improve perception, localization and operational readiness

After the validation programme, perception should expand beyond a second RGB fire detector. Calibrated thermal, particulate or gas sensing would supply a measurement with a different physical basis and reduce dependence on correlated visual errors. New data collection should cover smoke, larger standoff distances, small targets, glare, clouds, occlusion, motion blur and industrial backgrounds. Repeated frozen seeds and independent evaluation sources would separate model-selection effects from training and split variability.

The current target designation must also be converted from image-space metadata to a shared spatial estimate. This requires camera calibration, vehicle pose, time synchronization, frame transforms, uncertainty propagation and a validated safe-standoff planner. The Scout should use the resulting region or world estimate for directed observation rather than full-frame inference alone.

Operational readiness additionally requires hazard-specific alarm deadlines, human oversight, fail-safe policies, persistent artifact staging, radio planning, data governance, environmental qualification and compliance with applicable aviation and site procedures. These are requirements for future evaluation, not properties of the current implementation.

Evidence boundary. No new result is introduced. RQ2 has a bounded answer and RQ5 was not evaluated; physical smoke verification, geographic localization, autonomous industrial flight and universal reliability remain outside the evidence.

APPENDIX A

Reproducibility Configuration and Interfaces

This appendix identifies the repository contracts required to reproduce the accepted analyses. It is an index to the preserved source and evidence rather than a replacement for the raw bundles. Repository-relative paths are given so that a result can be traced without copying large logs or model artifacts into the manuscript. Every rerun should use a new output directory and must not overwrite the source evidence.

A.1 Evidence and configuration hierarchy

The reproduction hierarchy has four layers. Raw device, training and simulator outputs are the primary observations. Machine-readable manifests bind those outputs to file identities and hashes. Analysis scripts derive tables and figures from the manifests. The thesis reports the accepted values and their interpretation boundaries. When these layers disagree, the raw artifact and its contemporaneous manifest take precedence over prose summaries.

Table A.1: Principal frozen contracts used by the integrated thesis.

Repository-relative path	Role	Status represented in this thesis
`manifests/fire_only_curated_v7.freeze.json`	Dataset identity, counts and audit hashes	Frozen fire-only source-family split used for the final six-model study.
`config/six_model_validation_v1.json`	Common validation contract	Frozen 700-image validation evaluation and threshold-selection rule.
`config/model_deployment_registry_v1.json`	Native model registry	Six screened native artifacts; the strengthened platform matrix was not executed.
`config/exact_device_screening_v1.json`	Device-screen protocol	Registered 20-warm-up/200-measurement protocol; not completed as a six by two matrix.
`ros2_ws/src/drone_enhanced_nodes/config/runtime.yaml`	Node defaults and safety controls	Versioned defaults only; executed trials retain their own launch parameters and logs.
`config/formal_simulation_campaign_v1.json`	Formal simulation campaign	Preregistered campaign; development pilots are reported separately because the complete campaign was not executed.
`config/external_evaluation_sources_v1.json`	External-source gates	Readiness and eligibility contract; no external inference result was admitted.
`evidence/manifests/evidence_manifest_20260804.csv`	Evidence inventory	File size, UTC modification time and SHA-256 inventory for the preserved evidence snapshot.

The frozen dataset contains 4,991 records: 3,558 training, 700 validation and 733 test images. The record manifest SHA-256 is `478803efb7f8f4f6c3fe74d58658958bccfa2d0f58ec50a1a5c16535e393a7dd`. The validation COCO annotation SHA-256 recorded in the common contract is `1a93f800d63bf355bb586039da245cd612cbdd7052102cbfd6da1409312ab7c6`. These identities distinguish the final curated experiment from the earlier historical Sentinel and Scout datasets.

A.2 Reproduction stages and entry points

A complete rerun proceeds in the following order. Each stage consumes the preserved output of the preceding stage and writes a new derived bundle.

1. **Validate the contracts.** Use `tools/validate_strengthening_config.py`, `tools/validate_screening_contract.py` and `tools/validate_deployment_and_external_contracts.py` before training or evaluation.
2. **Materialize the frozen data.** Dataset construction and export are implemented by `tools/build_curated_research_manifest.py`, `tools/materialize_training_dataset.py`, `tools/export_training_coco.py` and `tools/export_training_tfrecord.py`. The generated counts and hashes must match the frozen manifest.
3. **Run model evaluation.** The framework-specific prediction entry points are `tools/predict_yolo_coco.py`, `tools/predict_tfod_coco.py` and `tools/predict_tflite_coco.py`. Common metric calculation uses `tools/evaluate_coco_predictions.py`; grouped F1 intervals use `tools/grouped_bootstrap_detection.py`. Thresholds are chosen on the validation split and must not be retuned on the opened test split.

4. **Validate conversion fidelity.** Candidate generation and checks use `tools/export_yolo_onnx_candidates.py`, `tools/export_ssd_tflite_candidate.py`, `tools/validate_yolo_onnx_fidelity.py` and `tools/validate_prediction_fidelity.py`. A converted artifact that fails the frozen fidelity rule is excluded from equivalent-model device ranking.
5. **Analyze retained device evidence.** Backend, communication and integrated-workflow bundles are processed by `tools/analyze_backend_benchmarks.py`, `tools/analyze_ros_roundtrip.py` and `tools/analyze_integrated_trial.py`. These tools operate on completed evidence; they do not turn an incomplete run into an accepted result.
6. **Check thesis transcription.** The numerical cross-check entry point is `tools/verify_thesis_numeric_contract.py`. The manuscript is then compiled with pdfLaTeX and Biber, and its pages are inspected visually.

The scripts expose their current arguments in the repository and should be run from the repository root in the environment recorded for the corresponding experiment. Environment locks are stored under `config/` and `simulation/`. This appendix intentionally does not prescribe a single global Python environment because Ultralytics, TensorFlow Object Detection API, EfficientDet and ROS 2 were executed in distinct compatible environments.

A.3 Deployment artifact identities

Table A.2 records the six native candidates used by the deployment-preparation register. A native training artifact and a converted runtime artifact are different identities; neither may inherit the other's performance result without a recorded conversion and fidelity check.

Table A.2: Native candidate identities from the deployment registry.

Model ID	Input	Native artifact SHA-256
`yolov8n_416`	416	`854482d0fbd320e51575f0141a479e517d721ff6cd512d12c60d0acf30a35ff3`
`yolov8n_640`	640	`1c49f90ee7e635b9011ae1ec34a9cfccfea75f9b08941d7475d031587a0c0a33`
`yolov8s_416`	416	`7bf6137add173f51d015c18841b5903300db909d31fb82ea3f51f3498ee366c5`
`yolov8s_640`	640	`53019aa916dcbfa11f3f38a3cc16579270436e44fa7bd80ccd1fc37afb60f0f0`
`ssd_mobilenet_v2_fpnlite_320`	320	`ee20414e69c0473e17989f19724709708c9a49303e2e26e8eabe4d771cf9e229` (checkpoint index; the registry also records the data shard).
`efficientdet_lite0_320`	320	`948e71fa53fa9a740a2bca45f169280a9c6e6596a49bfcab65d0ad9cfda2f12c`

The final deployed Sentinel engine is the 640-pixel TensorRT artifact with input shape $[1, 3, 640, 640]$ and output shape $[1, 6, 8400]$. The historical 416-pixel engine has output

shape $[1, 6, 3549]$. The two identities remain separate throughout Chapters 5–9; the older latency measurements are never assigned to the final engine.

A.4 Correlated ROS 2 interfaces

The enhanced message definitions are stored under `ros2_ws/src/drone_enhanced_interfaces/msg/`. Table A.3 groups the fields by analytical purpose. Field names and types remain defined by the message files, which are the authoritative interface source.

Table A.3: Analytical purpose of the enhanced ROS 2 message fields.

Message	Principal fields	Purpose
`HazardAlert`	`experiment_id`, `trial_id`, `event_id`, hazard type, confidence, temporal count, frame/alert monotonic times, model hash and normalized target box	Identifies the Sentinel request and preserves the detection context.
`HazardVerdict`	matching IDs, verdict, hazard type, Scout confidence, receipt/frame/verdict monotonic times, model hash and reason	Returns a correlated Scout outcome without claiming independent physical ground truth.
`HazardDecision`	matching IDs, final decision, both confidences, detection/alert/verdict/decision monotonic times and reason	Records the terminal system decision and supports latency decomposition.

Both physical nodes retained `enable_flight_commands=false` in the evaluated stationary workflow. Image-space target coordinates were logged for traceability; the Scout used its latest full frame and did not navigate to or crop around the transferred box.

A.5 Platform integration and environment inventory

Table A.4 records the physical integration facts admitted to the thesis and prevents an undocumented installation detail from becoming a quantitative claim. The topology is reproducible at the component level; the missing electrical and geometric measurements remain explicit.

Table A.4: Physical integration record and measurement boundary.

Item	Retained fact	Unmeasured or unverified detail
Sentinel airframe	Customized Holybro X500 V2 with Pixhawk 6C	Installed mass, dimensions, rotor spacing, centre of gravity and payload margin
Companion power	Jetson Nano supplied from airframe power distribution through an intermediate DC–DC regulator	Regulator identity, input/output time series, efficiency and energy per inference
Flight-controller link	Three-conductor physical link; retained service uses `/dev/ttyTHS1` at 57,600 baud	Connector-level pin map and electrical signal verification
Sentinel camera	D435i attached to the Sentinel and connected to the Jetson; colour endpoint exercised	Camera-to-body translation/rotation, depth use and vibration response
Scout platform	ModalAI development platform with integrated VOXL 2, camera and PX4 services	Independently measured mass, geometry and flight-performance parameters
Physical safety state	Stationary aircraft with propellers removed and flight commands disabled	No physical integrated flight result

Table A.5 freezes the software-generation facts used to interpret the deployment. An unavailable version is stated as such rather than reconstructed from a later vendor image.

Table A.5: Onboard operating-environment inventory.

Layer	Sentinel	Scout
Compute	Jetson Nano 4 GB	VOXL 2 / QRB5165
Host OS	Ubuntu 18.04.6 on Jetson Linux R32.7.6 / JetPack 4.6.6, kernel `4.9.337-tegra`	Exact base distribution/version not retained in the accepted inventory
ROS 2	Foxy inside `dustynv/ros:foxy-ros-base-l4t-r32.7.1`	Native Foxy at `/opt/ros/foxy`
Build/runtime boundary	Docker 20.10.21, NVIDIA runtime, host networking, bind-mounted workspace	Native `colcon` workspace and vendor services
Perception dependencies	D435i, GStreamer, CUDA, TensorRT and Python node	MPA camera server/bridge, YUV/RGB decoder, TensorFlow Lite and Python node
DDS configuration	Cyclone DDS; SwarmNet `wlan0`; multicast; isolated domain per experiment; matching enhanced interfaces	

Table A.6 maps the principal processes to their host and communication role. Helper probes used in individual experiments are excluded because they are measurement tools rather than required members of the deployed hazard-decision path.

Table A.6: Principal onboard node and service deployment.

Process or service	Host	Publishes	Subscribes / dependency
Sentinel perception/coordinator	Jetson Foxy container	`/swarm/hazard_alert`, `/swarm/hazard_decision`	D435i frame, `/swarm/hazard_verdict`, TensorRT engine
Scout verifier	VOXL 2 native Foxy	`/swarm/hazard_verdict`	`/swarm/hazard_alert`, latest MPA image, TFLite model
MPA-to-ROS 2 bridge	VOXL 2 vendor environment	Scout image topic	ModalAI camera pipe and camera-server service
MAVProxy telemetry service	Jetson host	Laboratory telemetry forwarding	Pixhawk serial endpoint; not a perception-decision dependency
Asynchronous evidence writers	Respective edge node	Device-local JSONL and manifest files	Local node-event queues
SwarmNet access point	Sentinel host	Local wireless connectivity	Physical radio and access-point service
Laboratory laptop	External observer	Test commands and retained copies	Not required for onboard inference or final decision

A.6 Reproduction limits

Reproducibility does not remove the validity limits reported in Chapter 8. The device operating systems, drivers and hardware state can affect latency; training used one controlled seed per configuration; and the physical trials contain dependent requests within stationary exposures. Repeating a script can verify the recorded computation, but it does not make the sample representative of industrial conditions.

APPENDIX B

Experiment and Claim Register

This appendix connects the thesis narrative to the authoritative experiment register and the evidence accepted at the 30 September cutoff. "Completed with limitation" means that usable evidence exists but its sample, duration or conditions restrict the claim. "Historical proof-of-function" denotes implementation evidence that was not collected under a controlled quantitative protocol. "Not executed" and "not evaluated" identify registered work that contributes no positive result to the final evidence set.

B.1 Final experiment status

Table B.1: Experiment register used by Chapters 6–9.

ID	Status	Accepted evidence	Permitted interpretation
DATA-V7	Completed	Frozen 4,991-record source-family split and annotation/duplicate audits.	Leakage-controlled internal dataset; no claim of population representativeness.
YOLO-N416	Completed	Single-seed YOLOv8n 416-pixel training and frozen validation/test predictions.	One screened candidate; early stopping is part of its recorded procedure.
YOLO-N640	Completed	Single-seed YOLOv8n 640-pixel training with full-state resume.	Offline candidate result; not the final Sentinel engine benchmark unless the engine identity also matches.
YOLO-S416	Completed	Single-seed YOLOv8s 416-pixel training and frozen evaluation.	Internal predictive result and fidelity-valid ONNX candidate; no completed device cell.

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Table B.1 continued

ID	Status	Accepted evidence	Permitted interpretation
YOLO-S640	Completed	Single-seed YOLOv8s 640-pixel training and frozen evaluation.	Internal predictive result; its tested ONNX conversion failed the frozen fidelity gate.
SSD-M320	Completed	SSD MobileNet V2 FPNLite training, validation and test bundles; corrected bounded-signature evaluation.	Internal predictive comparator; the prepared TFLite conversion failed fidelity.
EFFDET-L0	Completed	EfficientDet-Lite0 checkpoint lineage, completed recovery and native TFLite export.	Internal predictive comparator and native TFLite candidate; laptop CPU runtime is not a device benchmark.
SIX-VAL	Completed	Six candidates on 700 validation images with thresholds selected by maximum object-level F1.	Validation ranking and operating-point selection only.
SIX-TEST	Completed	One-time evaluation of 733 test images at validation-frozen thresholds.	Held-out internal result; the test ranking is not used for post-test reselection.
HW-BACKEND-20260803	Completed	Exact deployed Sentinel TensorRT and Scout TFLite 100-iteration camera-free backend records.	Runtime of the historical deployed pair; not the strengthened six-model/two-device matrix.
ROS-RTT-20260729	Completed	300 correlated request–reply events, including the retained 400.453 ms maximum.	Application-level RTT under the first stationary SwarmNet campaign.
ROS-RTT-20260803	Completed	A separate 300-request campaign with 300 replies.	Application-level RTT under the later stationary campaign; not one-way latency or proof of zero DDS packet loss.
CAM-INF-20260928	Completed with limitation	Ten-minute live Sentinel and Scout camera/inference coexistence, 108 correlated verdicts and 1 Hz resources.	Partial-stack operation without the Sentinel Pixhawk; the 2.617 s Scout p95 failed the registered 2 s target.
ROS-LOAD-20260928	Completed with limitation	1,000 correlated replies under concurrent live inference load plus 36 current-frame witness verdicts.	Stationary application RTT; not link-layer loss, one-way delay or complete-stack flight evidence.
SCOUT-NOFRAME-20260928	Completed with limitation	Five correlated ERROR/no_frame verdicts and zero inference.	ROS camera-interface fault, not physical camera-hardware failure.
SCOUT-STALE-20260928	Completed with limitation	Three live preflight inferences followed by five correlated ERROR/stale_frame verdicts and zero post-stop inference.	Controlled bridge stop after proven readiness; not a natural fault-frequency estimate.
SCOUT-PROC-20260928	Completed with limitation	Five Sentinel TIMEOUT decisions while the Scout verifier was absent.	Synthetic detections at the dispatch boundary; not camera or network-disconnect evidence.
LATE-DUP-20260928	Completed with limitation	Ten post-timeout CONFIRMED verdicts were rejected with zero additional final decisions.	Deterministic late/duplicate injection, not a natural delay distribution.
FULL-POS-STRONG	Completed with limitation	Thirteen correlated requests in a marked stronger visible-flame interval; all were confirmed.	Proof of function for that exposure, not a population recall estimate.
FULL-POS-SMALL	Completed with limitation	Ten correlated requests in a marked small unstable-flame interval; one was confirmed.	Condition-specific sensitivity to a difficult visible target.
FULL-NEG-FORCED	Completed with limitation	Six injected unlit Scout requests; all were rejected.	Rejection of those forced requests; not an environmental false-positive rate or tiered-workflow ablation.
HIST-LIGHTER	Historical proof-of-function	Earlier unsynchronized stationary lighter logs.	Implementation history only; script start is not ignition time and no detection-delay or success-rate estimate is allowed.

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Table B.1 continued

ID	Status	Accepted evidence	Permitted interpretation
FLIGHT-DRYRUN	Completed with limitation	Local-frame path preview, hold/resume logic and safety-locked command adapter.	Software dry-run behaviour only; no estimator, GPS, controller or physical-flight evidence.
M8 development pilots	Completed with limitation	Two nominal and four selected-fault retained pilots plus one continuous-video confirmed-hazard v17 run using provisional Gazebo airframe proxies; the accepted video and corrected v6 contact-sheet delivery were retained for presentation.	Existence of the exercised branches, corrected fault transition, identity-isolated PX4 command paths, and one visually recorded 4.5 m Scout-standoff sequence; no mission-success probability.
m8_dual_px4_formal_v1	Not executed	Preregistered 310-run campaign and readiness audit.	Protocol definition only; no formal reliability result.
Exact-device matrix	Not executed	Frozen fidelity and 20/200 benchmark contract.	No supported candidate-by-platform ranking beyond the historical deployed pair.
Full-service endurance	Completed with limitation	Thirty-minute synchronized stationary complete-stack run on both devices, plus 36/36 corrected correlated verdicts during a bounded 180.245 s probe.	The Scout p95 alert-to-verdict time was 2.618 s and Sentinel swap grew by 0.922 MiB; therefore the run did not satisfy every preregistered acceptance criterion and does not establish unlimited or airborne sustainability.
Common-stimulus ablation	Not executed	Registered single-frame, temporal-filter and Scout-verification comparison.	No filter/tiered-workflow effect size for event retention or false-alert reduction.
External evaluation	Not evaluated	FASDD admission remained blocked; MmodalFire acquisition/decode evidence existed but eligibility gates remained open.	No external transfer-performance conclusion.

B.2 Principal claim-to-evidence mapping

Table B.2 states the strongest form of each principal claim admitted by the thesis. More general wording is intentionally excluded.

Table B.2: Principal claim register and limiting condition.

Claim	Admitted statement	Evidence	Limiting condition
C1	The frozen internal dataset and source-family split support a controlled six-model comparison.	DATA-V7, frozen manifest and audit hashes.	Third-party sources, reconstructed groups and domain imbalance limit representativeness.
C2	The six candidates exhibit different predictive and laptop-runtime trade-offs; no universal winner was identified.	SIX-VAL and SIX-TEST.	One seed per model, framework differences and overlapping grouped intervals.
C3	Conversion success is not sufficient evidence of equivalent model behaviour.	Two 416-pixel YOLO ONNX passes and three recorded fidelity failures.	The fidelity sample and conversion paths are specific to the registered calibration protocol.
C4	The final 640-pixel Sentinel TensorRT path and the historical Scout TFLite path execute on their assigned devices.	HW-BACKEND-20260803.	Camera capture, communication and complete services were excluded from the 100-iteration backend timing.

Continued on next page

Table B.2 continued

Claim	Admitted statement	Evidence	Limiting condition
C5	The stationary workflow produces correlated Scout confirmations and rejections.	`FULL-POS-STRONG`, `FULL-POS-SMALL` and `FULL-NEG-FORCED`.	Small dependent samples, handheld lighter, stationary platforms and post-diagnostic Scout threshold.
C6	Application-level communication was typically short relative to Scout inference in the retained setup, while specific fail-closed coordinator paths rejected or finalized the registered faults.	Two 300-event RTT campaigns, the loaded 1,000-event campaign and four repeated fault cases.	Stationary SwarmNet conditions; no loaded-flight, natural fault-rate or interference generalization.
C7	Nominal and selected-fault mission branches are executable in development simulation.	Six retained pilots and pre-fix/post-fix fault evidence.	Provisional proxies and no completed repeated campaign or physical flight.
C8	The central hypothesis is supported only for role division in the tested stationary two-node workflow and development simulation.	C1–C7 as bounded by Chapters 7–8.	Sustainability, comparative tiered-workflow benefit, external transfer, localization and physical industrial operation remain unestablished.

B.3 Research-question closure

The final evidence supports bounded answers to RQ1–RQ4. RQ1 identifies different internal predictive and runtime trade-offs and demonstrates the historical deployed pair, but it does not provide a complete candidate-by-device ranking. RQ2 establishes more than 30 minutes of stationary concurrent-service operation, while retaining the failed Scout latency and strict Sentinel swap-growth criteria. RQ3 establishes correlated, condition-sensitive Scout verdicts and their decision latency, but not the effect size of the temporal filter or tiered workflow. RQ4 establishes executable nominal and selected-fault development paths in simulation, but not a mission-reliability estimate. RQ5 was not evaluated because neither external source reached a frozen eligible inference manifest. These dispositions are used consistently in the abstract, results, discussion and conclusion.

APPENDIX C

Supplementary Analyses and Failure Chronology

This appendix preserves the negative and incomplete outcomes that constrain the final interpretation. The chronology is analytical rather than exhaustive: raw timestamps, command transcripts and hashes remain in the repository evidence. A later successful attempt does not erase an earlier failure when that failure explains a design change or a validity boundary.

C.1 Failure and correction chronology

Table C.1: Retained failures, corrections and consequences for the thesis.

Date/phase	Observation	Correction or disposition	Consequence for interpretation
29 July device audit	The earlier Sentinel benchmark artifact had 416-pixel input and output shape [1, 6, 3549], while the final ROS workspace used a 640-pixel engine with output shape [1, 6, 8400].	Both engine identities were retained and the final 640-pixel path was benchmarked separately.	Historical 416-pixel latency is never reported as a measurement of the final deployed engine.
Scout camera bring-up	The camera published UYVY/YUV data and the model returned normalized coordinates while the shared decoder initially assumed a different representation. The mismatch produced repeated zero-confidence outcomes.	The image decode and coordinate convention were inspected and corrected before the accepted trials.	Early zero-confidence records are configuration failures, not evidence of detector incapability.

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Table C.1 continued

Date/phase	Observation	Correction or disposition	Consequence for interpretation
Scout calibration	A 0.05 live threshold was selected after diagnostic illuminated and unlit observations.	The threshold was recorded and used for the bounded proof-of-function trials.	It was not preregistered on an independent Scout validation set; live counts are not unbiased accuracy estimates.
Scout power and storage	The Scout battery depleted and the device rebooted, clearing volatile deployment files. Three of five repeated negative JPEG payloads were unavailable after their hashes had been logged.	Structured event records and the two preserved images were retained; the missing images were not reconstructed.	The accepted scores remain usable, but later visual inspection of three payloads is impossible.
YOLO ONNX fidelity	YOLOv8n-640 obtained a 0.9496 match rate against a frozen minimum of 0.95; YOLOv8s-640 also failed.	Both outcomes remained failures; the criterion was not rounded or relaxed after observation.	Neither failed ONNX artifact enters equivalent-model device ranking.
SSD TFLite fidelity	The exported SSD artifact achieved a 0.8714 match rate and failed the registered fidelity rule.	The artifact and report were preserved and the candidate was excluded from device ranking through that conversion.	Export and runtime allocation are insufficient evidence of equivalent behaviour.
Decoder profiling	Python post-processing dominated the early Sentinel pipeline and masked much of the TensorRT benefit.	Vectorized decoding was implemented and compared against the retained baseline.	Complete-pipeline profiling is required; backend inference time alone can misidentify the bottleneck.
Simulation prefix pilots	Two fault pilots assigned `TAKEOFF` before Scout actuation when the system should have degraded from preflight.	The state assignment was corrected; v2 runs transitioned directly from `PREFLIGHT` to `DEGRADED`.	Pre-fix and post-fix evidence are both retained; only corrected runs support the repaired behaviour.
Simulation arming variability	A retained nominal scenario required six mode/arm attempts after temporary PX4 health rejections.	No safety bypass was introduced; the final accepted attempt and preceding variability were retained.	A final pass does not imply deterministic mission availability.
Visual-simulation revision	The earlier v7 screenshot-derived delivery was rejected for ambiguous role presentation, intrusive annotations and poor camera framing. A later v16 run completed the principal hazard branch but failed its 30 s landing bound.	Both attempts were retained. The v17 run used an explicitly recorded 60 s landing bound and a continuous native video; it passed identity, topic, terminal-state and capture gates. The video was qualitatively reviewed and accepted for thesis/defence presentation; the v6 derived delivery retained its hash and corrected the contact-sheet evidence.	v17 supports only one bounded development observation; visual acceptance adds no simulator-fidelity, physical-flight or reliability claim.
9 September readiness	The formal simulation matrix and external evaluations had not passed their respective readiness/eligibility gates.	Protocols and blockers remained visible at the evidence cutoff; no substitute run was admitted.	RQ4 has a bounded development answer; RQ5 was not evaluated.

C.2 Excluded and deferred evidence

Table C.2: Evidence excluded from quantitative conclusions or deferred beyond the cutoff.

Evidence	Reason for exclusion or deferral	Permitted use
Unsynchronized historical lighter logs	Ignition intervals and ground-truth frame labels are unavailable.	Implementation history and qualitative proof of function only.
Failed/unbounded retracing attempt	Evaluation did not satisfy the corrected bounded-signature execution contract.	Failure chronology; the corrected run supplies the accepted SSD result.
Fidelity-failed conversions	Converted predictions did not satisfy the preregistered equivalence gate.	Conversion-failure result, not device performance of an equivalent model.
Camera-free device loops	Camera acquisition, ROS 2 traffic and complete services were absent.	Backend/pipeline timing only; no sustainability conclusion.
Forced unlit Scout requests	Requests bypassed Sentinel detection and were not natural false alerts.	Verification-branch rejection behaviour only.
Pre-fix simulation pilots	Controller state assignment was known to be incorrect.	Defect evidence and comparison with the repaired v2 path.
Incomplete formal simulation campaign	The preregistered denominator was not executed.	Protocol and readiness status only.
MmodalFire preliminary readiness outputs	Timestamp consolidation, event-label and overlap gates remained open.	Acquisition and readiness evidence only; no transfer metric.
FASDD-UAV	Local acquisition and licence/admission requirements were not completed.	External-source plan only.
Physical flight and industrial operation	No accepted integrated two-vehicle flight or industrial-site campaign exists.	Future work; no present-tense performance claim.

C.3 Supplementary interpretation of timing

The timing layers in the evidence are intentionally non-interchangeable. Laptop inference-call latency excludes device runtime effects. The historical device backend benchmarks exclude camera capture, communication and final decision handling. ROS 2/DDS RTT measures an application request and reply on the Sentinel monotonic clock. Alert-to-final-decision latency includes the Scout verification path for a correlated live request. A ratio between two layers is reported only when its limited meaning is explicit.

Within the retained stationary trials, Scout inference dominated the final decision delay, while the median communication RTT was much smaller. The earlier 400.453 ms communication maximum remains relevant to timeout design, and the absence of loaded or moving-network conditions prevents a broader communication guarantee. Likewise, the short backend loops cannot establish steady-state temperature, power or memory behaviour.

C.4 Preservation and revision rule

Raw observations remain immutable. Cleaning, grouping, metric calculation and figure generation create new derived files with a recorded parent. Failed, interrupted and rejected attempts remain distinguishable from accepted runs. The evidence inventory records repository-relative path, byte size, UTC modification time and SHA-256 so that later revisions can identify exactly which artifact changed.

Results arriving after the cutoff may be incorporated only by revising every affected layer: experiment register, method, result table or figure, interpretation, abstract and conclusion. This rule prevents a later isolated number from silently changing the answer to a research question.

Evidence boundary. The supplementary chronology documents why particular records were accepted, limited, excluded or deferred. It does not add an industrial deployment, physical-flight, smoke-verification or external-transfer result.

Bibliography

[1] G. Vazquez, S. Zhai, and M. Yang, "Edge-friendly uav wildfire smoke and flame detection using transfer learning-enhanced lightweight deep learning models," *Sensors*, vol. 26, no. 10, p. 3197, 2026. DOI: 10.3390/s26103197

[2] Y. Qu et al., "Elastic collaborative edge intelligence for UAV swarm: Architecture, challenges, and opportunities," *IEEE Communications Magazine*, vol. 62, no. 1, pp. 62–68, 2024. DOI: 10.1109/MCOM.002.2300129

[3] X. Fan, F. Lei, and K. Yang, "Real-time detection of smoke and fire in the wild using unmanned aerial vehicle remote sensing imagery," *Forests*, vol. 16, no. 2, p. 201, 2025. DOI: 10.3390/f16020201

[4] I. E. Tampu, A. Eklund, and N. Haj-Hosseini, "Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images," *Scientific Data*, vol. 9, p. 580, 2022. DOI: 10.1038/s41597-022-01618-6

[5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788.

[6] G. Jocher, A. Chaurasia, and J. Qiu, *Ultralytics yolov8*, version 8.0.0, 2023. Accessed: Aug. 9, 2026. [Online]. Available: https://github.com/ultralytics/ultralytics

[7] W. Liu et al., "Ssd: Single shot multibox detector," in *Computer Vision – ECCV 2016*, Springer, 2016, pp. 21–37. DOI: 10.1007/978-3-319-46448-0_2

[8] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 4510–4520. DOI: 10.1109/CVPR.2018.00474

[9] M. Tan, R. Pang, and Q. V. Le, "Efficientdet: Scalable and efficient object detection," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 10 778–10 787. DOI: 10.1109/CVPR42600.2020.01079

[10] NVIDIA Corporation. "NVIDIA TensorRT 8.6.1 Developer Guide," Accessed: Aug. 4, 2026. [Online]. Available: https://archive.docs.nvidia.com/tensorrt/tensorrt-861/developer-guide/index.html

[11] TensorFlow Authors. "Post-training quantization," TensorFlow Model Optimization, Accessed: Aug. 9, 2026. [Online]. Available: https://www.tensorflow.org/model_optimization/guide/quantization/post_training

[12] NVIDIA Corporation. "Jetson nano technical specifications," Accessed: Aug. 4, 2026. [Online]. Available: https://developer.nvidia.com/embedded/jetson-nano

[13] ModalAI, Inc. "VOXL 2 Feature Matrix," Accessed: Aug. 4, 2026. [Online]. Available: https://docs.modalai.com/voxl2-feature-matrix/

[14] D. Qiao and F. Zulkernine, "Adaptive feature fusion for cooperative perception using LiDAR point clouds," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 1186–1195.

[15] H. Kang, X. Chang, J. Misic, V. B. Misic, J. Fan, and Y. Liu, "Cooperative UAV resource allocation and task offloading in hierarchical aerial computing systems: A MAPPO-based approach," *IEEE Internet of Things Journal*, vol. 10, no. 12, pp. 10 497–10 509, 2023. DOI: 10.1109/JIOT.2023.3240173

[16] S. G. Kong, D. Jin, S. Li, and H. Kim, "Fast fire flame detection in surveillance video using logistic regression and temporal smoothing," *Fire Safety Journal*, vol. 79, pp. 37–43, 2016. DOI: 10.1016/j.firesaf.2015.11.015

[17] Open Robotics. "Quality of service settings," ROS 2 Documentation, Accessed: Aug. 4, 2026. [Online]. Available: https://docs.ros.org/en/humble/Concepts/Intermediate/About-Quality-of-Service-Settings.html

[18] Open Robotics. "Different ROS 2 middleware vendors," ROS 2 Documentation, Accessed: Sep. 29, 2026. [Online]. Available: https://docs.ros.org/en/ros2_documentation/kilted/Concepts/Intermediate/About-Different-Middleware-Vendors.html

[19] S. Lee, T. Kim, J. Chae, and K.-J. Park, "Optimizing ROS 2 communication for wireless robotic systems," *arXiv preprint arXiv:2508.11366*, 2025. DOI: 10.48550/arXiv.2508.11366

[20] MAVLink Project. "MAVLink protocol overview," Accessed: Aug. 9, 2026. [Online]. Available: https://mavlink.io/en/about/overview.html

[21] Intel Corporation. "Intel RealSense D400 Series Product Family Datasheet," Accessed: Aug. 4, 2026. [Online]. Available: https://dev.intelrealsense.com/docs/intel-realsense-d400-series-product-family-datasheet

[22] yangin. "Drone-fire-smoke computer vision dataset." Roboflow Universe, version 1, CC BY 4.0, Accessed: Sep. 18, 2026. [Online]. Available: https://universe.roboflow.com/mohammadamins-workspace/drone-fire-smoke-o58av-nj8ag/dataset/1

[23] J. Bhathena. "Weather image recognition." Kaggle dataset, CC0, Accessed: Sep. 18, 2026. [Online]. Available: https://www.kaggle.com/datasets/jehanbhathena/weather-dataset

[24] V. Gupta. "Weather classification." Kaggle dataset, CC0, Accessed: Sep. 18, 2026. [Online]. Available: https://www.kaggle.com/datasets/vijaygiitk/multiclass-weather-dataset

[25] H. Xiao, *Weather phenomenon database (WEAPD)*, version V1, Harvard Dataverse, 2021. DOI: 10.7910/DVN/M8JQCR [Online]. Available: https://doi.org/10.7910/DVN/M8JQCR

[26] T.-Y. Lin et al., "Microsoft coco: Common objects in context," in *Computer Vision – ECCV 2014*, Springer, 2014, pp. 740–755. DOI: 10.1007/978-3-319-10602-1_48

[27] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979. DOI: 10.1214/aos/1176344552

[28] J. J. Harden, "A bootstrap method for conducting statistical inference with clustered data," *State Politics & Policy Quarterly*, vol. 11, no. 2, pp. 223–246, 2011. DOI: 10.1177/1532440011406233